

AIRPORT FWD DATA ANALYZER

Backcalculation

Training Presentation

FWD/HWD data review and pavement layer backcalculation

Branded edition of the supplied BAKFAA 3.4 presentation (March 3, 2023).
Technical source content is retained for engineering traceability.
Independent branded distribution - not an official FAA release.

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Presentation Outline

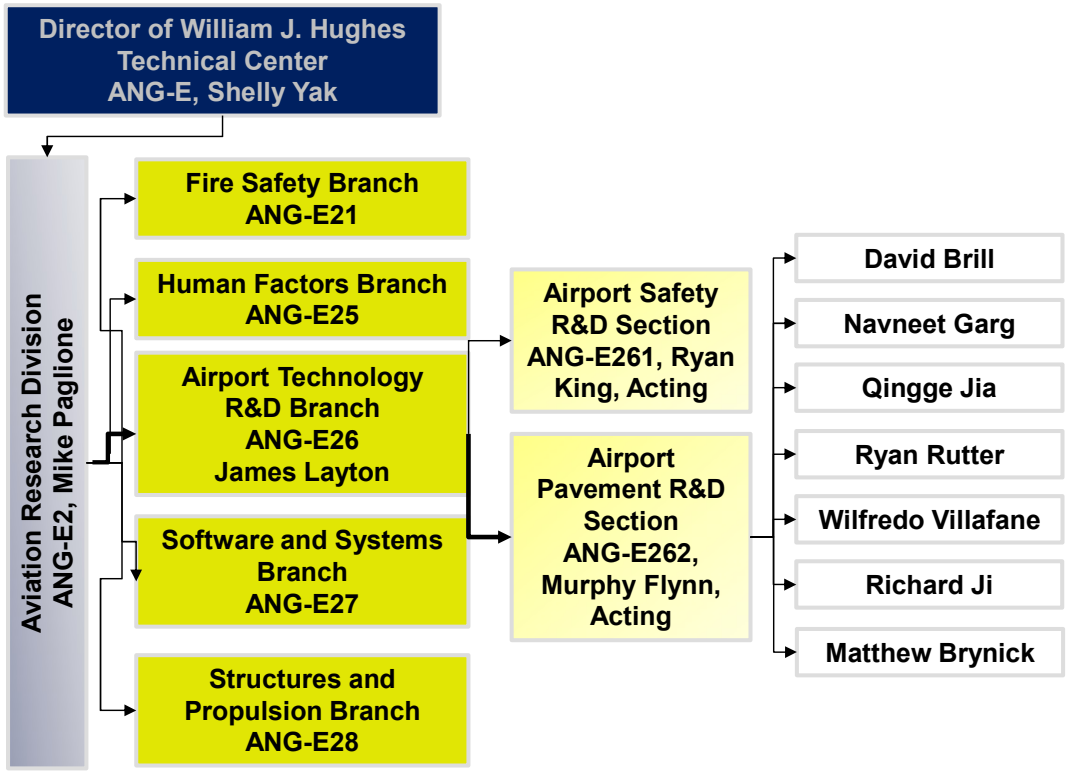
- FAA Airport Pavement R&D Overview
- BAKFAA Introduction
- BAKFAA Main Features
- BAKFAA Modular Interface
  - FWD Data
  - BAKFAA Inputs
  - Backcalculation Analysis
  - Load Transfer Efficiency (LTE) Calculation
  - Layered Elastic Analysis Program (LEAF)
- Software Demo
  - Flexible Pavement Example
  - Rigid Pavement Example
  - Composite Pavement Example
  - LTE Example
- Contact Us

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FAA Airport Pavement R&D Overview

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Who We Are

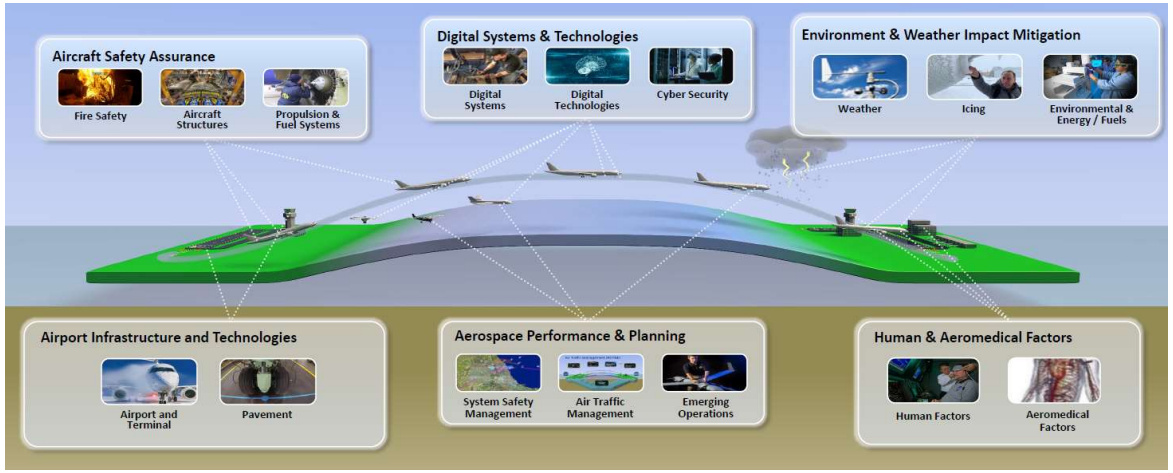


Pavement Consultant Support – ARA
Materials Lab O&M – ARA
Facilities O&M - GDIT

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What We Do

- **Aviation Research Division Mission**
  - Develop scientific solutions to current and future air transportation challenges by conducting applied research and development in collaboration with industry, academia, and government.



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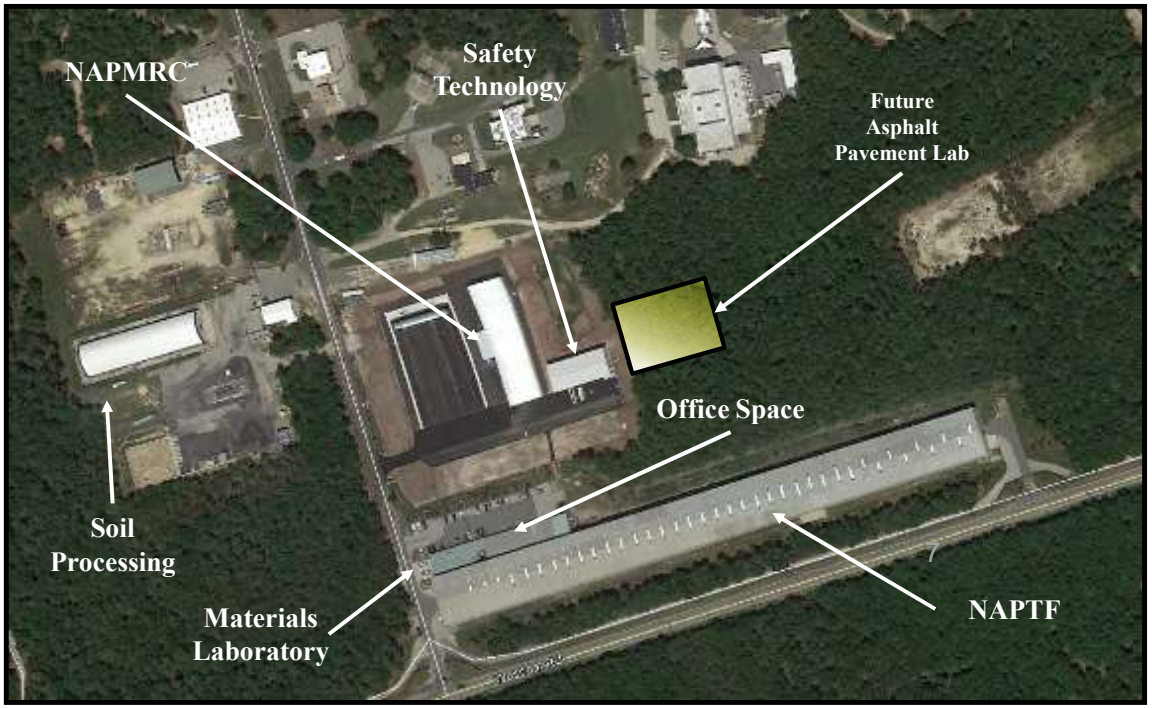
ANG-E262 Airport Pavement R&D

- **Research Conducted at the FAA William J. Hughes Technical Center, Atlantic City International Airport, NJ, USA.**
  - ANG-E262 Airport Pavement R&D Section
- **Sponsor: FAA Office of Airport Safety and Standards, Washington, DC.**
  - AAS-100 Airport Engineering Division (Michael Meyers)
  - AAS-110 Airport Engineering Division (Alberto Cruz, Harold Honey, Jeffrey Crislip)
  - AAS-120 Airport Data and Airspace
  - AAS-200 Airports Emerging Entrants Division
- **Provide Support for Development of FAA Pavement Standards**
  - Advisory Circulars
  - Engineering Briefs



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Airport Pavement R&D Section Facility Layout



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National Airport Pavement Test Facility (NAPTF)

NAPTF Facts:

- FAA / Boeing (CRDA) Partnership at $21M
- Opened April 1999
- Fully Enclosed Facility
- Accelerated Traffic Testing
- 900 ft. x 65 ft. of Test Pavement Surface
- Full-scale Pavement Structures and Landing Gear Loads





NAPTV Facts:

- Fully Automated & Programmed Wander Patterns
- Up to 5-dual wheel configuration
- Roughly 1.3 million lbs.
- Up to 75,000 lbs. per wheel

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National Airport Pavement Materials Research Center (NAPMRC)

NAPMRC Facts:

- Dedication Ceremony August 2015
- Indoor and Outdoor Testing Capability
- Accelerated Traffic Testing
- Outdoor: 150ft. × 300ft. & Indoor: 72ft. × 300ft.
- Accelerated resurfacing

Heavy Vehicle Simulator – (HVS-A) Facts:

- Temperature Control Capability up to 150°F
- Wheel Load Capacity 10,000 to 100,000 lbs.
- Single Wheel – radial aircraft tire size 52×21.0R22
- Dual Wheel (B737-800)
  - Fully Automated & Programmed Wander Patterns up to 6 ft.
- Test Speeds 0.17 to 5 mph



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FAA NextGen Pavement Materials Lab

- **2010: Laboratory Opened**
- **2013: AASHTO Material Reference Laboratory (AMRL)**
- **2013: Cement and Concrete Reference Laboratory (CCRL)**
- **Full Test Capabilities: Asphalt, Concrete, Aggregate, Soils**
- **Advanced Test Capabilities:**
  - Asphalt Pavement Analyzer (APA)
  - Asphalt and Concrete Flexural Beam Fatigue
  - Semi-Circular Beam (SCB)
  - Disk-Shaped Compact Tension (DCT)
  - High Temperature Indirect Tension (IDT)
  - Overlay Tester
- **Benefits to the NAPTF:**
  - Quality Control of Testing
  - Expedient Testing of Materials During Construction
  - Perform Advanced Materials Characterization On-site
  - Development of Performance Based Specification



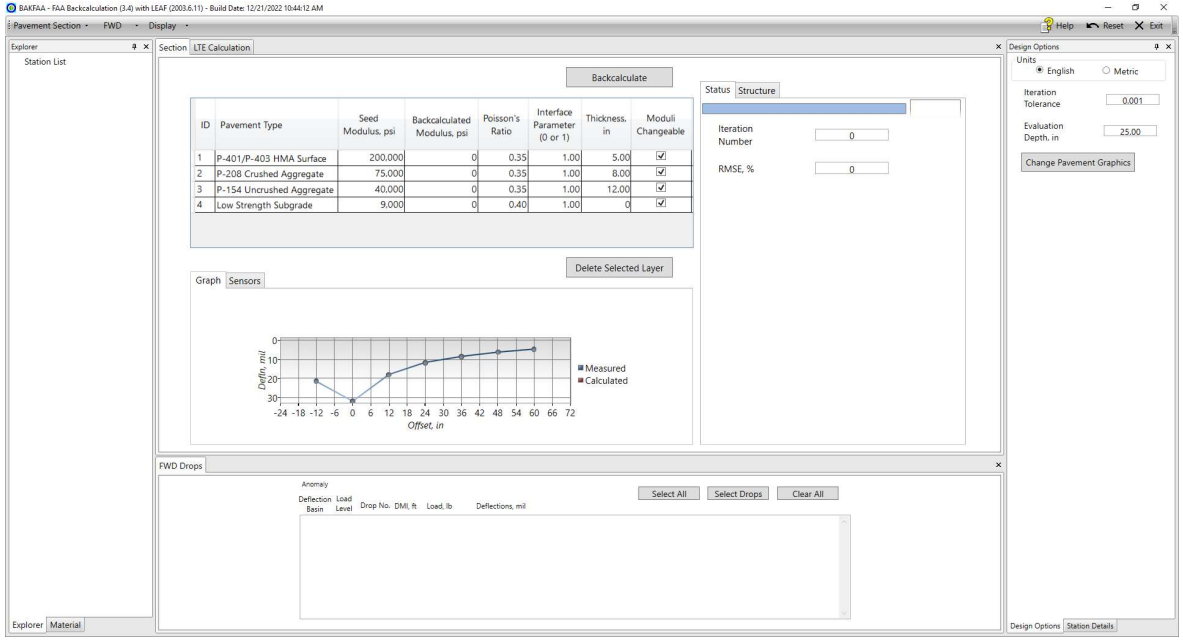
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BAKFAA Introduction

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About BAKFAA

- Computer program developed by the FAA for backcalculation of pavement layer modulus values using HWD/FWD data.
- Publicly available and free to use (source code available).
- Descriptions of use included in FAA AC 150/5370-11B 'Use of Nondestructive Testing in the Evaluation of Airport Pavements'.



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BAKFAA Main Features

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BAKFAA Main Features

- Designed as a Visual Basic .Net application
- Ability to switch between English and Metric units
- Includes a library of FAA standard pavement material
- Ability to import different FWD file formats from leading FWD manufacturers
- Converts a proprietary manufacturer's FWD file into PDDX format file
- FWD File Parser: file processing class to read proprietary manufacturer's file
- Ability to identify and flag FWD drops with anomalies
- Allows for editing FWD data and deleting FWD test points
- Ability to perform backcalculation analysis in single analysis or batch analysis modes
- Includes graphs of backcalculated modulus values and measured and calculated deflections
- Calculates and displays Load Transfer Efficiency (LTE)
- LEAF analysis to calculate pavement responses under aircraft gear or FWD load

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BAKFAA Modular Interface

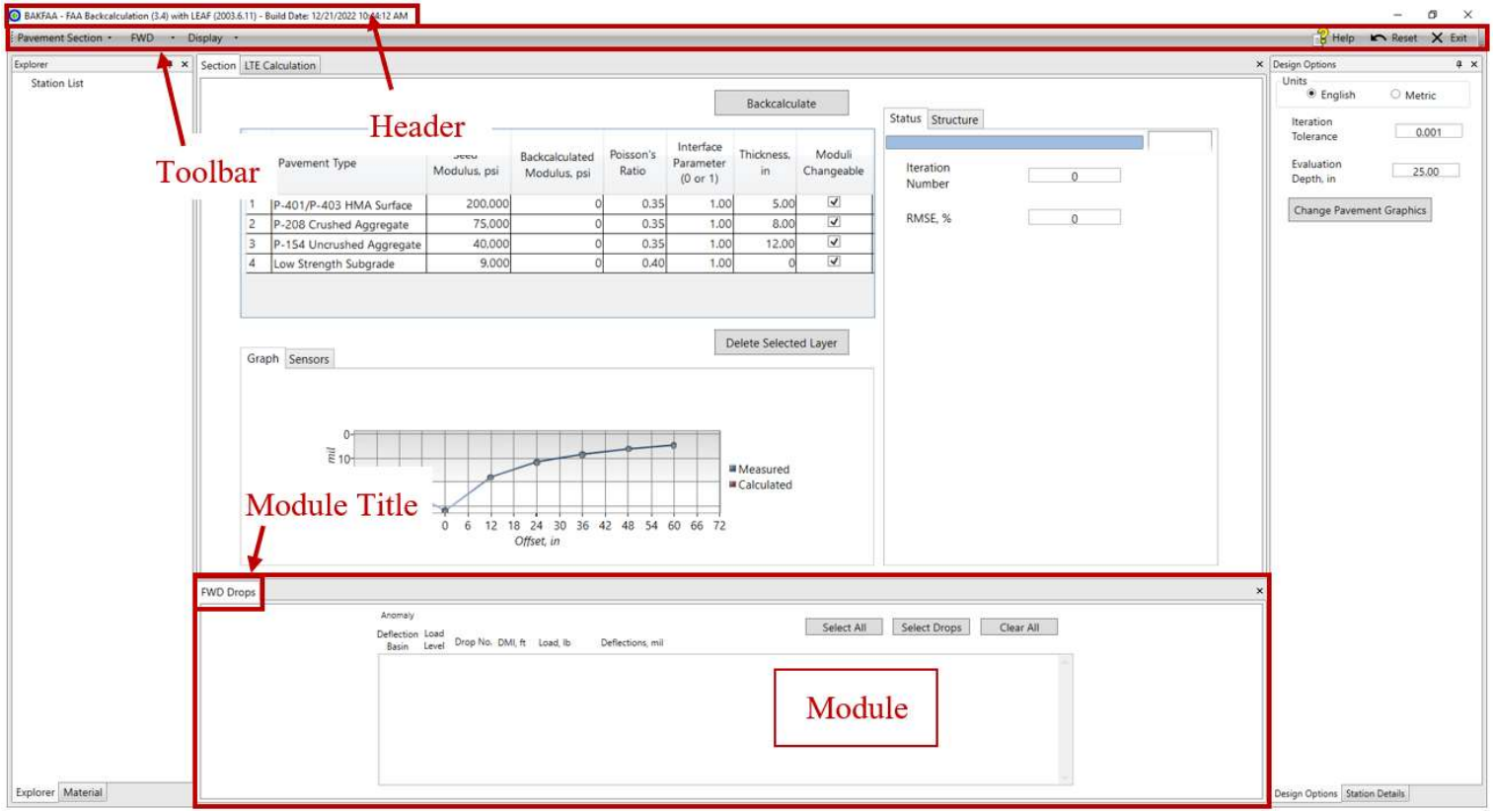
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BAKFAA Modular Interface

- Graphical user interface was overhauled
- Program functions are divided into ‘modules’ that serve a specific purpose
- Modules are user-configurable: resize, reposition, or remove as desired
- Main program controls relocated and organized in a toolbar
- BAKFAA Modules:
  - Section
  - FWD Drops/Select Load
  - Design Options
  - LTE Calculation
  - (FWD) Explorer
  - Material
  - Station Details

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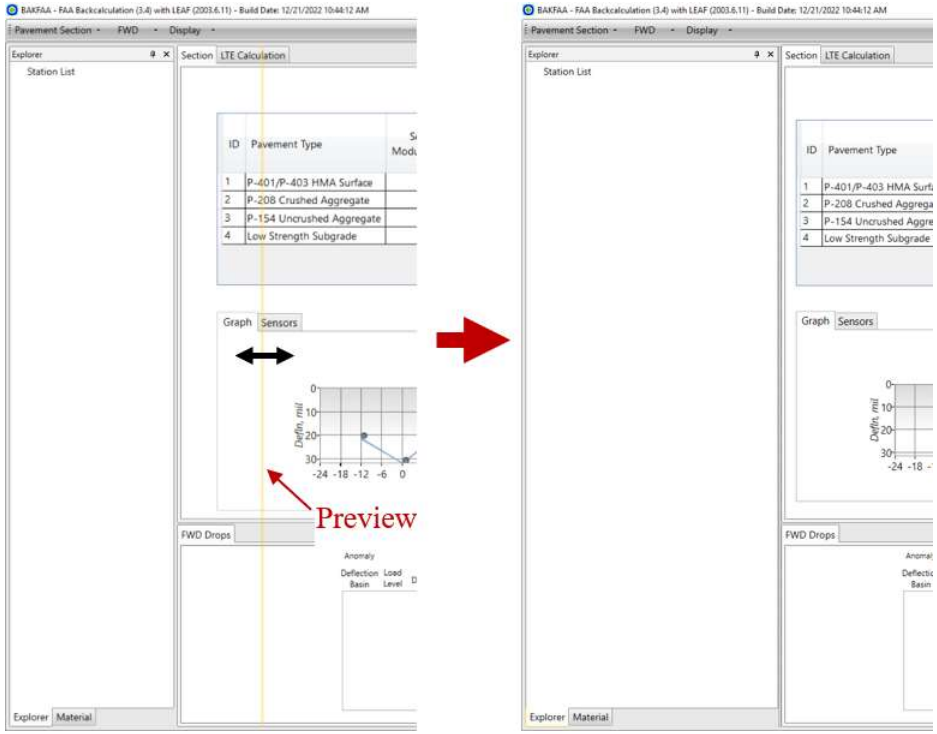
BAKFAA Modular Interface



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BAKFAA Modular Interface

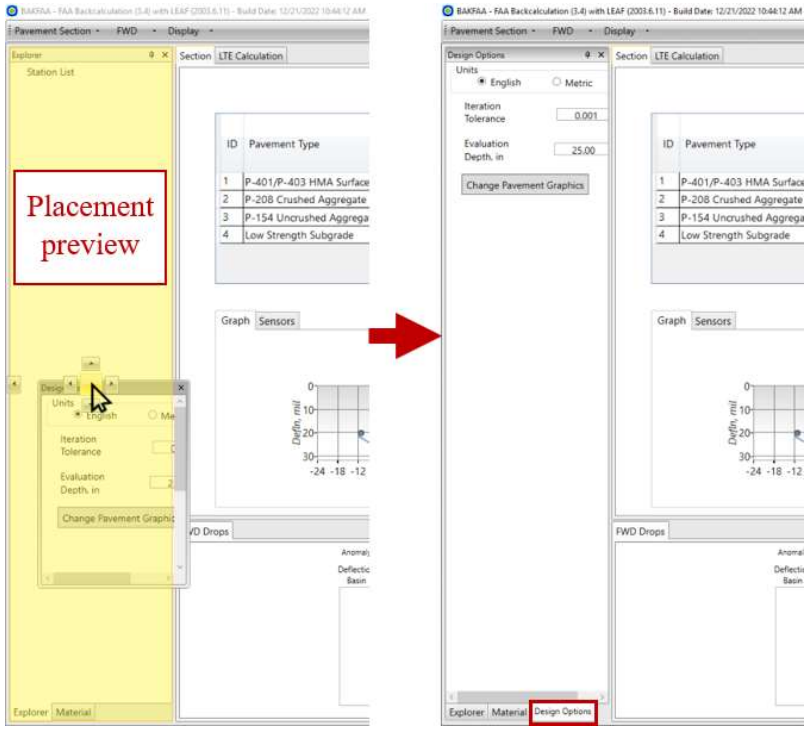
Module Adjustment



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BAKFAA Modular Interface

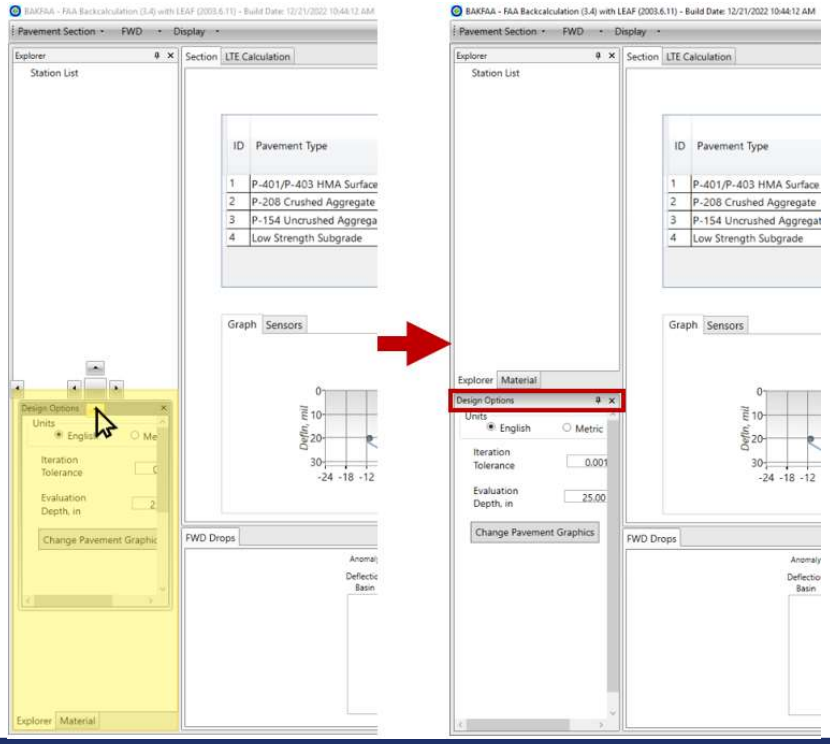
Module Adjustment



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BAKFAA Modular Interface

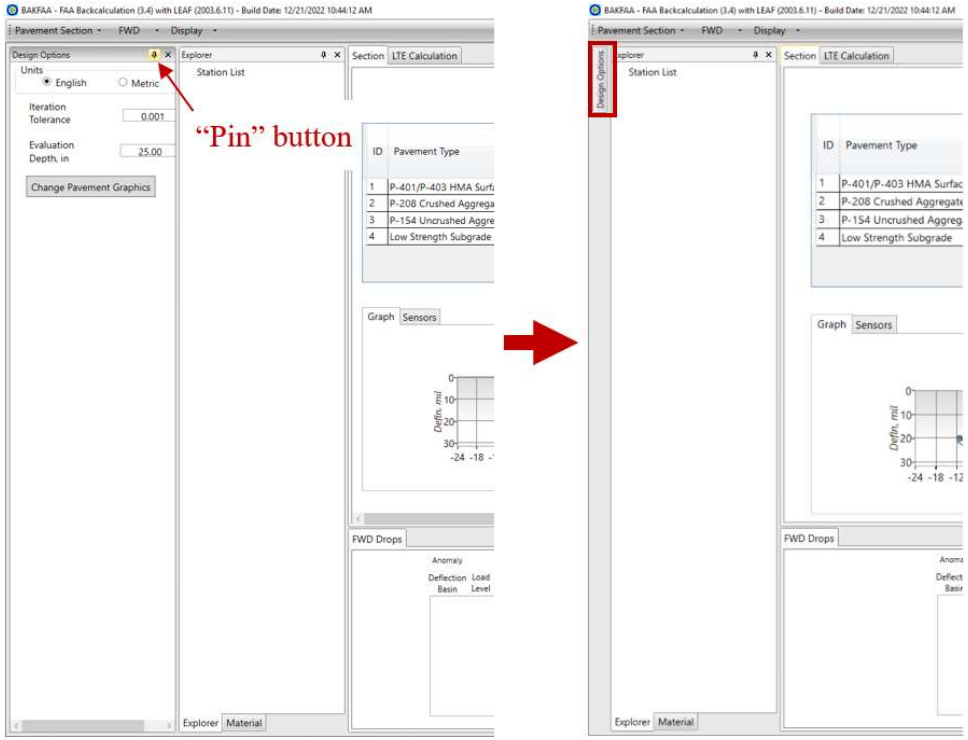
Module Adjustment



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BAKFAA Modular Interface

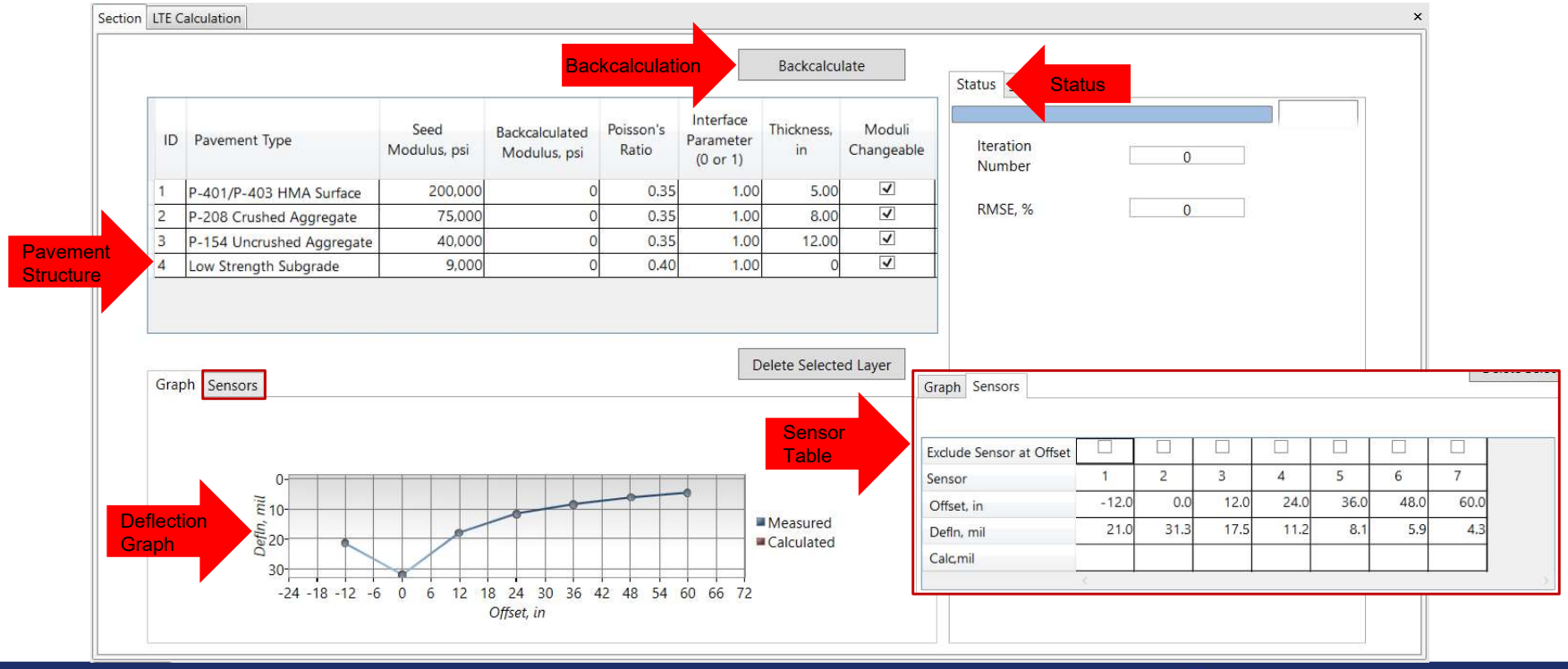
Module Adjustment



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BAKFAA Modular Interface

Section Module



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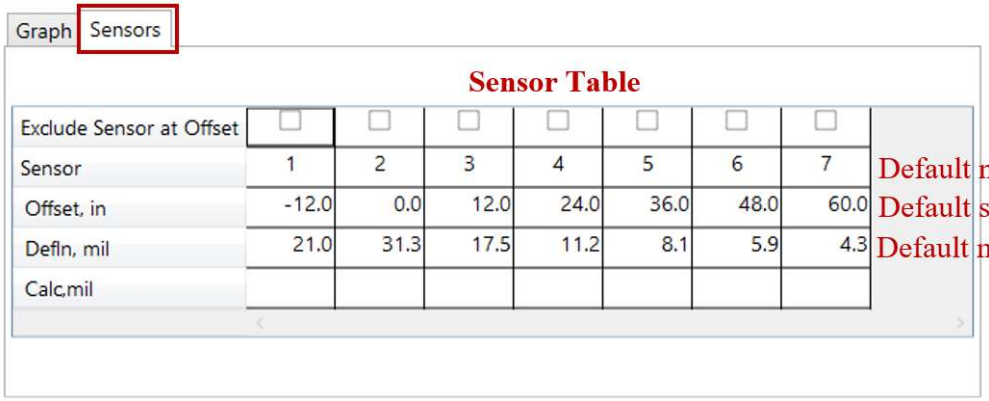
FWD Data

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FWD Data

FWD Test Information

- Sensor: sensor number
- Offset: sensor offset from FWD load plate
- Defln: measured FWD sensor deflection

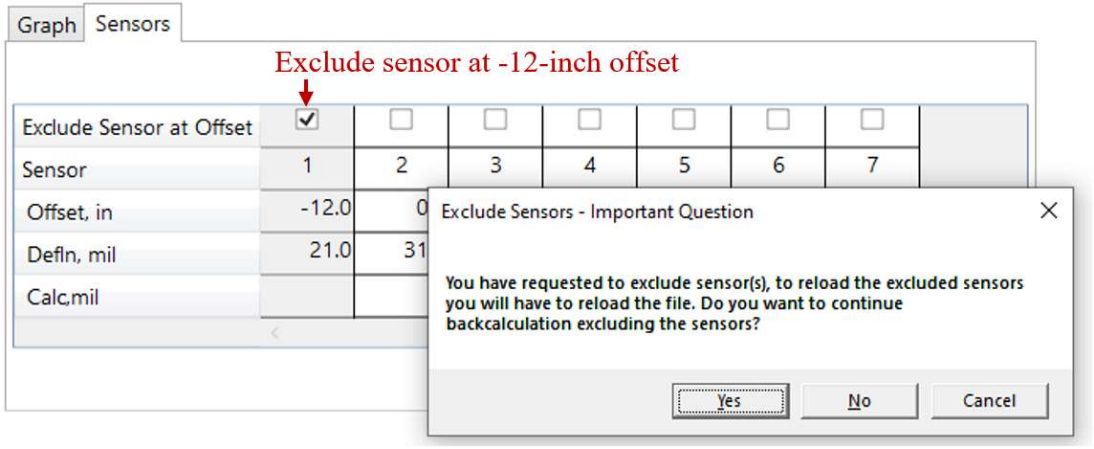


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FWD Data

FWD Information updates after Backcalculation Analysis

- Exclude Sensor at Offset: allows user to exclude any of the sensors in backcalculation analysis.

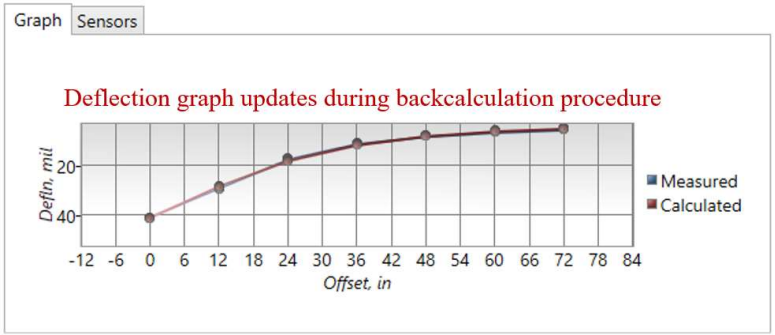


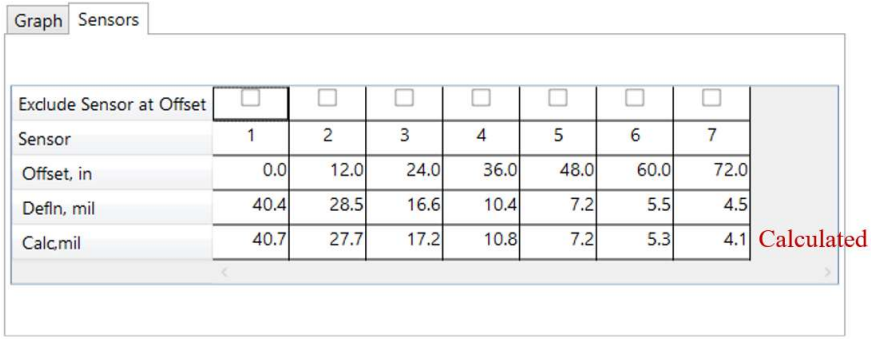
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FWD Data

FWD Information updates after Backcalculation Analysis

- Deflection Graph: plot of measured and calculated deflections.
- Calc: calculated deflections. Will update after the backcalculation is done.



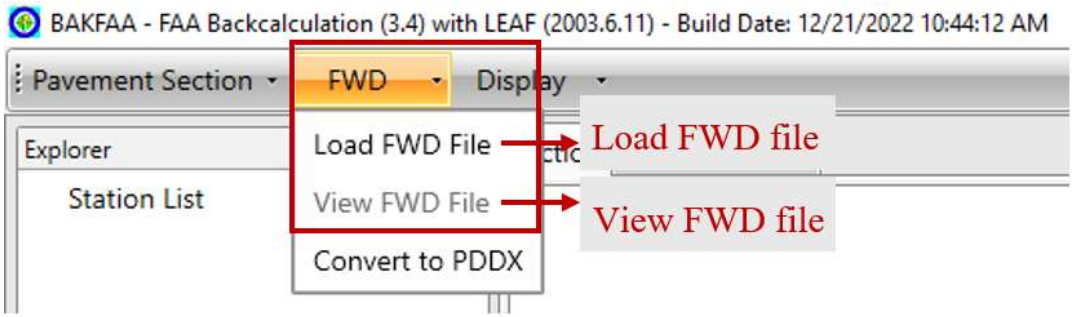


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FWD Data

Load/View FWD File

- Load FWD File: import FWD file
- View FWD File: open FWD file viewer



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FWD Data

FWD File Formats

- FWD File Formats:

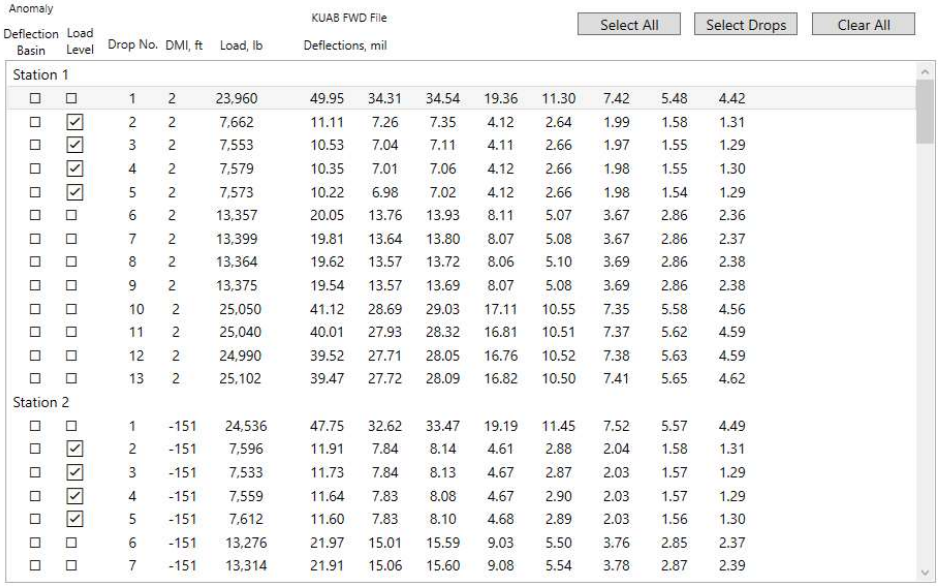
Equipment Manufacturer	Associated Data Format	File Extensions
Dynatest	R80, R32	.fwd, .hwd
	F20, F25	.fwd, .F20, .F25
	PDDX, XML	.ddx, .xml
KUAB	KUAB	.fwd, .hwd
JILS	M3, M5	.fwd, .hwd, .dat
PaveTest	PaveTest	.csv
CarlBro	CarlBro	.fwd, .txt
AASHTO	Pavement Deflection Data Exchange	.ddx

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FWD Data

FWD File Information

- Anomaly Check: FWD drops are flagged if anomalies are identified.
  - Load levels are flagged if FWD plate load is smaller than 9,900 lbs. or larger than 60,000 lbs. (44 kN – 267 kN)
  - Deflection basins are flagged if the recorded deflections are one of the following:
    - Showing a non-decreasing sequence
    - Larger than 100 mils (2540 mc)
    - Zero



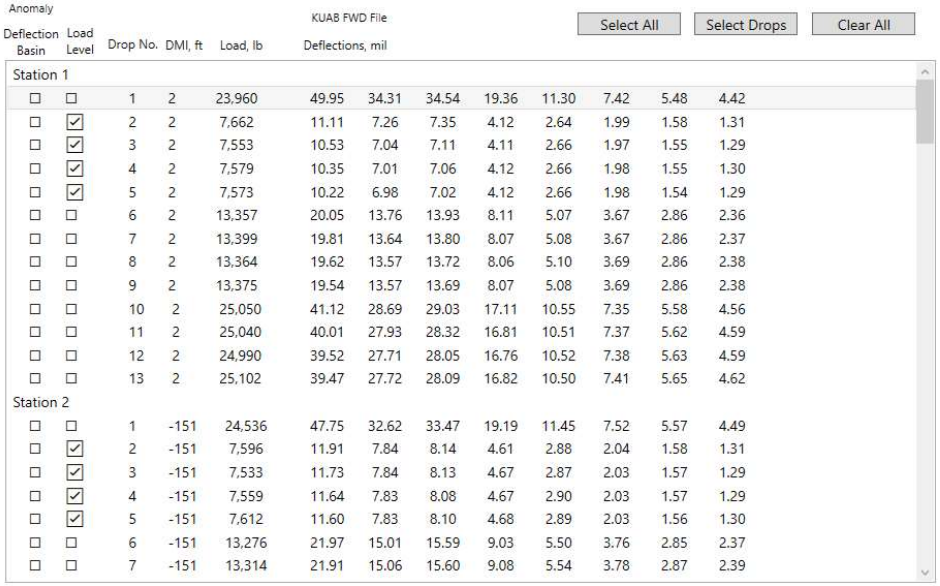
Deflection Basin	Load Level	Drop No.	DMI, ft	Load, lb	Deflections, mil							
Station 1												
☐	☐	1	2	23,960	49.95	34.31	34.54	19.36	11.30	7.42	5.48	4.42
☐	☑	2	2	7,662	11.11	7.26	7.35	4.12	2.64	1.99	1.58	1.31
☐	☑	3	2	7,553	10.53	7.04	7.11	4.11	2.66	1.97	1.55	1.29
☐	☑	4	2	7,579	10.35	7.01	7.06	4.12	2.66	1.98	1.55	1.30
☐	☑	5	2	7,573	10.22	6.98	7.02	4.12	2.66	1.98	1.54	1.29
☐	☐	6	2	13,357	20.05	13.76	13.93	8.11	5.07	3.67	2.86	2.36
☐	☐	7	2	13,399	19.81	13.64	13.80	8.07	5.08	3.67	2.86	2.37
☐	☐	8	2	13,364	19.62	13.57	13.72	8.06	5.10	3.69	2.86	2.38
☐	☐	9	2	13,375	19.54	13.57	13.69	8.07	5.08	3.69	2.86	2.38
☐	☐	10	2	25,050	41.12	28.69	29.03	17.11	10.55	7.35	5.58	4.56
☐	☐	11	2	25,040	40.01	27.93	28.32	16.81	10.51	7.37	5.62	4.59
☐	☐	12	2	24,990	39.52	27.71	28.05	16.76	10.52	7.38	5.63	4.59
☐	☐	13	2	25,102	39.47	27.72	28.09	16.82	10.50	7.41	5.65	4.62
Station 2												
☐	☐	1	-151	24,536	47.75	32.62	33.47	19.19	11.45	7.52	5.57	4.49
☐	☑	2	-151	7,596	11.91	7.84	8.14	4.61	2.88	2.04	1.58	1.31
☐	☑	3	-151	7,533	11.73	7.84	8.13	4.67	2.87	2.03	1.57	1.29
☐	☑	4	-151	7,559	11.64	7.83	8.08	4.67	2.90	2.03	1.57	1.29
☐	☑	5	-151	7,612	11.60	7.83	8.10	4.68	2.89	2.03	1.56	1.30
☐	☐	6	-151	13,276	21.97	15.01	15.59	9.03	5.50	3.76	2.85	2.37
☐	☐	7	-151	13,314	21.91	15.06	15.60	9.08	5.54	3.78	2.87	2.39

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FWD Data

FWD File Information

- Drop No: Number of FWD drops at each test point.
- DMI: the station or location of each FWD test point.
- Load: the applied FWD load at each drop.
- Deflections: measured deflections by each FWD sensor.
- Station: operator comments are identified and replaced with a station number.



Anomaly — KUAB FWD File — Select All | Select Drops | Clear All

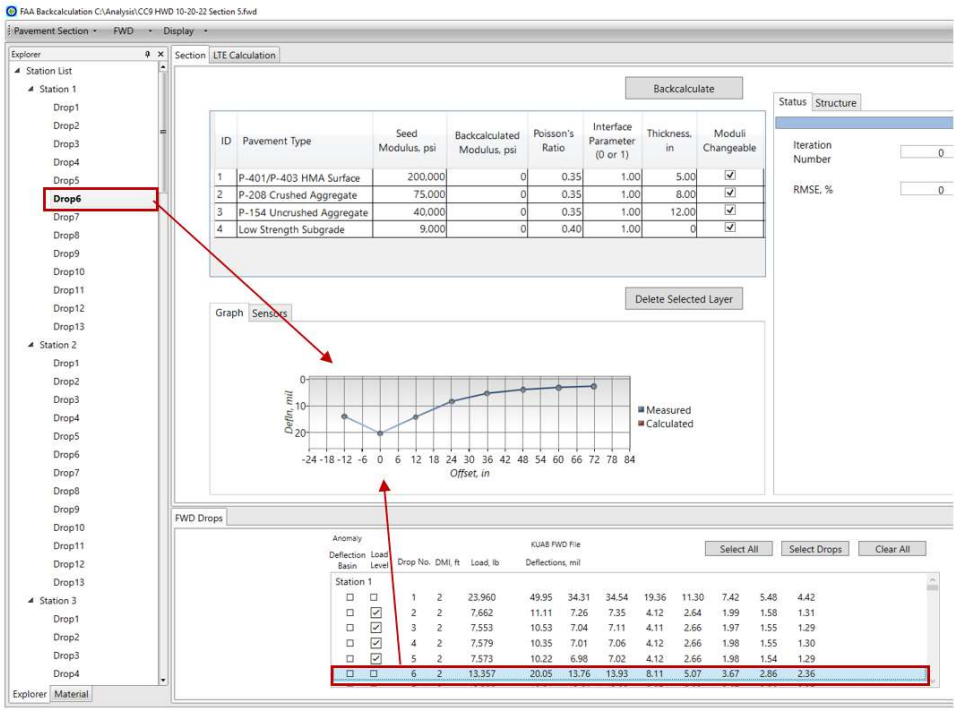
Deflection Basin	Load Level	Drop No.	DMI, ft	Load, lb	Deflections, mil							
Station 1												
☐	☐	1	2	23,960	49.95	34.31	34.54	19.36	11.30	7.42	5.48	4.42
☐	☑	2	2	7,662	11.11	7.26	7.35	4.12	2.64	1.99	1.58	1.31
☐	☑	3	2	7,553	10.53	7.04	7.11	4.11	2.66	1.97	1.55	1.29
☐	☑	4	2	7,579	10.35	7.01	7.06	4.12	2.66	1.98	1.55	1.30
☐	☑	5	2	7,573	10.22	6.98	7.02	4.12	2.66	1.98	1.54	1.29
☐	☐	6	2	13,357	20.05	13.76	13.93	8.11	5.07	3.67	2.86	2.36
☐	☐	7	2	13,399	19.81	13.64	13.80	8.07	5.08	3.67	2.86	2.37
☐	☐	8	2	13,364	19.62	13.57	13.72	8.06	5.10	3.69	2.86	2.38
☐	☐	9	2	13,375	19.54	13.57	13.69	8.07	5.08	3.69	2.86	2.38
☐	☐	10	2	25,050	41.12	28.69	29.03	17.11	10.55	7.35	5.58	4.56
☐	☐	11	2	25,040	40.01	27.93	28.32	16.81	10.51	7.37	5.62	4.59
☐	☐	12	2	24,990	39.52	27.71	28.05	16.76	10.52	7.38	5.63	4.59
☐	☐	13	2	25,102	39.47	27.72	28.09	16.82	10.50	7.41	5.65	4.62
Station 2												
☐	☐	1	-151	24,536	47.75	32.62	33.47	19.19	11.45	7.52	5.57	4.49
☐	☑	2	-151	7,596	11.91	7.84	8.14	4.61	2.88	2.04	1.58	1.31
☐	☑	3	-151	7,533	11.73	7.84	8.13	4.67	2.87	2.03	1.57	1.29
☐	☑	4	-151	7,559	11.64	7.83	8.08	4.67	2.90	2.03	1.57	1.29
☐	☑	5	-151	7,612	11.60	7.83	8.10	4.68	2.89	2.03	1.56	1.30
☐	☐	6	-151	13,276	21.97	15.01	15.59	9.03	5.50	3.76	2.85	2.37
☐	☐	7	-151	13,314	21.91	15.06	15.60	9.08	5.54	3.78	2.87	2.39

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FWD Data

FWD File Information

- Deflection information, deflection graph and sensor table will update as the user clicks on an FWD drop.

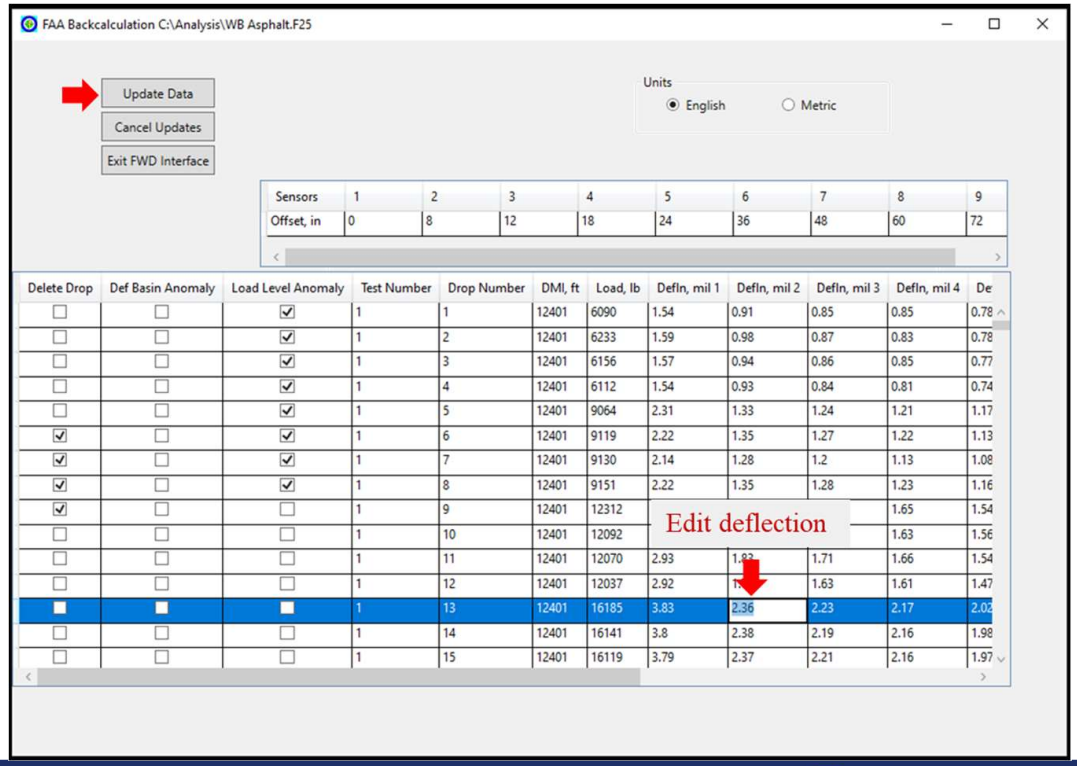


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FWD Data

View FWD File

- Edit sensor data

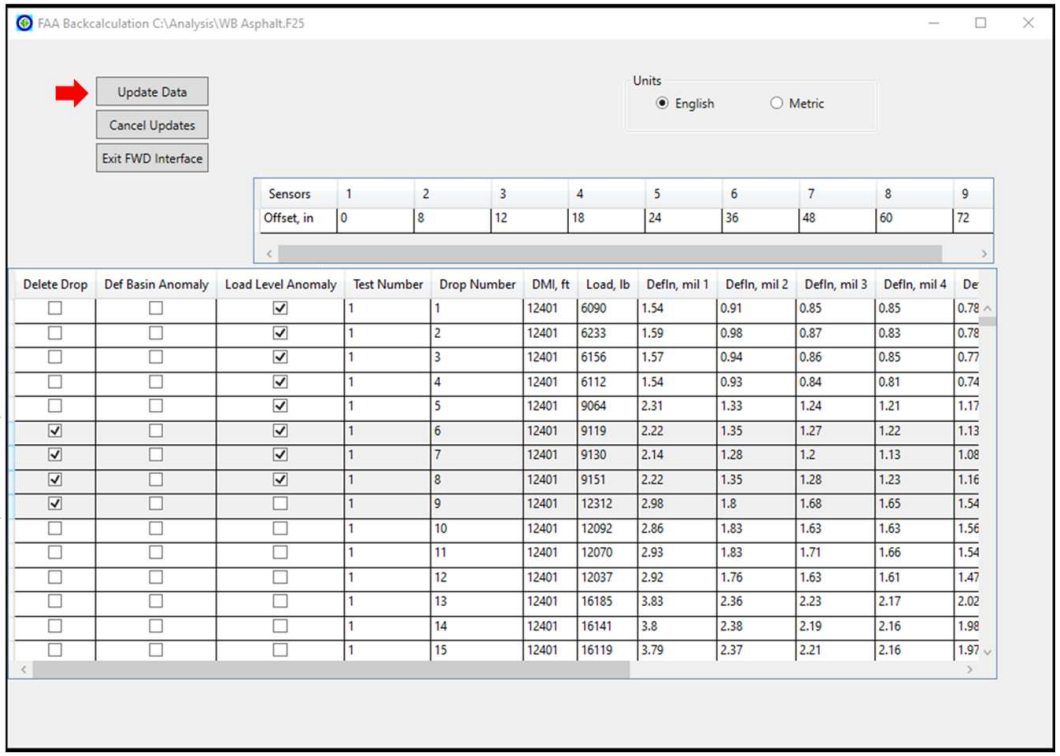


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FWD Data

View FWD File

- Delete drops

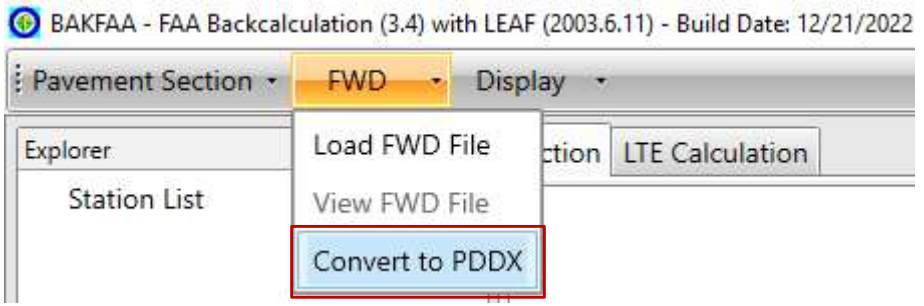


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FWD Data

Convert FWD File to PDDX Format

- Convert FWD file format supported by BAKFAA to Pavement Deflection Data Exchange (PDDX) format file
- PDDX format specification has a modular structure in which data elements of similar characteristics are grouped in individual sections.
- The PDDX files can be loaded to BAKFAA



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FWD Data

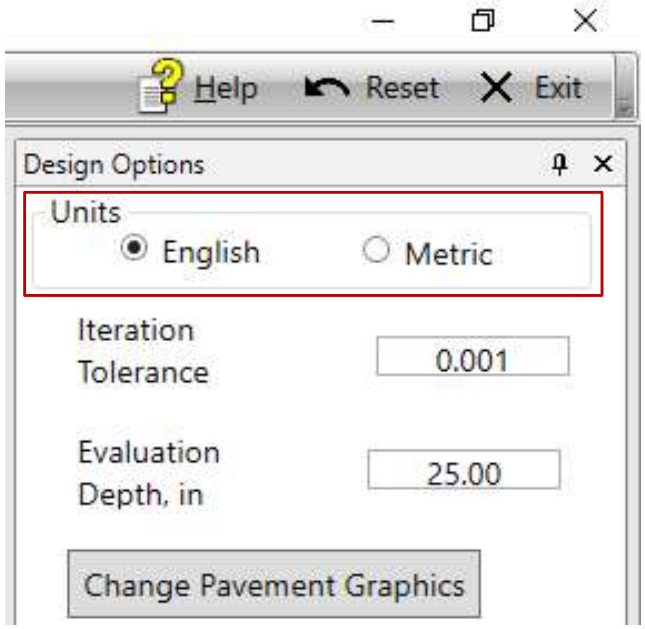
FWD File Parser

- A file processing class used to read FWD files.
- This class uses the file format specifications from FWD manufacturers and generates a set of properties to retrieve data required for functions in the backcalculation and LEAF analysis.
- The advantage of this design is that when a new data format is introduced by a manufacturer, only a file specification needs to be created for the new layout.
- BAKFAA will read the specification and parses the file without requiring any reprograming in the source code.

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Units

- BAKFAA can display values in English or metric units
- Radio button under Design Options toggles between the options
- Selection updates all FWD and pavement information, and dictates the units used in the output file



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Units

Entity	Unit	
	English	Metric
Pavement Layer Thickness	Inch (in.)	Millimeter (mm)
Distance	Feet (ft.)	Meter (m)
FWD Sensor Offset	Inch (in.)	Millimeter (mm)
FWD Sensor Deflection	Mil	Micron (mc)
FWD Plate Radius	Inch (in.)	Millimeter (mm)
FWD Plate Contact Area	Square inch (in^2)	Square millimeter (mm^2)
FWD Plate Contact Pressure	Pound per square inch (psi)	Megapascal (MPa)
Load	Pound (lbs.)	Kilonewton (kN)
Modulus	Pound per square inch (psi)	Megapascal (MPa)

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Units

English

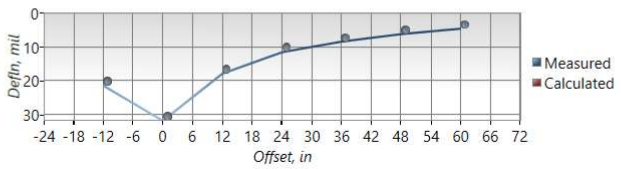
Section | LTE Calculation

Backcalculate

ID	Pavement Type	Seed Modulus, psi	Backcalculated Modulus, psi	Poisson's Ratio	Interface Parameter (0 or 1)	Thickness, in	Moduli Changeable
1	P-401/P-403 HMA Surface	200,000	0	0.35	1.00	5.00	☑
2	P-208 Crushed Aggregate	75,000	0	0.35	1.00	8.00	☑
3	P-154 Uncrushed Aggregate	40,000	0	0.35	1.00	12.00	☑
4	Low Strength Subgrade	9,000	0	0.40	1.00	0	☑

Delete Selected Layer

Graph | Sensors



Metric

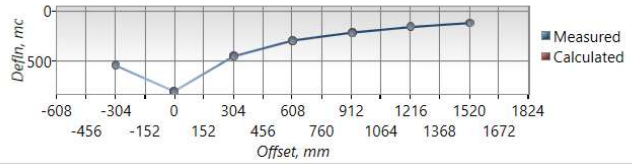
Section | LTE Calculation

Backcalculate

ID	Pavement Type	Seed Modulus, MPa	Backcalculated Modulus, MPa	Poisson's Ratio	Interface Parameter (0 or 1)	Thickness, mm	Moduli Changeable
1	P-401/P-403 HMA Surface	1,379	0	0.35	1.00	127	☑
2	P-208 Crushed Aggregate	517	0	0.35	1.00	203	☑
3	P-154 Uncrushed Aggregate	276	0	0.35	1.00	305	☑
4	Low Strength Subgrade	62	0	0.40	1.00	0	☑

Delete Selected Layer

Graph | Sensors



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Units

English

Graph | Sensors

Exclude Sensor at Offset	☐	☐	☐	☐	☐	☐	☐
Sensor	1	2	3	4	5	6	7
Offset, in	0.0	12.0	24.0	36.0	48.0	60.0	72.0
Defln, mil	20.1	13.9	8.1	5.1	3.7	2.9	2.4
Calc.mil	20.2	13.5	8.4	5.3	3.6	2.7	2.2

Metric

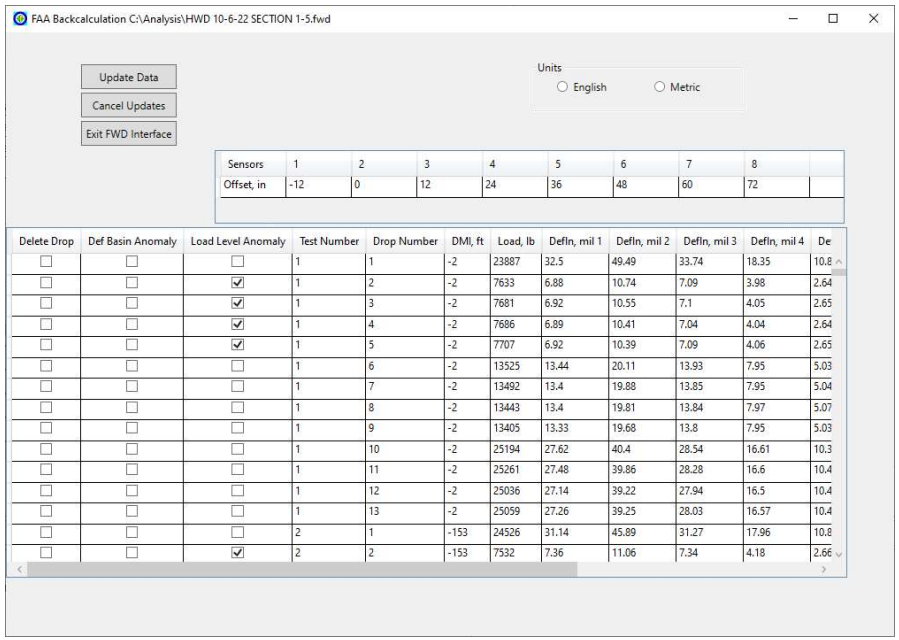
Graph | Sensors

Exclude Sensor at Offset	☐	☐	☐	☐	☐	☐	☐
Sensor	1	2	3	4	5	6	7
Offset, mm	0	305	610	914	1,219	1,524	1,829
Defln, mc	509	354	206	129	93	73	60
Calc, mc	513	343	213	136	92	69	55

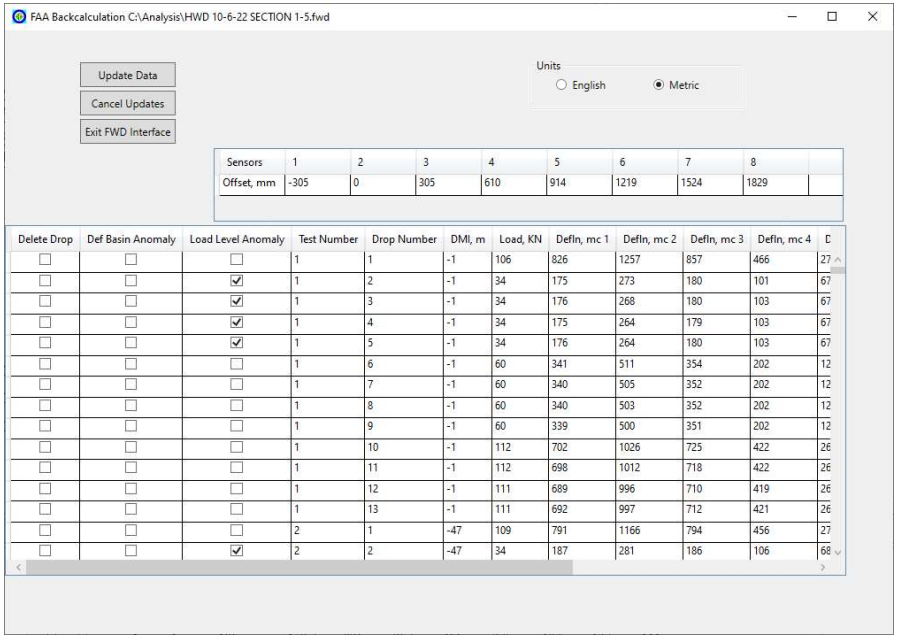
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Units

English



Metric



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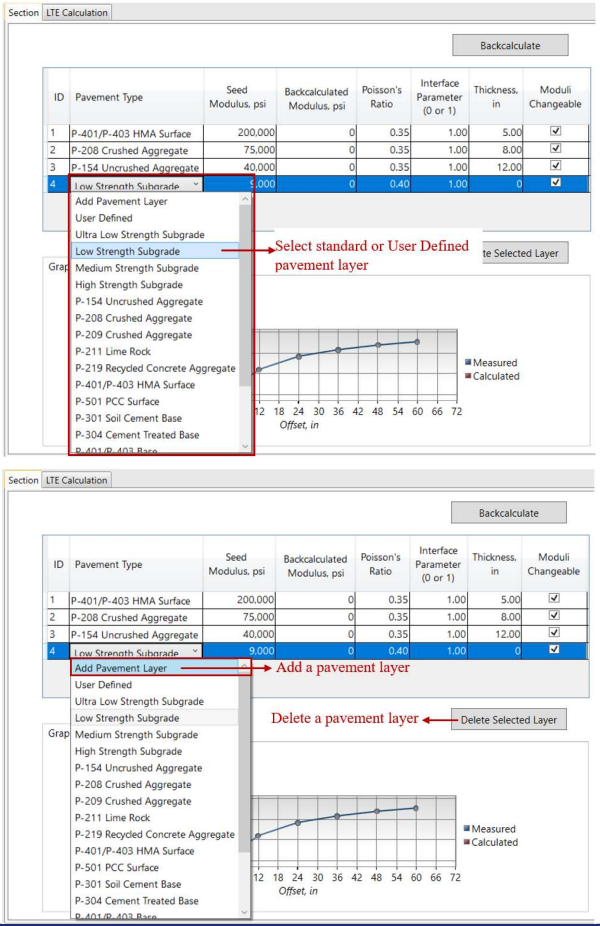
BAKFAA Inputs

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BAKFAA INPUTS

Pavement Layer Properties

- Pavement Layer Type:
  - Minimum 3 layers and maximum 6 layers
  - Select layer type from drop down menu.
  - Library of standard FAA materials (AC 150/5370-10)
  - Four subgrade types with different strengths
  - “User Defined” layer type
  - Add or remove a layer from any location in the structure.
  - Last layer is a semi-infinite layer

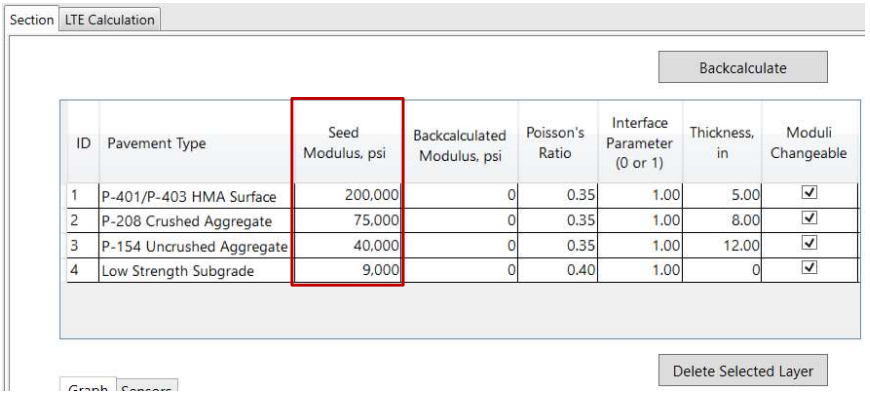


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BAKFAA INPUTS

Pavement Layer Properties

- Seed Modulus:
  - Initial modulus value used in backcalculation iterative procedure
  - BAKFAA has default seed modulus values for FAA standard materials
  - Default seed modulus values are editable, but BAKFAA will automatically change the pavement type to "User Defined"



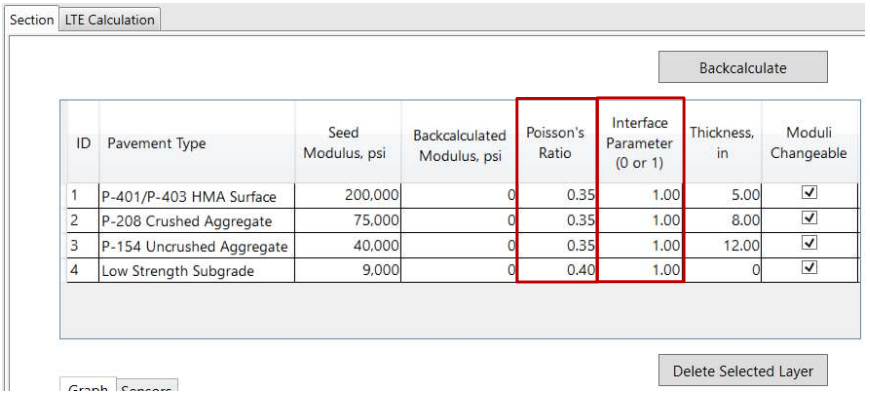
ID	Pavement Type	Seed Modulus, psi	Backcalculated Modulus, psi	Poisson's Ratio	Interface Parameter (0 or 1)	Thickness, in	Moduli Changeable
1	P-401/P-403 HMA Surface	200,000	0	0.35	1.00	5.00	☑
2	P-208 Crushed Aggregate	75,000	0	0.35	1.00	8.00	☑
3	P-154 Uncrushed Aggregate	40,000	0	0.35	1.00	12.00	☑
4	Low Strength Subgrade	9,000	0	0.40	1.00	0	☑

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BAKFAA INPUTS

Pavement Layer Properties

- Poisson's Ratio:
  - BAKFAA has default Poisson's ratio values for FAA standard materials
  - Numerical value between 0.0 to 0.5
- Interface Parameter:
  - Accepts values of 0 or 1
  - 0 indicates no bond
  - 1 indicates fully bonded layers



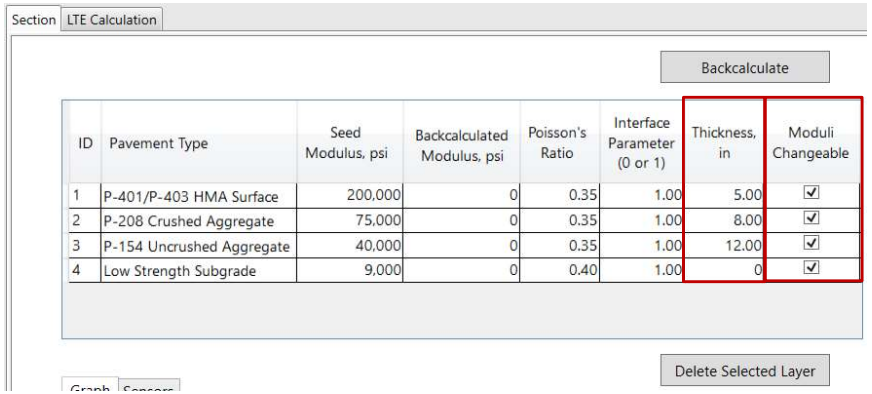
ID	Pavement Type	Seed Modulus, psi	Backcalculated Modulus, psi	Poisson's Ratio	Interface Parameter (0 or 1)	Thickness, in	Moduli Changeable
1	P-401/P-403 HMA Surface	200,000	0	0.35	1.00	5.00	☑
2	P-208 Crushed Aggregate	75,000	0	0.35	1.00	8.00	☑
3	P-154 Uncrushed Aggregate	40,000	0	0.35	1.00	12.00	☑
4	Low Strength Subgrade	9,000	0	0.40	1.00	0	☑

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BAKFAA INPUTS

Pavement Layer Properties

- Thickness:
  - BAKFAA requires a non-zero thickness for each pavement layer
  - The thickness of the last layer in a pavement structure must be zero to represent a semi-infinite layer
- Moduli Changeable:
  - Determines whether the layer modulus will change in the backcalculation analysis
  - Checkbox toggled off indicates the layer modulus will remain fixed in the backcalculation analysis

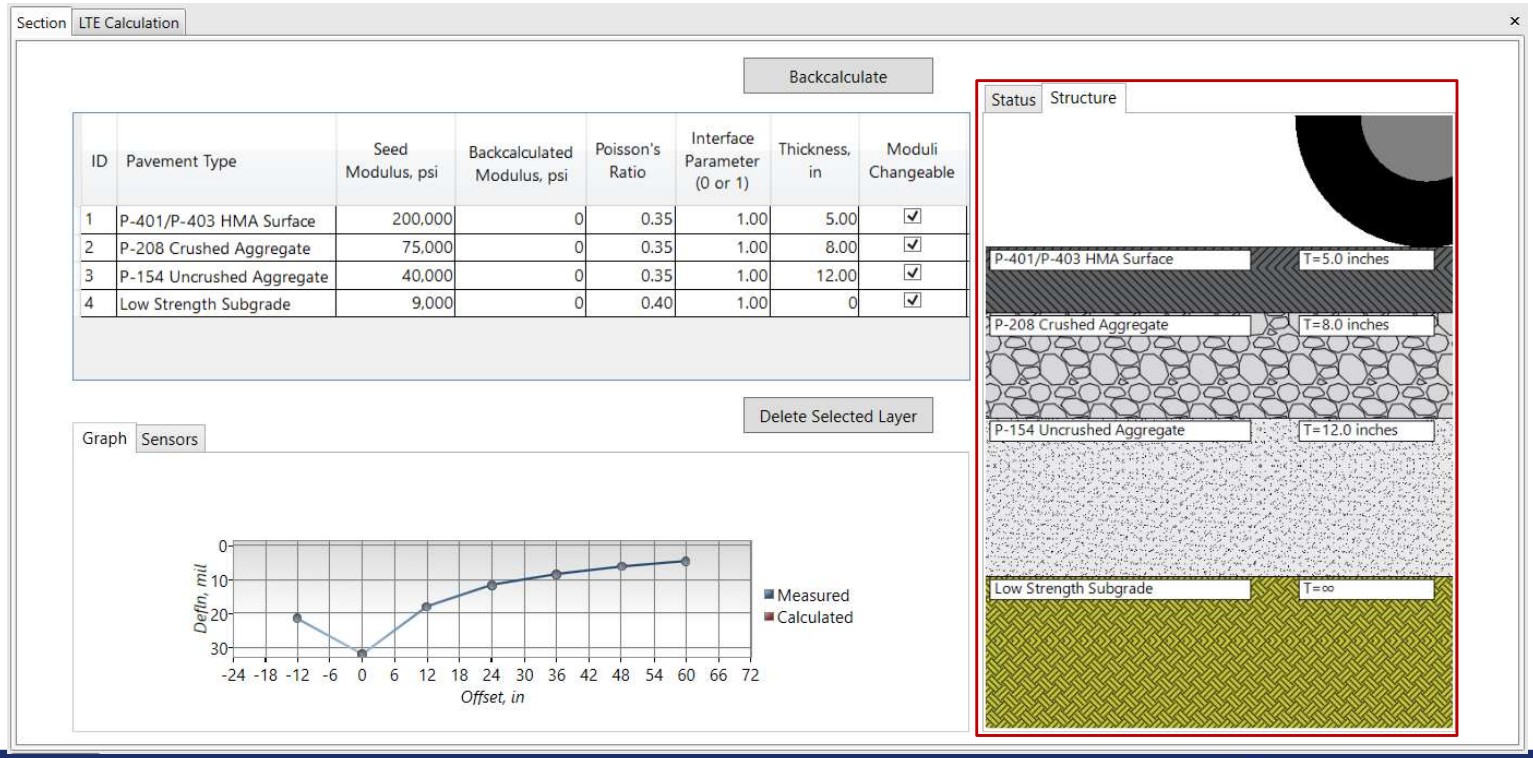


ID	Pavement Type	Seed Modulus, psi	Backcalculated Modulus, psi	Poisson's Ratio	Interface Parameter (0 or 1)	Thickness, in	Moduli Changeable
1	P-401/P-403 HMA Surface	200,000	0	0.35	1.00	5.00	☑
2	P-208 Crushed Aggregate	75,000	0	0.35	1.00	8.00	☑
3	P-154 Uncrushed Aggregate	40,000	0	0.35	1.00	12.00	☑
4	Low Strength Subgrade	9,000	0	0.40	1.00	0	☑

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BAKFAA INPUTS

Pavement Layer Structure



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BAKFAA INPUTS

Backcalculation Input

- Iteration Tolerance:
  - Dimensionless parameter that controls the stoppage criteria in the iterative process of backcalculation analysis
  - At the end of each iteration

$$Err = \sum_{i=1}^{N} (MD_i - CD_i)^2$$

MD_i and CD_i : measured and calculated deflections at sensor i
N : the number of FWD sensors

$$rtol = 2 \times \frac{(Err)_j - (Err)_{j+1}}{(Err)_j + (Err)_{j+1}}$$

$(Err)_j$ and $(Err)_{j+1}$ are the errors at two consecutive iterations j and $j+1$

  - BAKFAA stops backcalculation when the *rtol* is less than the user defined Iteration Tolerance
  - Typical range is 0.001-0.1. Iterate through this range to achieve desired accuracy.

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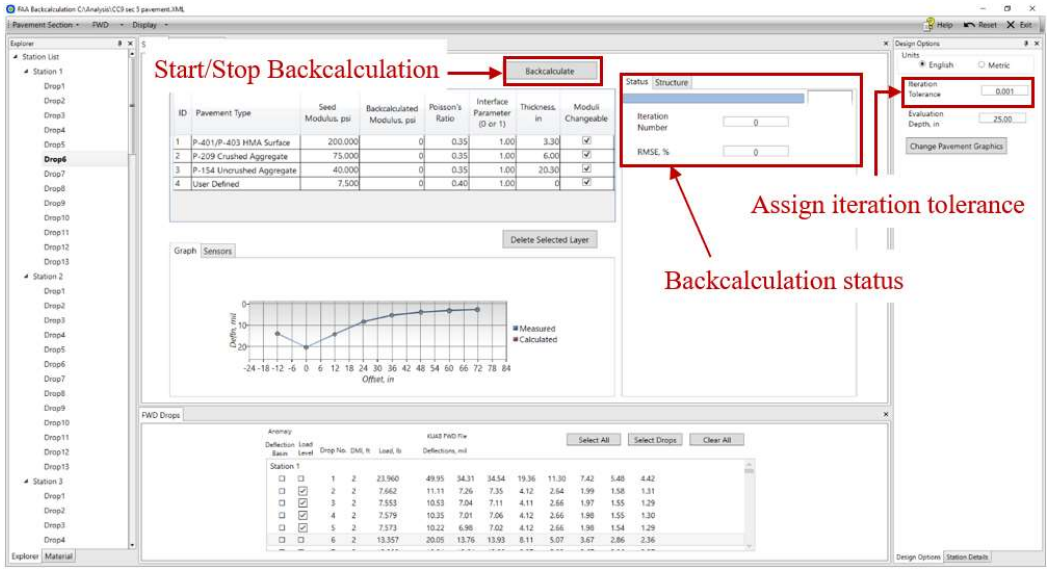
Backcalculation Analysis

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Backcalculation

Analysis Mode

- Single Analysis Mode: when a single FWD drop is selected. Results will be shown in the program.
- Batch Analysis Mode: when more than one drop is selected. Results will be saved in .csv file.



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Backcalculation

Select FWD File

- Select individual drops



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Backcalculation

Select FWD File

- Select All button



Anomaly

KUAB FWD File

Select All | Select Drops | Clear All

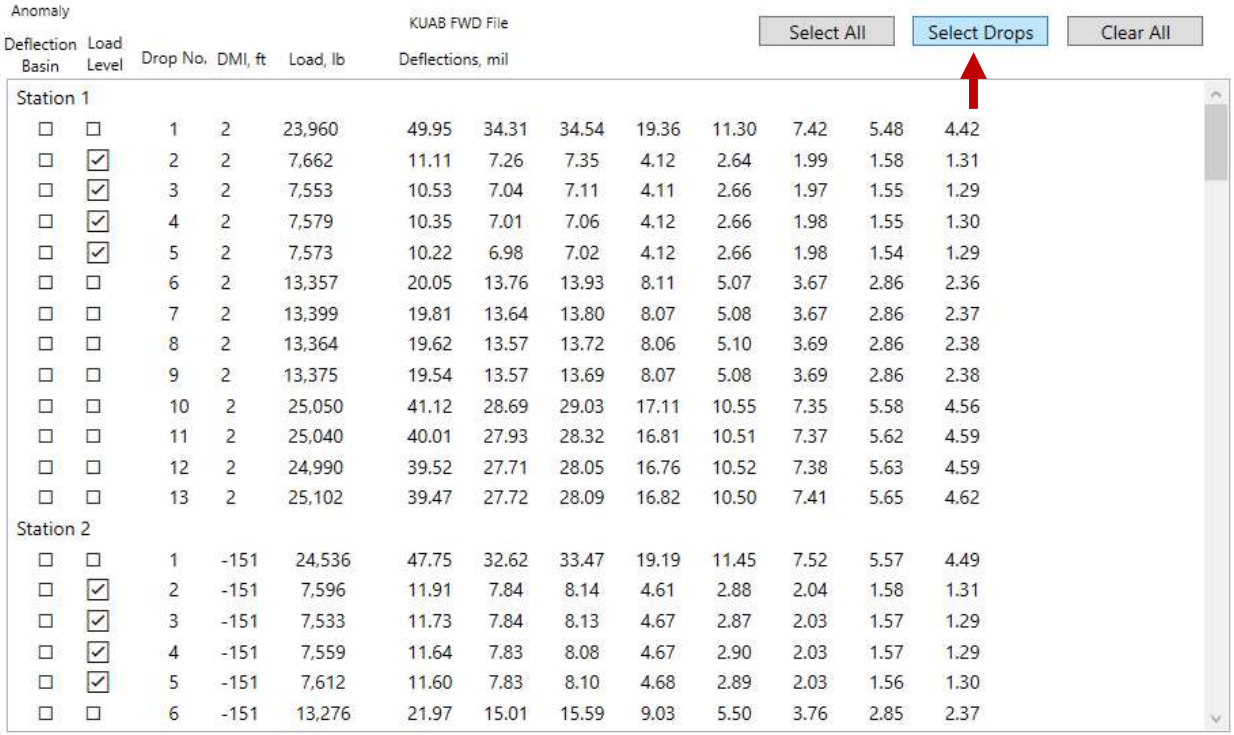
Deflection Basin	Load Level	Drop No.	DMI, ft	Load, lb	Deflections, mil							
Station 1												
☐	☐	1	2	23,960	49.95	34.31	34.54	19.36	11.30	7.42	5.48	4.42
☐	☑	2	2	7,662	11.11	7.26	7.35	4.12	2.64	1.99	1.58	1.31
☐	☑	3	2	7,553	10.53	7.04	7.11	4.11	2.66	1.97	1.55	1.29
☐	☑	4	2	7,579	10.35	7.01	7.06	4.12	2.66	1.98	1.55	1.30
☐	☑	5	2	7,573	10.22	6.98	7.02	4.12	2.66	1.98	1.54	1.29
☐	☐	6	2	13,357	20.05	13.76	13.93	8.11	5.07	3.67	2.86	2.36
☐	☐	7	2	13,399	19.81	13.64	13.80	8.07	5.08	3.67	2.86	2.37
☐	☐	8	2	13,364	19.62	13.57	13.72	8.06	5.10	3.69	2.86	2.38
☐	☐	9	2	13,375	19.54	13.57	13.69	8.07	5.08	3.69	2.86	2.38
☐	☐	10	2	25,050	41.12	28.69	29.03	17.11	10.55	7.35	5.58	4.56
☐	☐	11	2	25,040	40.01	27.93	28.32	16.81	10.51	7.37	5.62	4.59
☐	☐	12	2	24,990	39.52	27.71	28.05	16.76	10.52	7.38	5.63	4.59
☐	☐	13	2	25,102	39.47	27.72	28.09	16.82	10.50	7.41	5.65	4.62
Station 2												
☐	☐	1	-151	24,536	47.75	32.62	33.47	19.19	11.45	7.52	5.57	4.49
☐	☑	2	-151	7,596	11.91	7.84	8.14	4.61	2.88	2.04	1.58	1.31
☐	☑	3	-151	7,533	11.73	7.84	8.13	4.67	2.87	2.03	1.57	1.29
☐	☑	4	-151	7,559	11.64	7.83	8.08	4.67	2.90	2.03	1.57	1.29
☐	☑	5	-151	7,612	11.60	7.83	8.10	4.68	2.89	2.03	1.56	1.30
☐	☐	6	-151	13,276	21.97	15.01	15.59	9.03	5.50	3.76	2.85	2.37

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Backcalculation

Select FWD File

- Select Drops button



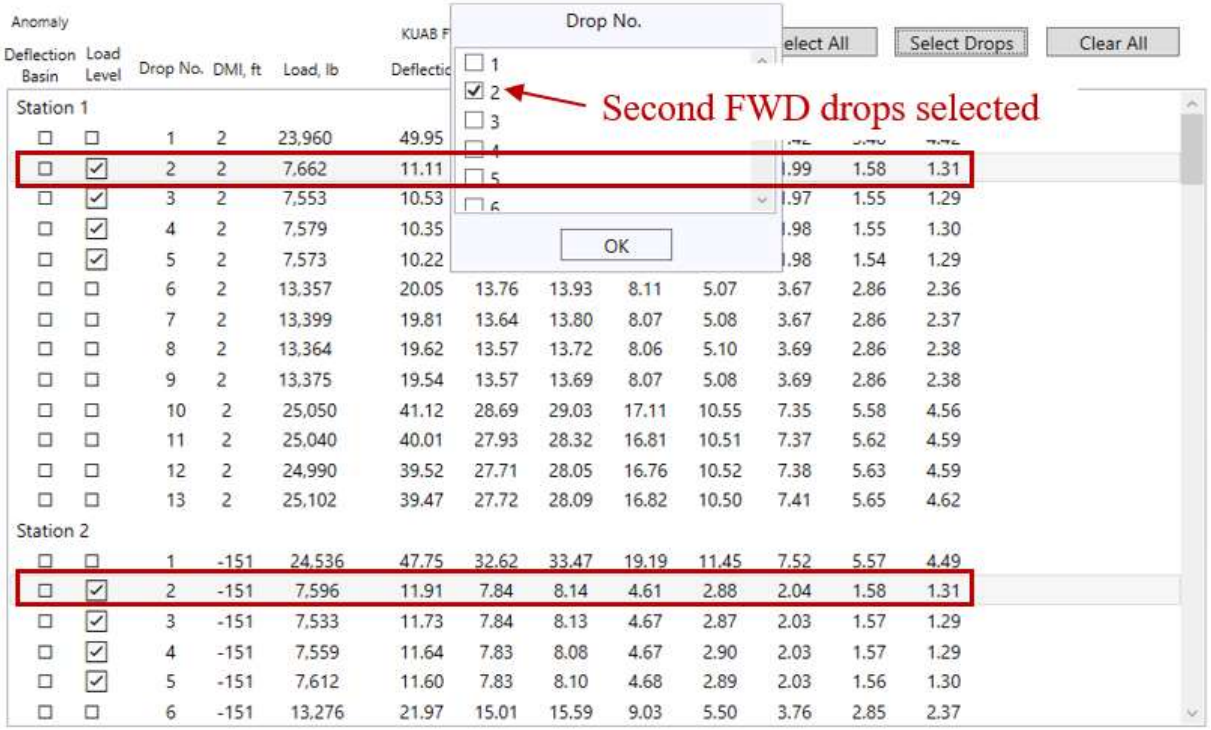
Deflection Basin	Load Level	Drop No.	DMI, ft	Load, lb	Deflections, mil							
Station 1												
☐	☐	1	2	23,960	49.95	34.31	34.54	19.36	11.30	7.42	5.48	4.42
☐	☑	2	2	7,662	11.11	7.26	7.35	4.12	2.64	1.99	1.58	1.31
☐	☑	3	2	7,553	10.53	7.04	7.11	4.11	2.66	1.97	1.55	1.29
☐	☑	4	2	7,579	10.35	7.01	7.06	4.12	2.66	1.98	1.55	1.30
☐	☑	5	2	7,573	10.22	6.98	7.02	4.12	2.66	1.98	1.54	1.29
☐	☐	6	2	13,357	20.05	13.76	13.93	8.11	5.07	3.67	2.86	2.36
☐	☐	7	2	13,399	19.81	13.64	13.80	8.07	5.08	3.67	2.86	2.37
☐	☐	8	2	13,364	19.62	13.57	13.72	8.06	5.10	3.69	2.86	2.38
☐	☐	9	2	13,375	19.54	13.57	13.69	8.07	5.08	3.69	2.86	2.38
☐	☐	10	2	25,050	41.12	28.69	29.03	17.11	10.55	7.35	5.58	4.56
☐	☐	11	2	25,040	40.01	27.93	28.32	16.81	10.51	7.37	5.62	4.59
☐	☐	12	2	24,990	39.52	27.71	28.05	16.76	10.52	7.38	5.63	4.59
☐	☐	13	2	25,102	39.47	27.72	28.09	16.82	10.50	7.41	5.65	4.62
Station 2												
☐	☐	1	-151	24,536	47.75	32.62	33.47	19.19	11.45	7.52	5.57	4.49
☐	☑	2	-151	7,596	11.91	7.84	8.14	4.61	2.88	2.04	1.58	1.31
☐	☑	3	-151	7,533	11.73	7.84	8.13	4.67	2.87	2.03	1.57	1.29
☐	☑	4	-151	7,559	11.64	7.83	8.08	4.67	2.90	2.03	1.57	1.29
☐	☑	5	-151	7,612	11.60	7.83	8.10	4.68	2.89	2.03	1.56	1.30
☐	☐	6	-151	13,276	21.97	15.01	15.59	9.03	5.50	3.76	2.85	2.37

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Backcalculation

Select FWD File

- Select Drops button
  - Selection menu



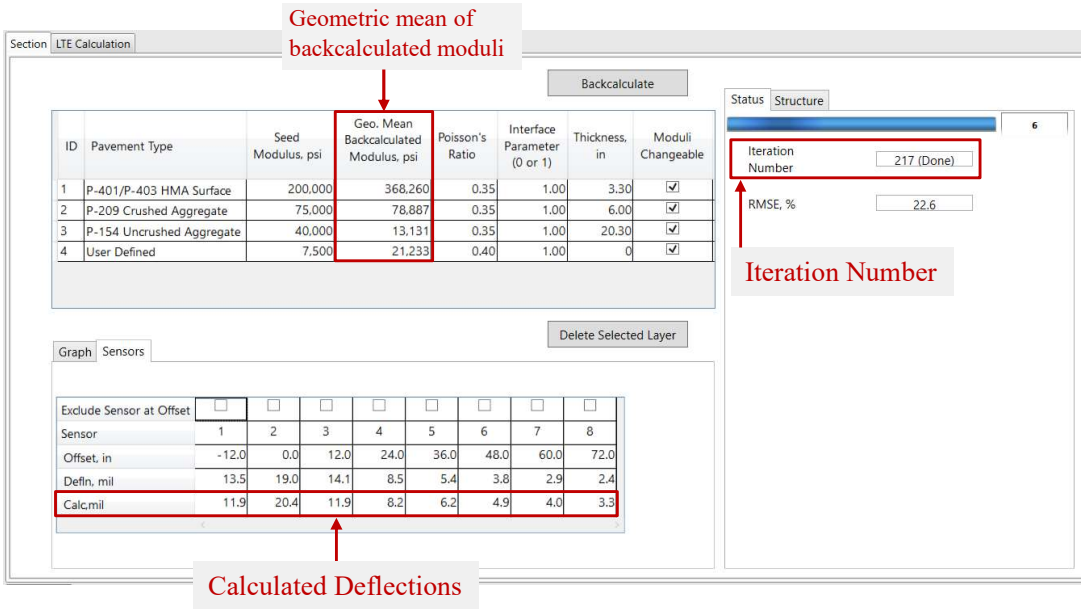
Anomaly												
Deflection Basin	Load Level	Drop No.	DMI, ft	Load, lb	KUAB F Deflectic							
Station 1												
☐	☐	1	2	23,960	49.95							
☐	☑	2	2	7,662	11.11					.99	1.58	1.31
☐	☑	3	2	7,553	10.53					.97	1.55	1.29
☐	☑	4	2	7,579	10.35					.98	1.55	1.30
☐	☑	5	2	7,573	10.22					.98	1.54	1.29
☐	☐	6	2	13,357	20.05	13.76	13.93	8.11	5.07	3.67	2.86	2.36
☐	☐	7	2	13,399	19.81	13.64	13.80	8.07	5.08	3.67	2.86	2.37
☐	☐	8	2	13,364	19.62	13.57	13.72	8.06	5.10	3.69	2.86	2.38
☐	☐	9	2	13,375	19.54	13.57	13.69	8.07	5.08	3.69	2.86	2.38
☐	☐	10	2	25,050	41.12	28.69	29.03	17.11	10.55	7.35	5.58	4.56
☐	☐	11	2	25,040	40.01	27.93	28.32	16.81	10.51	7.37	5.62	4.59
☐	☐	12	2	24,990	39.52	27.71	28.05	16.76	10.52	7.38	5.63	4.59
☐	☐	13	2	25,102	39.47	27.72	28.09	16.82	10.50	7.41	5.65	4.62
Station 2												
☐	☐	1	-151	24,536	47.75	32.62	33.47	19.19	11.45	7.52	5.57	4.49
☐	☑	2	-151	7,596	11.91	7.84	8.14	4.61	2.88	2.04	1.58	1.31
☐	☑	3	-151	7,533	11.73	7.84	8.13	4.67	2.87	2.03	1.57	1.29
☐	☑	4	-151	7,559	11.64	7.83	8.08	4.67	2.90	2.03	1.57	1.29
☐	☑	5	-151	7,612	11.60	7.83	8.10	4.68	2.89	2.03	1.56	1.30
☐	☐	6	-151	13,276	21.97	15.01	15.59	9.03	5.50	3.76	2.85	2.37

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Backcalculation

Analysis Output

- Backcalculated Moduli:
  - In batch mode: geometric mean ("Geo. Mean")
- Calculated Deflections:
  - In batch mode: displayed only for the last drop selected
- Iteration Number:
  - In batch mode: displayed only for the last drop selected



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Backcalculation

Analysis Output

- Root Mean Square Error (RMSE): between measured and calculated deflections

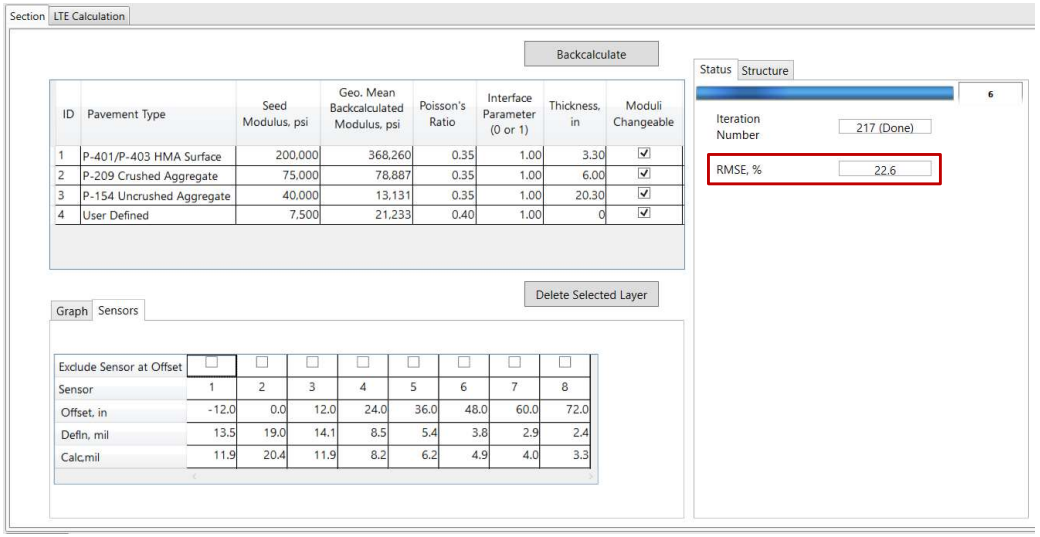
$$RMSE = 100 \times \left[\frac{1}{N}\sum_{i=1}^{N}\left(\frac{d_i^c - d_i^m}{d_i^m}\right)^2\right]^{1/2}$$

where:

d_i^c : Calculated deflection for ith sensor

d_i^m : Measured deflection for ith sensor

N: Number of sensors



ID	Pavement Type	Seed Modulus, psi	Geo. Mean Backcalculated Modulus, psi	Poisson's Ratio	Interface Parameter (0 or 1)	Thickness, in	Moduli Changeable
1	P-401/P-403 HMA Surface	200,000	368,260	0.35	1.00	3.30	☑
2	P-209 Crushed Aggregate	75,000	78,887	0.35	1.00	6.00	☑
3	P-154 Uncrushed Aggregate	40,000	13,131	0.35	1.00	20.30	☑
4	User Defined	7,500	21,233	0.40	1.00	0	☑

Sensor	1	2	3	4	5	6	7	8
Offset, in	-12.0	0.0	12.0	24.0	36.0	48.0	60.0	72.0
Defln, mil	13.5	19.0	14.1	8.5	5.4	3.8	2.9	2.4
Calc.mil	11.9	20.4	11.9	8.2	6.2	4.9	4.0	3.3

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Backcalculation

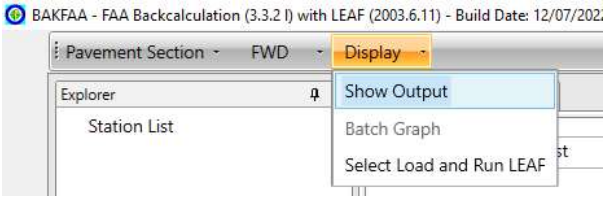
Analysis Output

- .CSV Output File content:
  - Pavement structure (pavement type, seed modulus, Poisson's ratio, interface parameter, thickness, and moduli changeable)
  - FWD (DMI, drop no., load, and operator comments or remarks)
  - Iteration tolerance
  - Backcalculated modulus
  - RMSE
  - Number of iterations completed
  - LTE values based on the user selection
  - Measured deflections for each selected FWD drop
  - Calculated deflections for each selected FWD drop

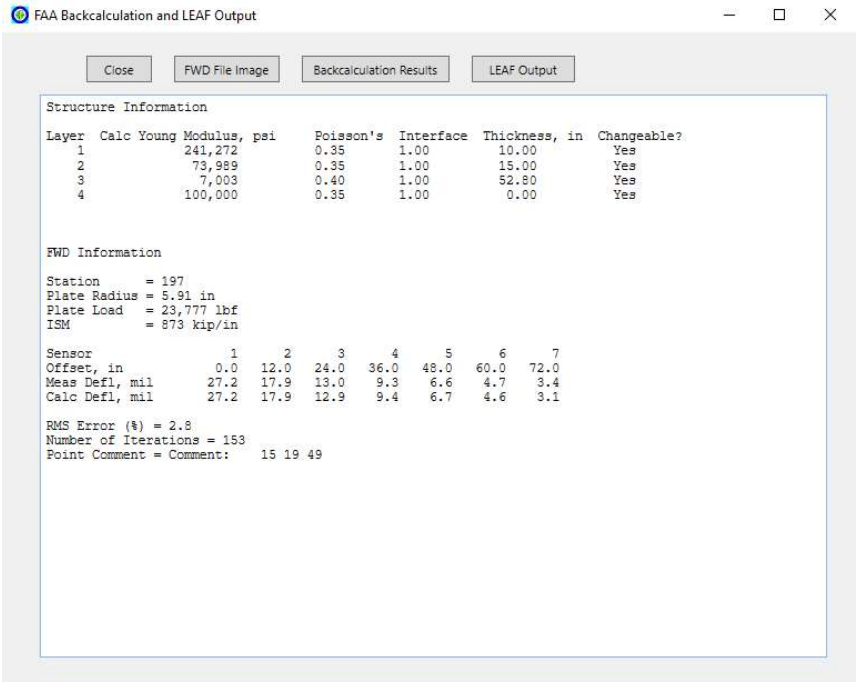
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Backcalculation

Show Output



- Show Output – Single Analysis Mode
- Displays backcalculation results for that drop, including:
  - Calculated deflections
  - RMSE
  - Iteration number
  - FWD information
  - Impulse Stiffness Modulus ($ISM=L/D_0$)



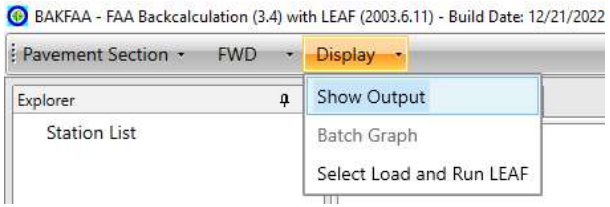
L is the applied FWD load;

D_0 is the deflection under FWD load plate.

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Backcalculation

Show Output

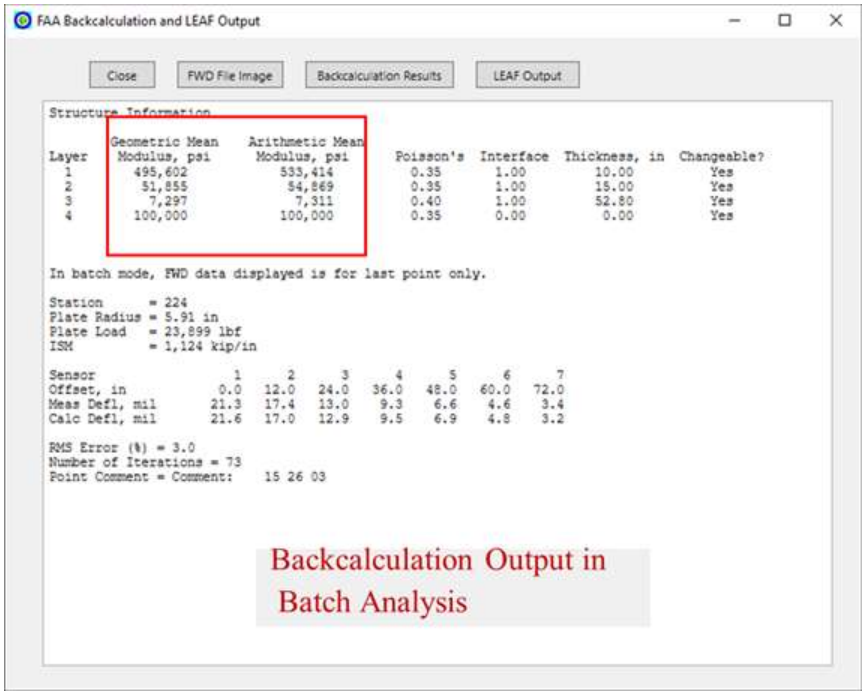


- Show Output – Batch Analysis Mode
  - Backcalculation outputs, FWD information, and ISM are displayed for the last drop selected
  - Arithmetic and geometric means of backcalculated modulus of each layer

$$Arithmetic\ Mean\ (E_1) = \frac{1}{n}(E_{11} + E_{12} + \cdots + E_{1n})$$

$$Geometric\ Mean\ (E_1) = (E_{11} \times E_{12} \times \ldots \times E_{1n})^{1/n}$$

n is the number of test points selected for backcalculation; E_{11}, E_{12} and E_{1n} are the backcalculated modulus of layer 1 at test points 1, 2 and n.

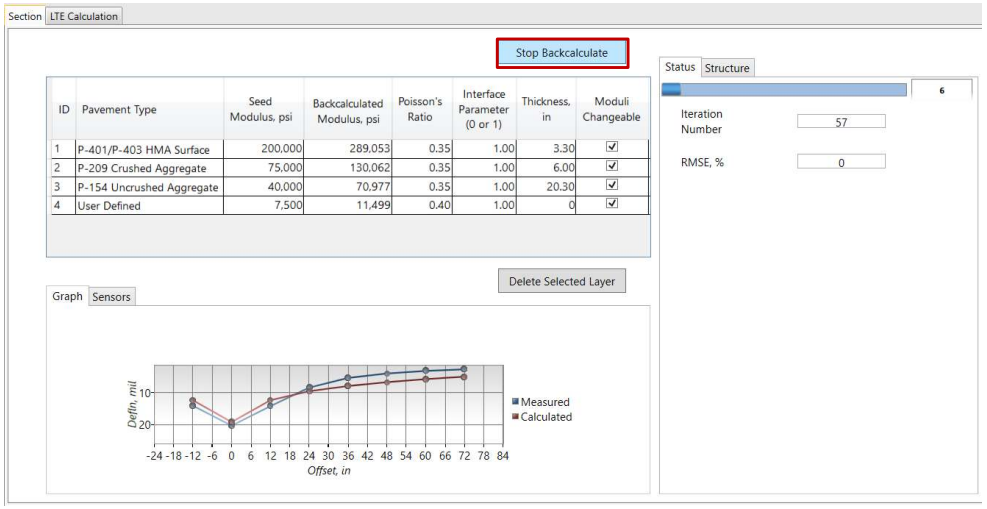


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Backcalculation

Stop Backcalculation

- Stop Backcalculation
- Aborts in-progress backcalculation
- Backcalculation results can still be found up to the point of termination:
  - In “Show Output” for single mode
  - In the .csv file for batch mode

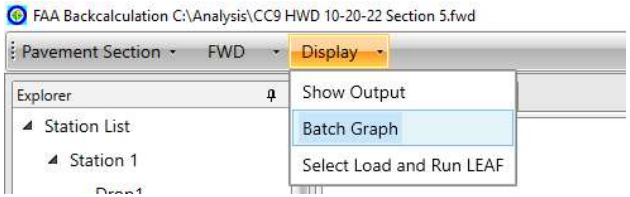


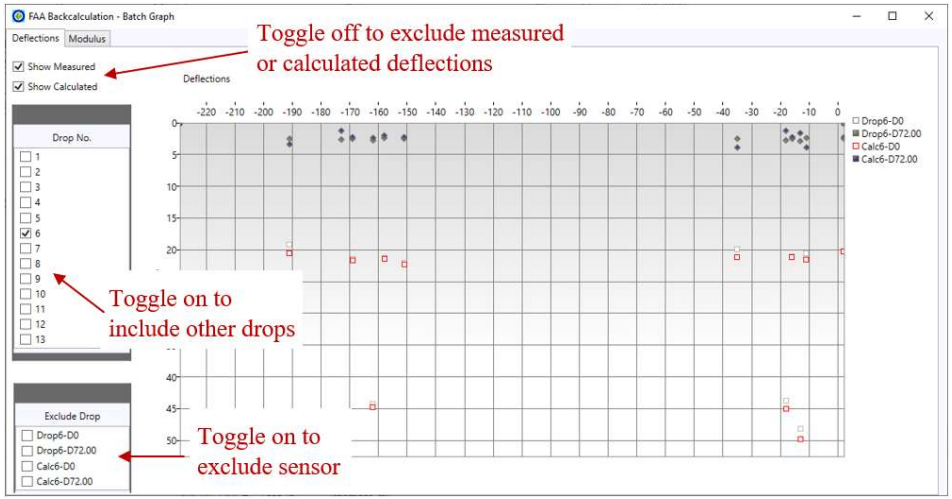
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Backcalculation

Batch Graph

- Batch Graph
- Deflections tab: graphs measured and calculated deflections for two sensors:
  - Sensor under the FWD load plate (D_0)
  - The outermost sensor (D_{72})
- "Show Measured" displays measured deflections.
- "Show Calculated" displays backcalculated deflections.
- X-axis is DMI location of selected FWD test points.



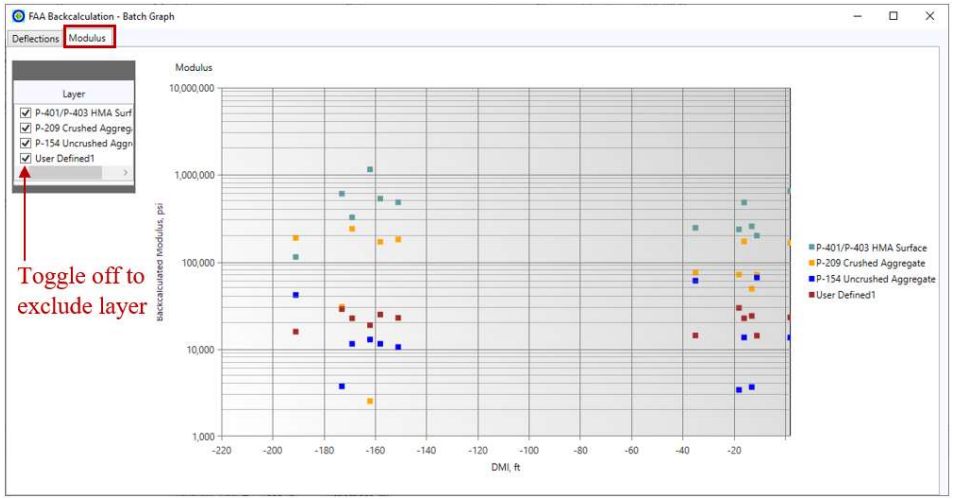


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Backcalculation

Batch Graph

- Batch Graph
- Modulus tab: displays backcalculated modulus values in logarithmic scale.
- X-axis is DMI location of selected FWD test points.
- User can toggle off any layer of pavement in the backcalculated modulus graph.



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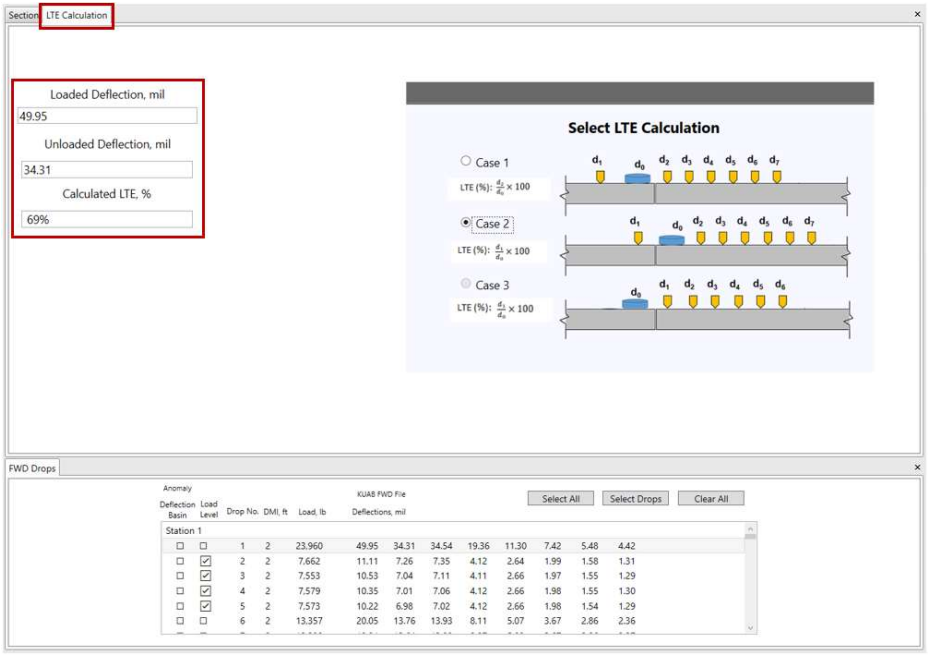
Load Transfer Efficiency (LTE) Calculation

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LTE Calculation

Calculate LTE between Two Jointed Concrete Slabs

- Select between case 1, 2, or 3 based on FWD sensor configuration and FWD test layout at slab joints.
  - BAKFAA disables case 1 and 2 for FWD tests with no negative offset sensor (-12 in. offset) or when the user excludes the negative offset sensor.
  - BAKFAA disables case 3 for FWD tests with a negative offset sensor.



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Layered Elastic Analysis Program (LEAF)

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Run LEAF

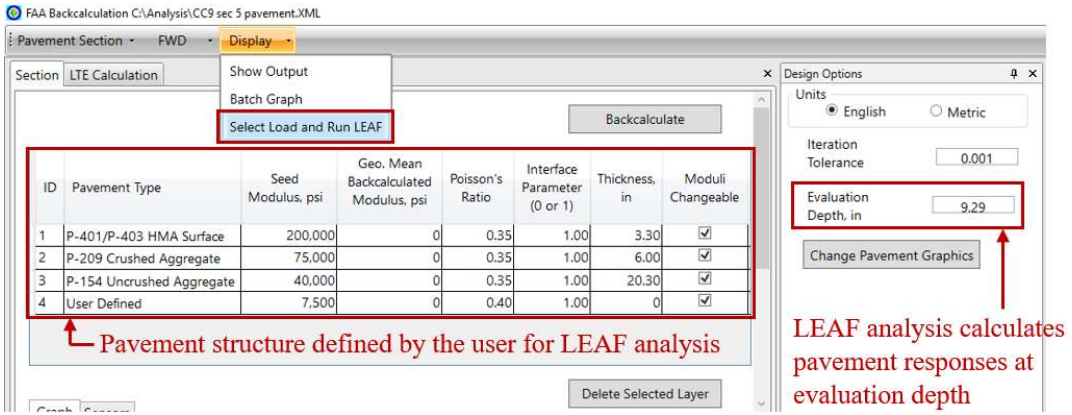
- LEAF is built into BAKFAA.
- Compute pavement responses (stress, strain, and deflection) under defined load types at different evaluation points within the pavement structure.
- LEAF is based on the layered elastic model.
- The model is based on the Boussinesq mathematical model which assumes:
  - Pavement layers extend infinitely in the horizontal direction.
  - Bottom layer extends infinitely downward.
  - Materials are not stressed beyond their elastic ranges.

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Run LEAF

Inputs Required for LEAF Analysis

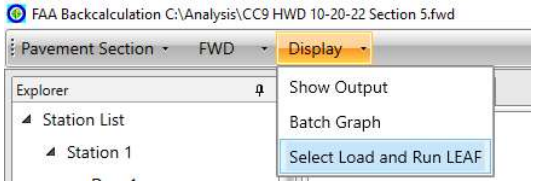
- Inputs required for LEAF analysis:
  - Modulus of elasticity (either seed modulus or backcalculated modulus)
  - Poisson's ratio
  - Layer thickness
  - Layer interface condition
  - Depth of evaluation point
- Continuity in responses across pavement layers depends on their interface condition.
  - Depth of evaluation must represent the target layer



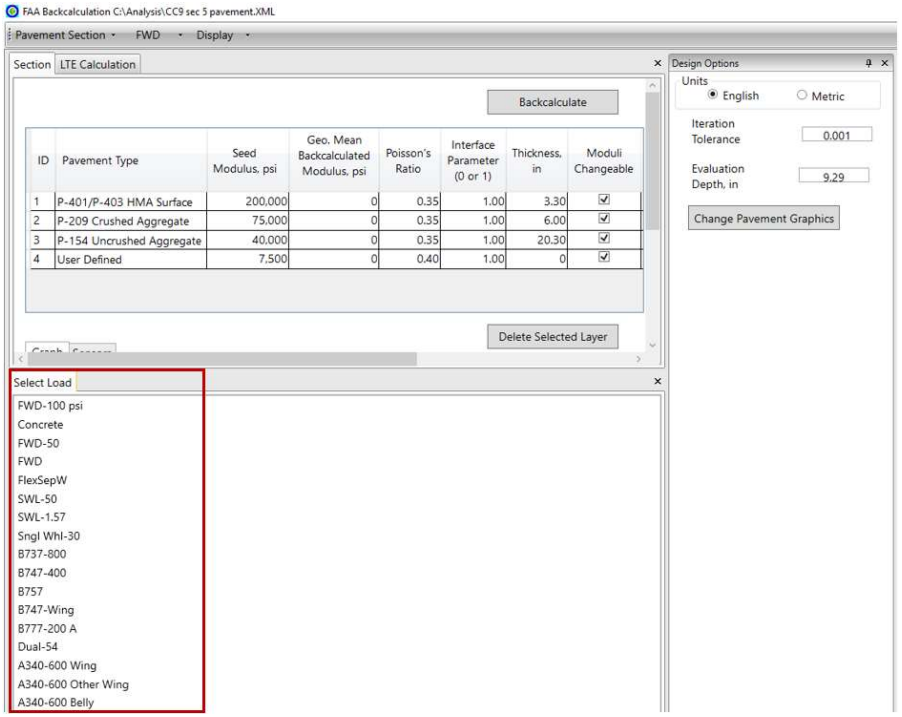
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Run LEAF

Select Load and Gear Configuration



- Select load from library
  - Generic gear geometry
  - Defined aircraft
  - Defined FWD loading configurations
  - Library information is in LEAFAircraft.Ext file located in BAKFAA directory (C:\Program Files\FAA\BAKFAA)
- User can modify LEAFAircraft.Ext file or add user-defined landing gear geometry or FWD configuration.
- User can modify coordinates of evaluation points



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Run LEAF

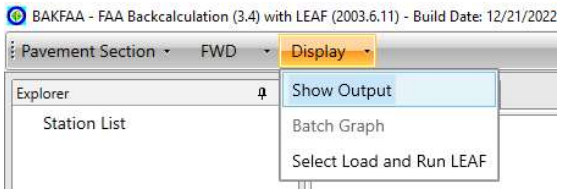
Library Example (B-747-400)

B-747-400		Aircraft Name
873000		Gross Weight of Aircraft, lbs
23.75		Percent of Gross Weight on evaluation gear
200		Tire pressure, PSI
J		Gear ID letter, military designation
4		Gear ID number
44		TT, Dual Spacing, inches
98		TS, Gear track between inside wheels, inches
106		TG, Four-gear spacing, inches
58		B, Tandem spacing, inches
4		Number of tires on evaluation gear, NTires
-22	0	
22	0	TX TY (for NTires) in inches
-22	58	
22	58	
8		Number of tire tracks, NTTrack
-239		
-195		
-97		
-53		XAC (for NTTracks) in inches
53		
97		
195		
239		
8		Number of response evaluation points, NEVPTS
22	0	
17.6	1.87	
13.2	3.75	
8.8	5.62	Evaluation point X and Y coordinates (for NEVPTS) in inches
4.4	7.49	
0.0	9.363	
0.0	19.18	
0.0	29.0	

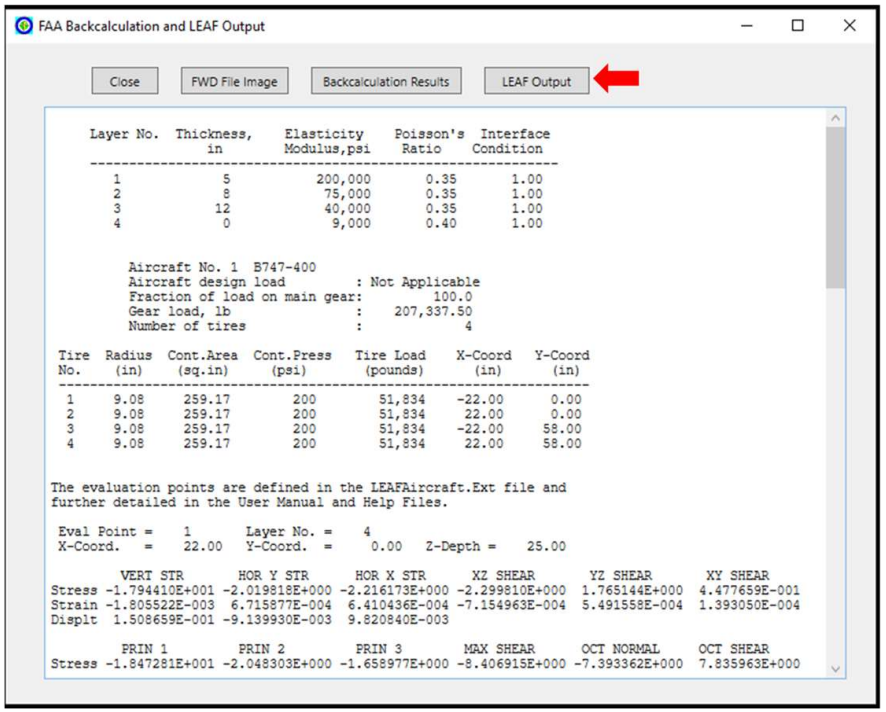
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Run LEAF

LEAF Outputs



- Show Output – LEAF Output
- Pavement layer structure table
- Aircraft table
- Tires data table
- Pavement responses at evaluation points including eval point no., layer no. (determined by evaluation depth), X-coord, Y-coord, and Z-depth



Layer No.	Thickness, in	Elasticity Modulus,psi	Poisson's Ratio	Interface Condition
1	5	200,000	0.35	1.00
2	8	75,000	0.35	1.00
3	12	40,000	0.35	1.00
4	0	9,000	0.40	1.00

Aircraft No. 1 B747-400
Aircraft design load : Not Applicable
Fraction of load on main gear: 100.0
Gear load, lb : 207,337.50
Number of tires : 4

Tire No.	Radius (in)	Cont.Area (sq.in)	Cont.Press (psi)	Tire Load (pounds)	X-Coord (in)	Y-Coord (in)
1	9.08	259.17	200	51,834	-22.00	0.00
2	9.08	259.17	200	51,834	22.00	0.00
3	9.08	259.17	200	51,834	-22.00	58.00
4	9.08	259.17	200	51,834	22.00	58.00

The evaluation points are defined in the LEAFAircraft.Ext file and further detailed in the User Manual and Help Files.

Eval Point = 1 Layer No. = 4
X-Coord. = 22.00 Y-Coord. = 0.00 Z-Depth = 25.00

	VERT STR	HOR Y STR	HOR X STR	XZ SHEAR	YZ SHEAR	XY SHEAR
Stress	-1.794410E+001	-2.019818E+000	-2.216173E+000	-2.299810E+000	1.765144E+000	4.477659E-001
Strain	-1.805522E-003	6.715877E-004	6.410436E-004	-7.154963E-004	5.491558E-004	1.393050E-004
Displt	1.508659E-001	-9.139930E-003	9.820840E-003			

	PRIN 1	PRIN 2	PRIN 3	MAX SHEAR	OCT NORMAL	OCT SHEAR
Stress	-1.847281E+001	-2.048303E+000	-1.658977E+000	-8.406915E+000	-7.393362E+000	7.835963E+000

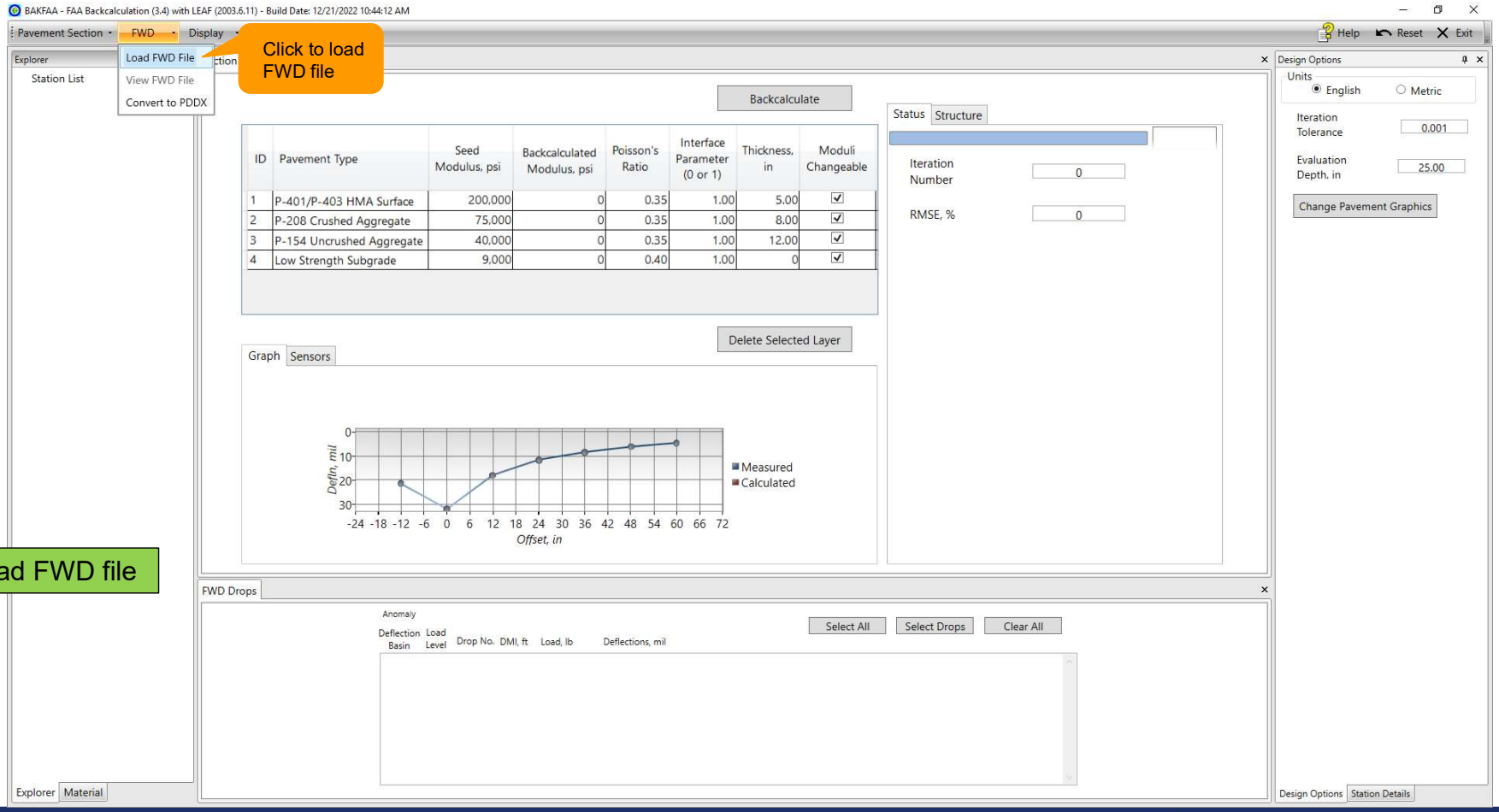
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Software Demo

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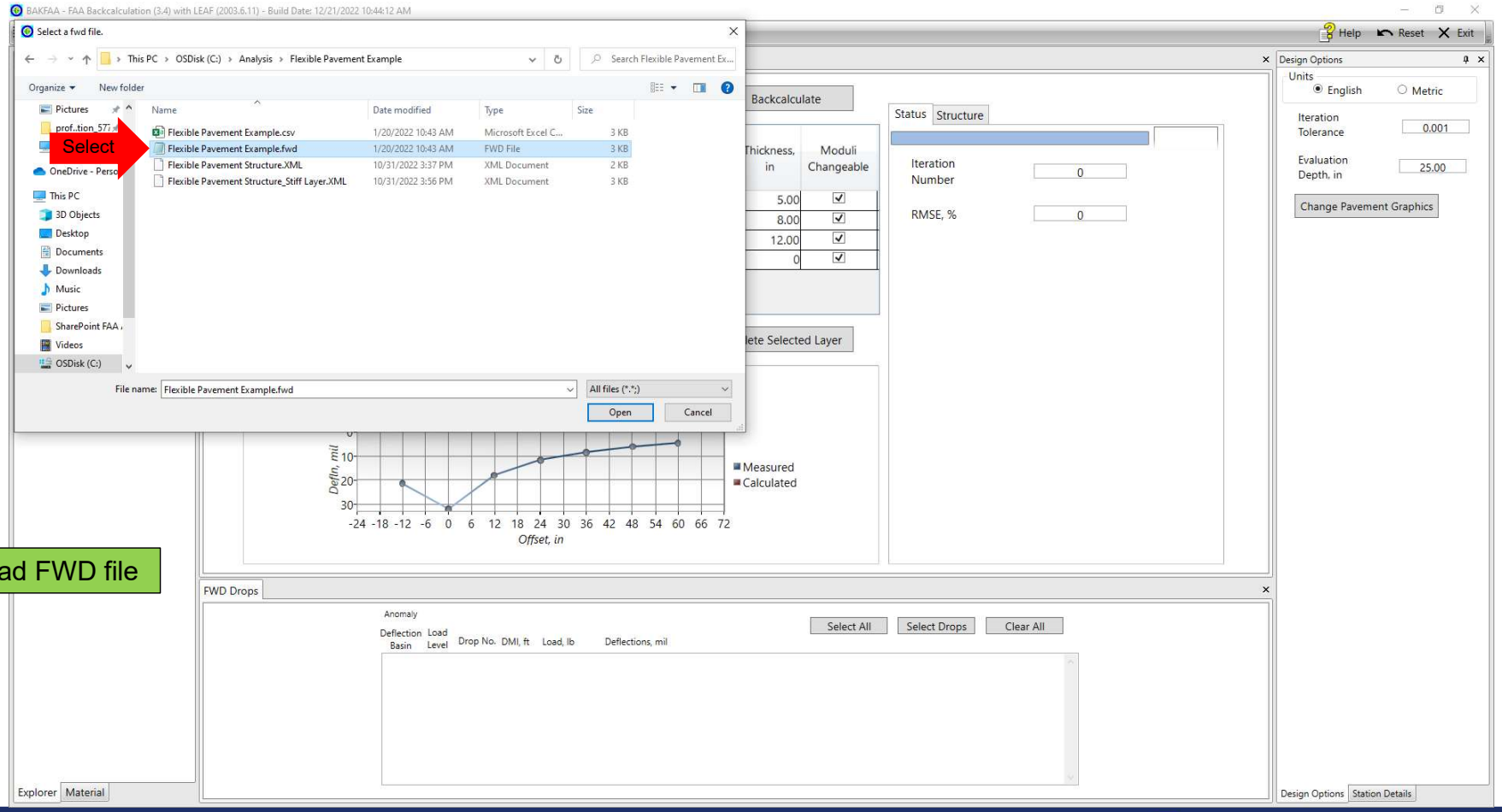
Flexible Pavement Example

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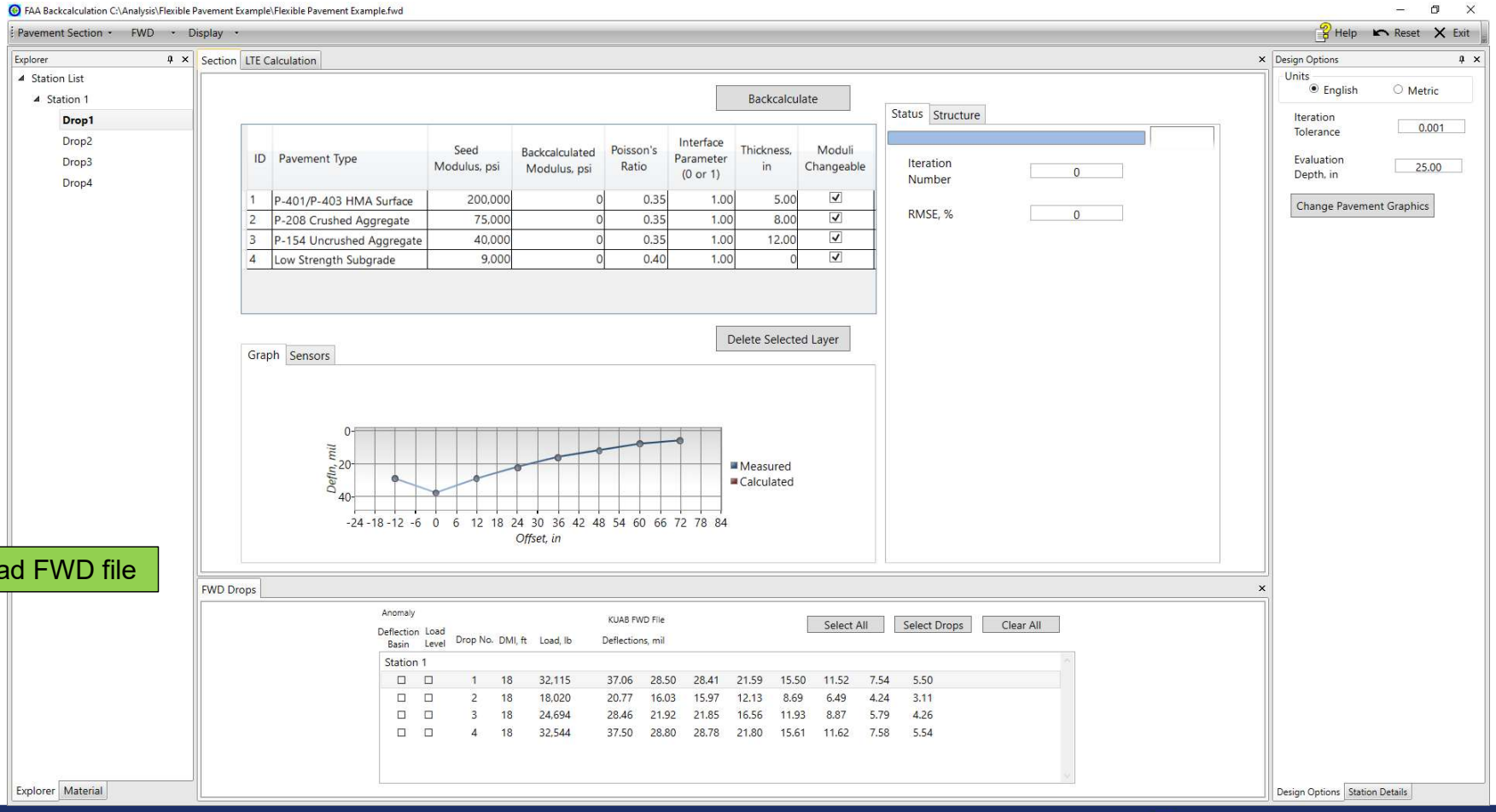
Load FWD file

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Load FWD file

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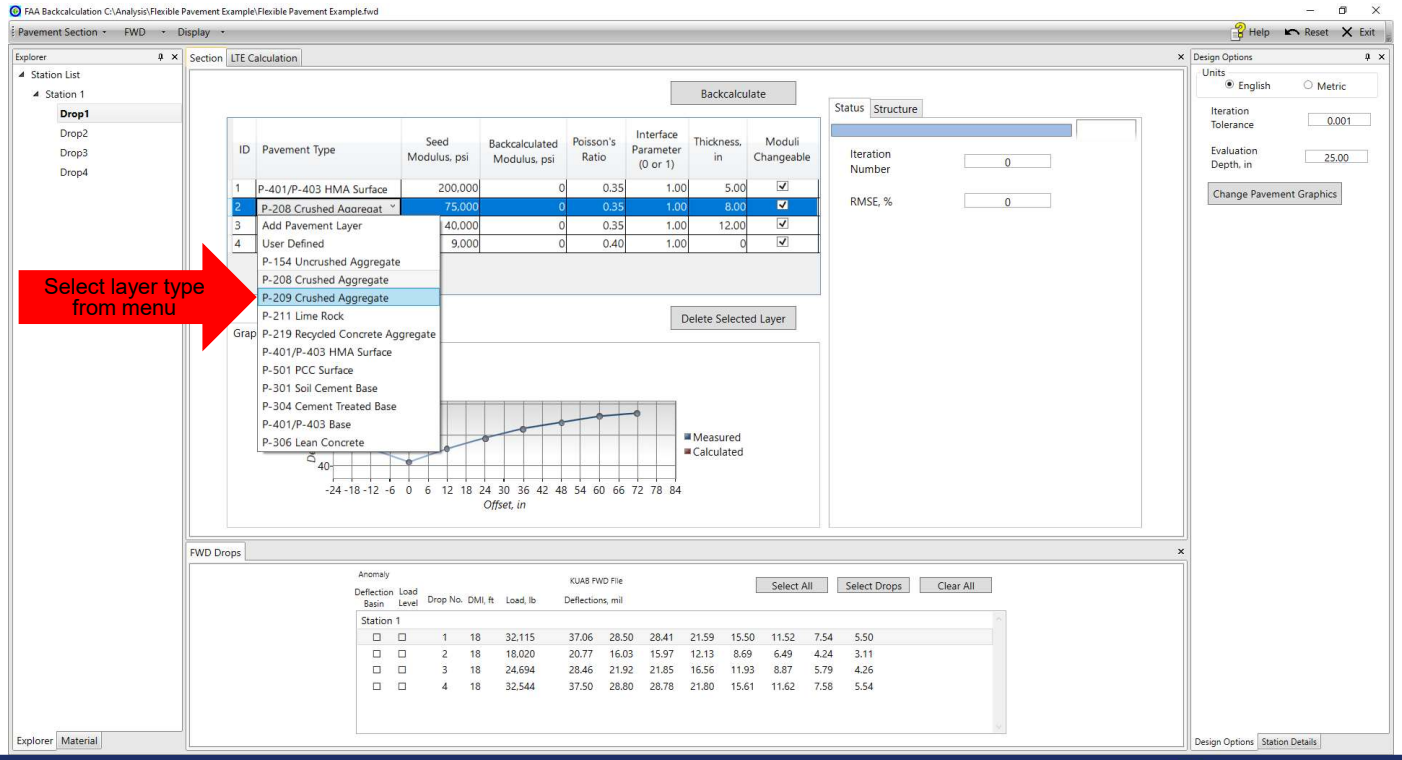


Load FWD file

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Assign pavement structure

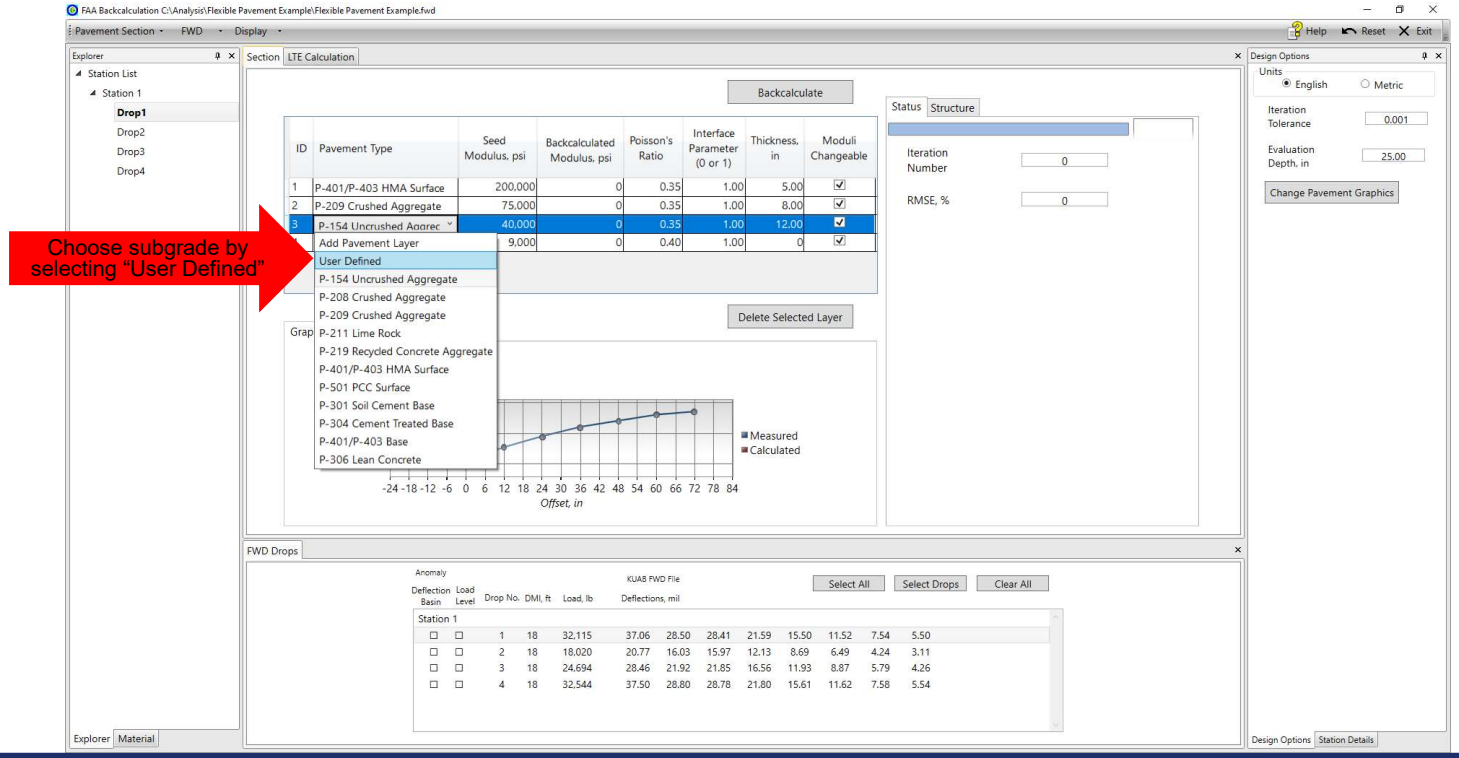
Material Type	Thickness (in.)
P-401/P-403 hot mix asphalt	6
P-209 crushed stone base	12
Subgrade with a CBR of 10	



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Assign pavement structure

Material Type	Thickness (in.)
P-401/P-403 hot mix asphalt	6
P-209 crushed stone base	12
Subgrade with a CBR of 10	

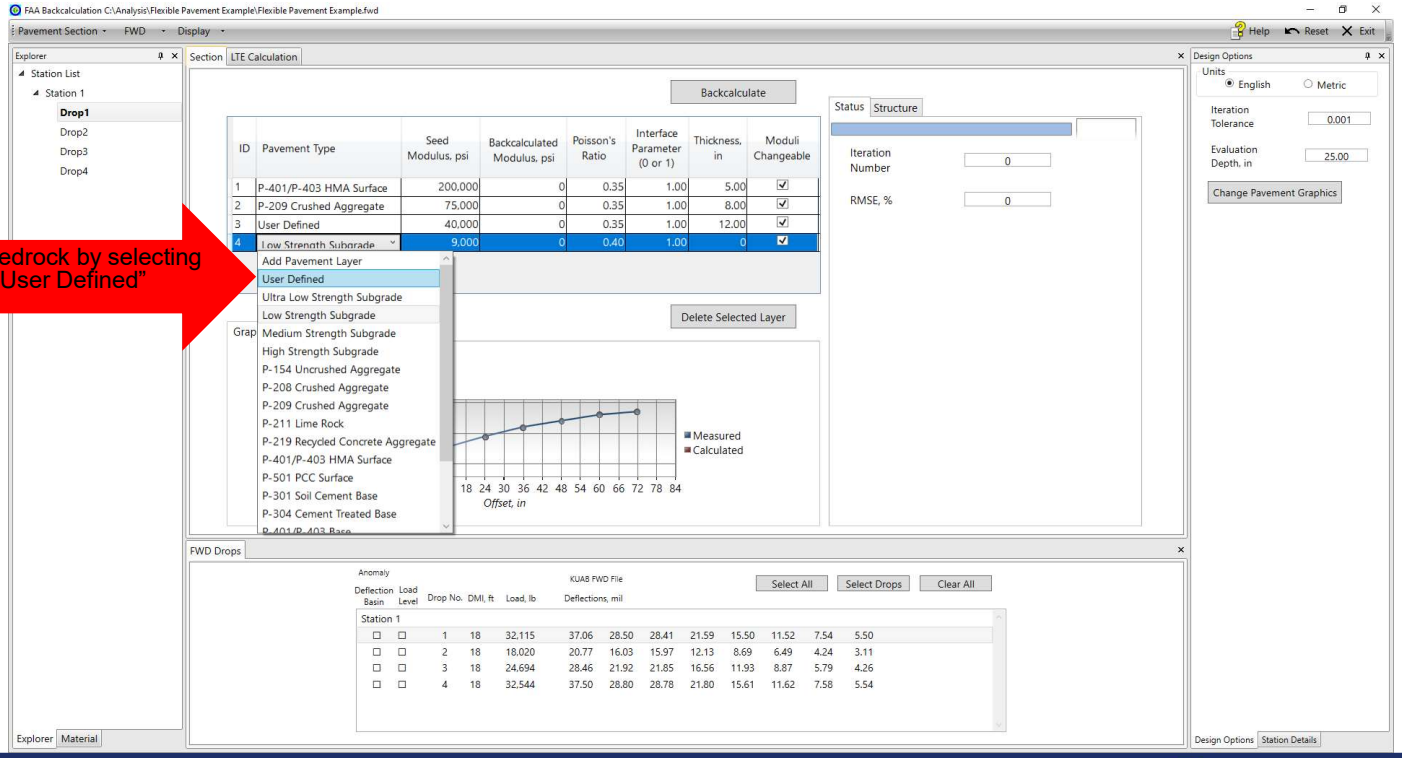


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Assign pavement structure

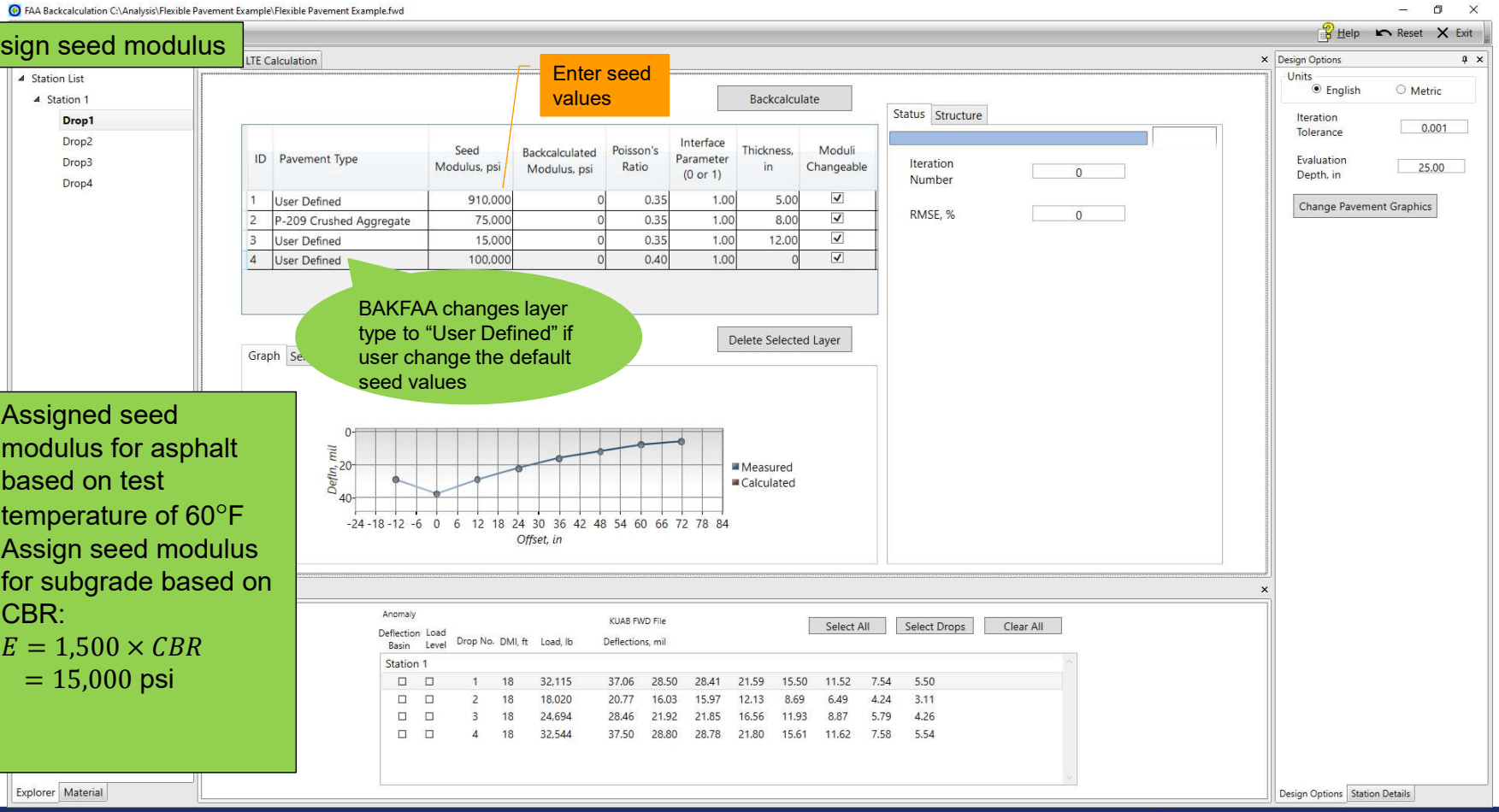
Material Type	Thickness (in.)
P-401/P-403 hot mix asphalt	6
P-209 crushed stone base	12
Subgrade with a CBR of 10	

Add bedrock by selecting "User Defined"



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Assign seed modulus



- Assigned seed modulus for asphalt based on test temperature of 60°F
- Assign seed modulus for subgrade based on CBR:

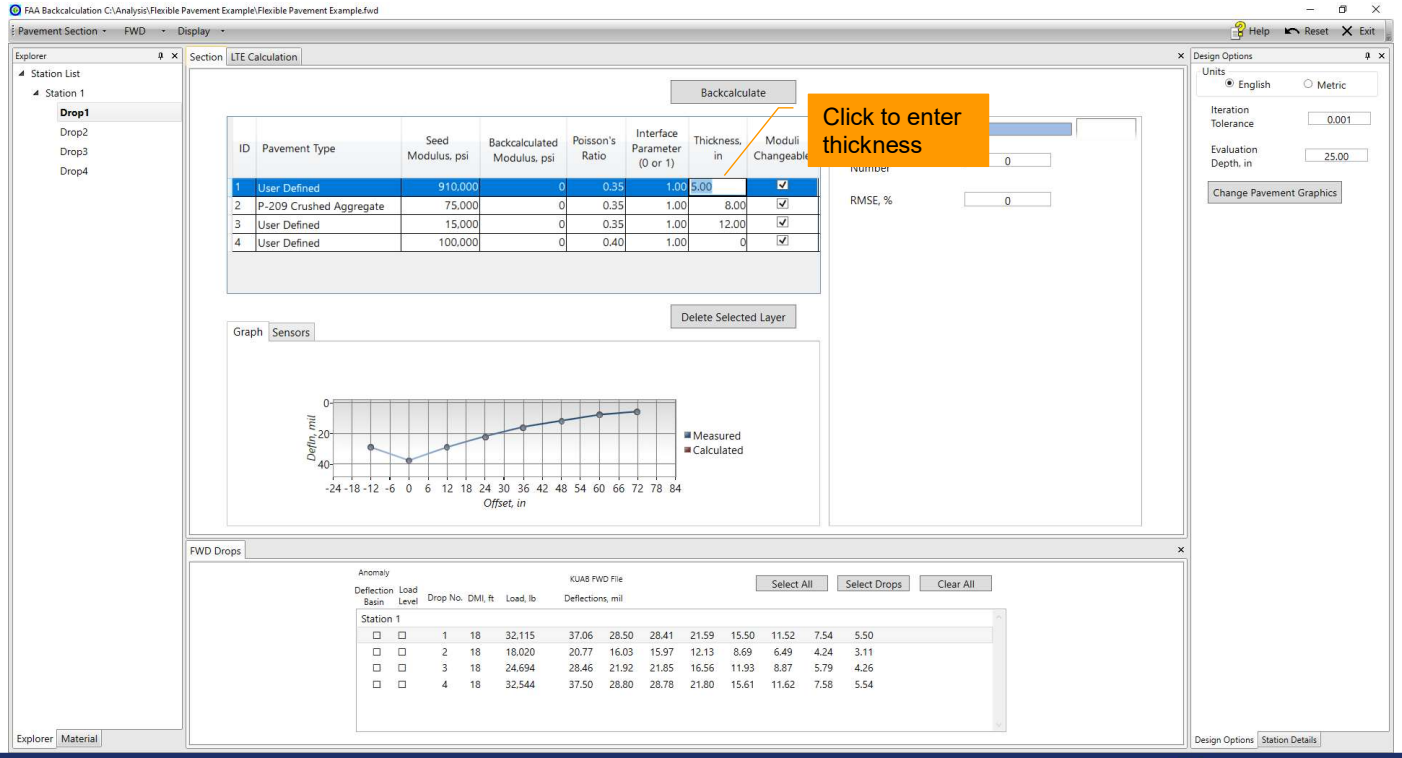
$E = 1{,}500 \times CBR$

$= 15{,}000$ psi

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Assign layer thickness

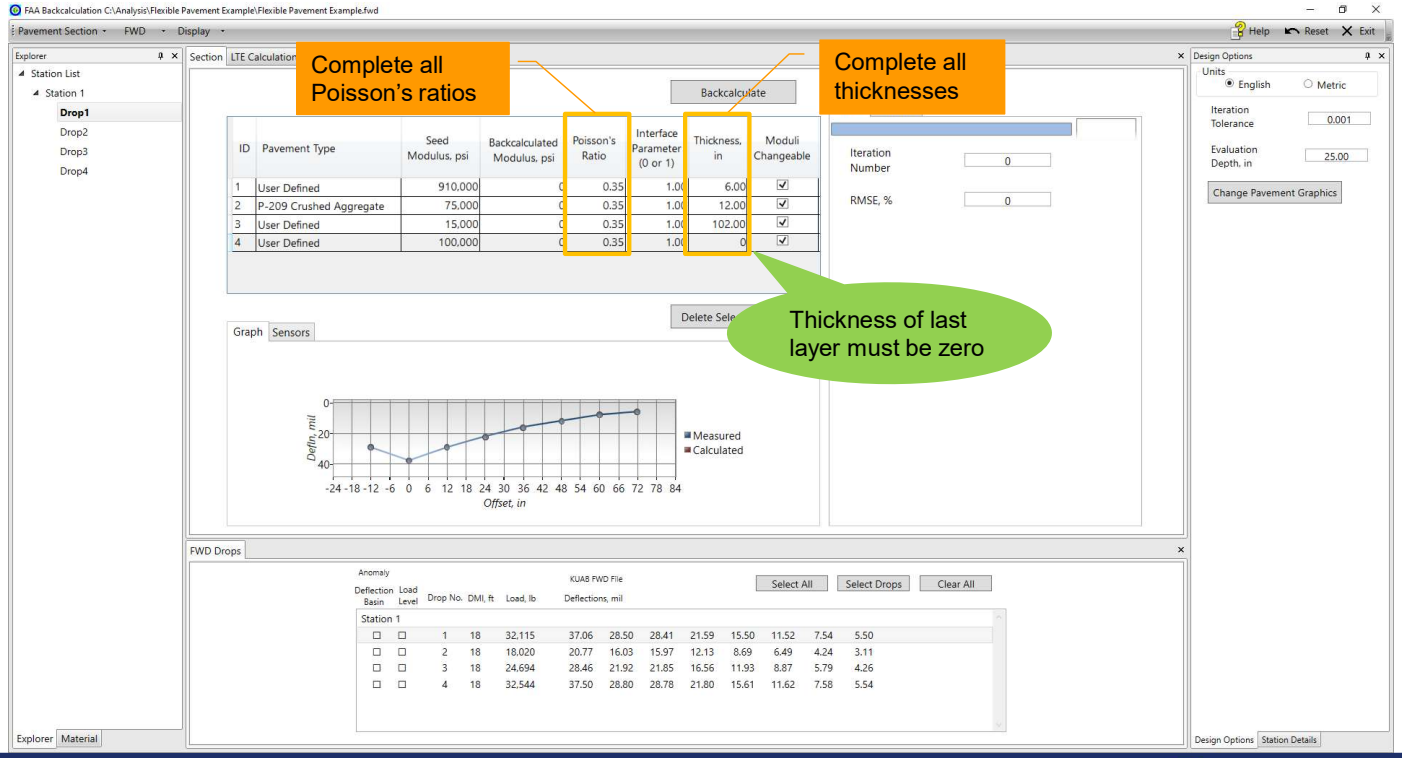
Material Type	Thickness (in.)
P-401/P-403 hot mix asphalt	6
P-209 crushed stone base	12
Subgrade with a CBR of 10	



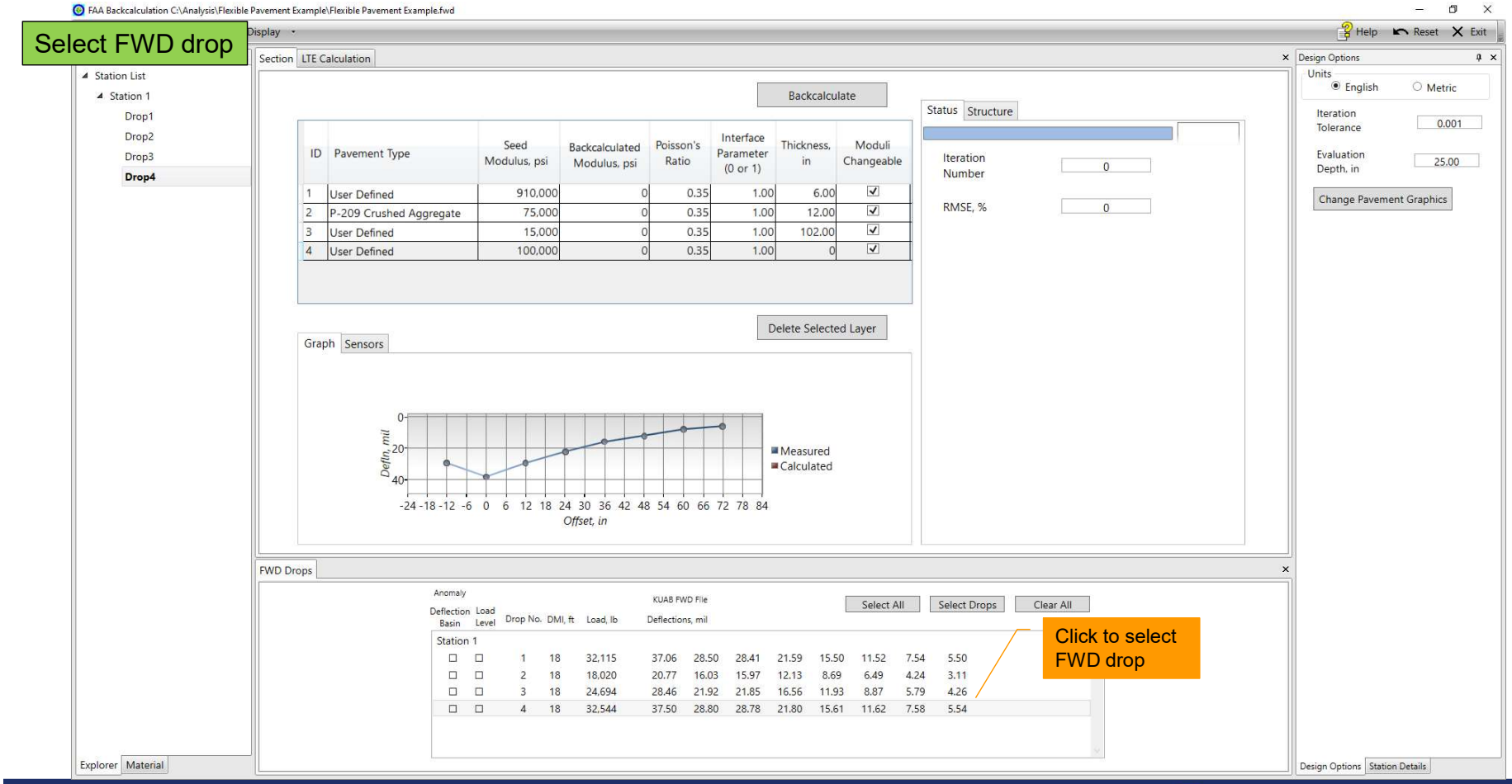
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Assign layer thickness

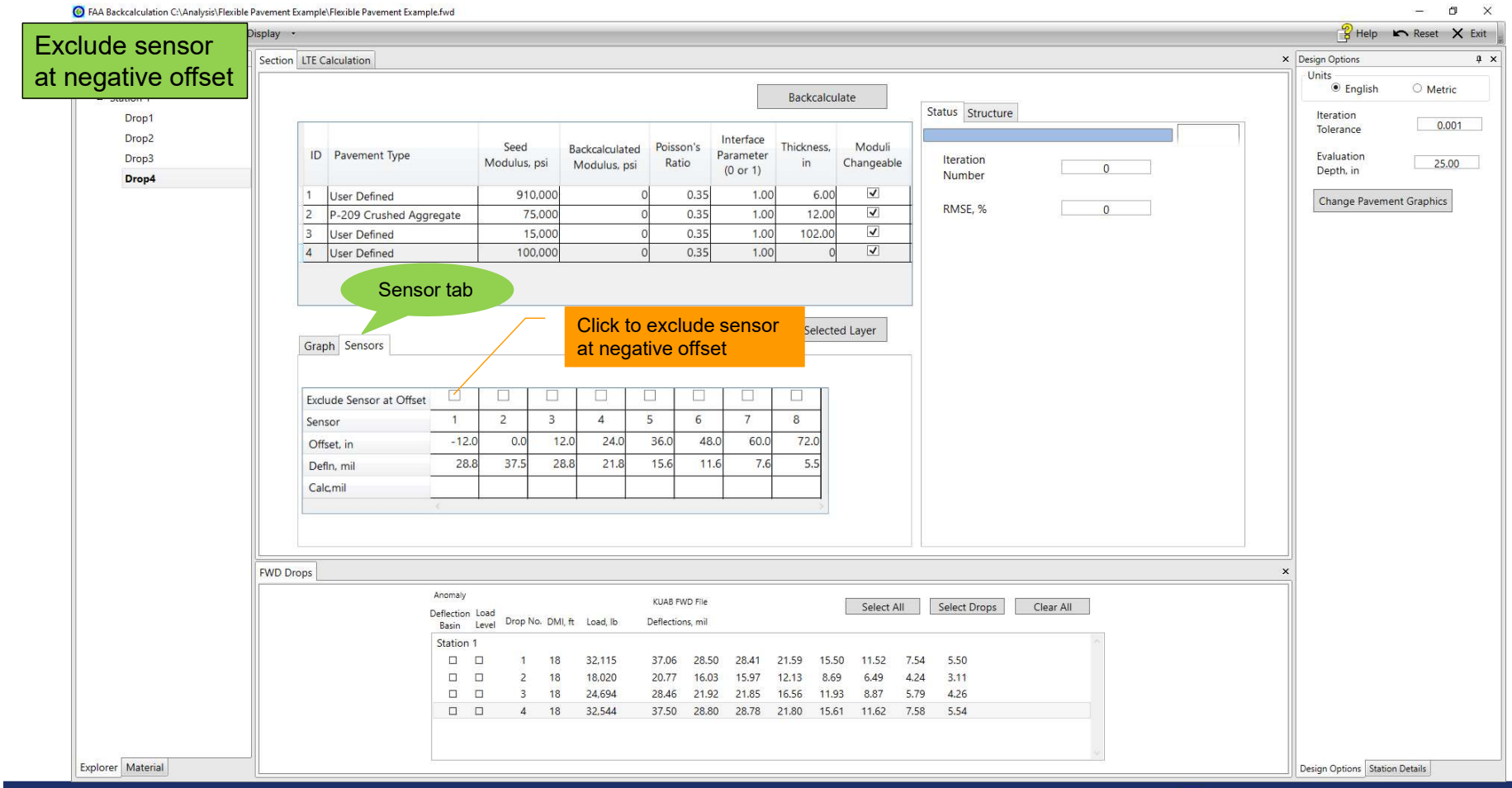
Material Type	Thickness (in.)
P-401/P-403 hot mix asphalt	6
P-209 crushed stone base	12
Subgrade with a CBR of 10	



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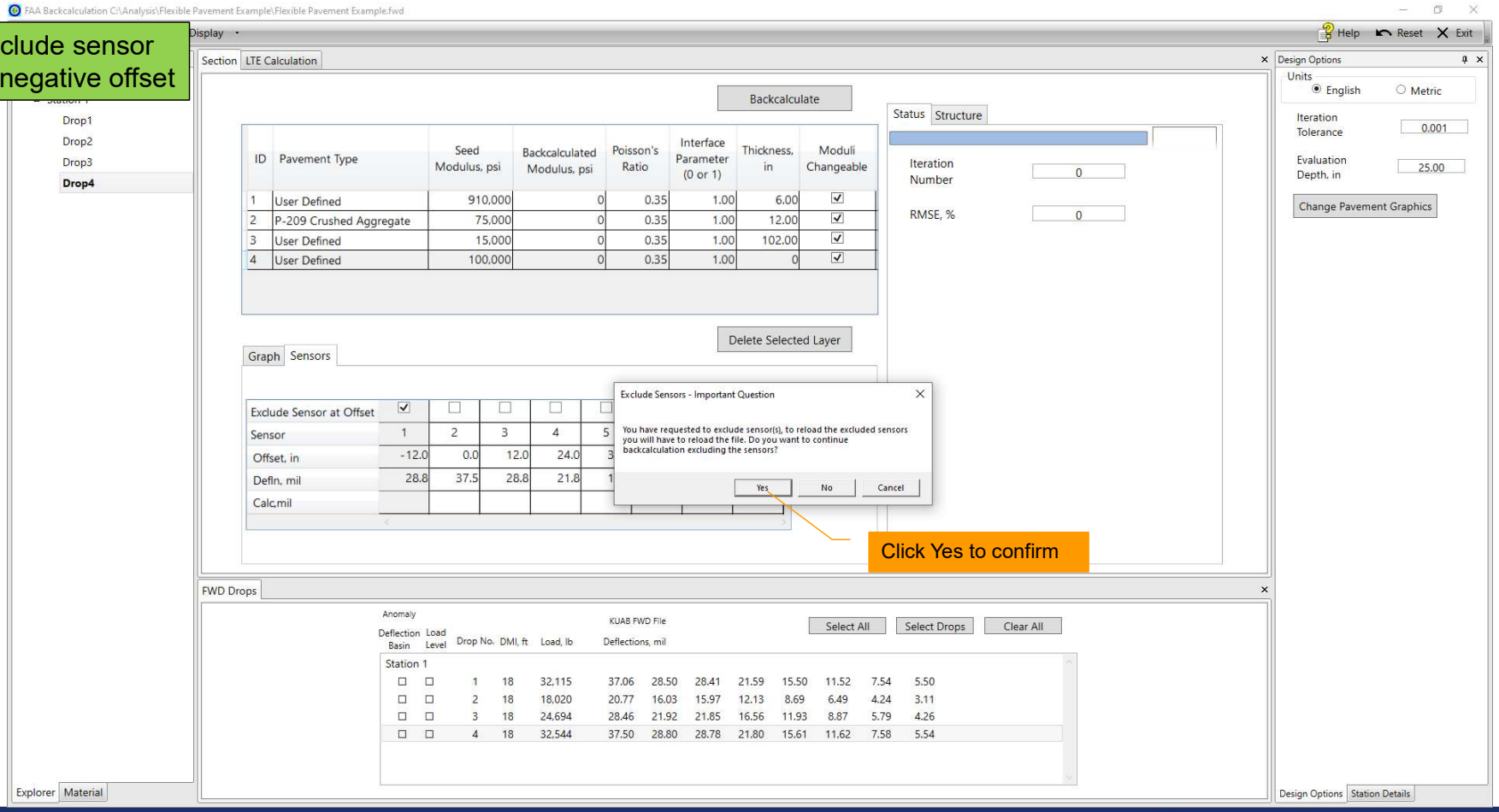


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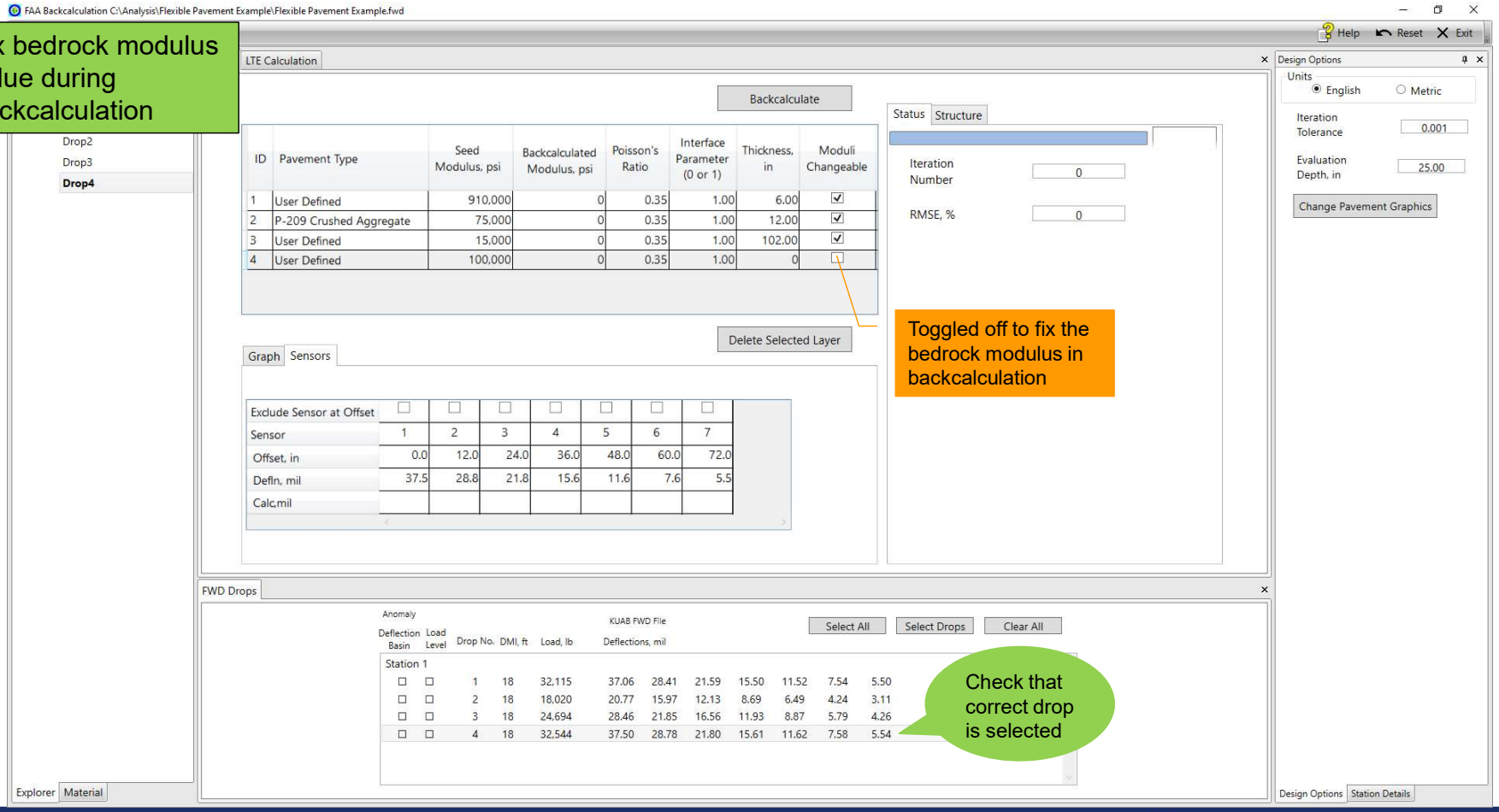
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Exclude sensor at negative offset



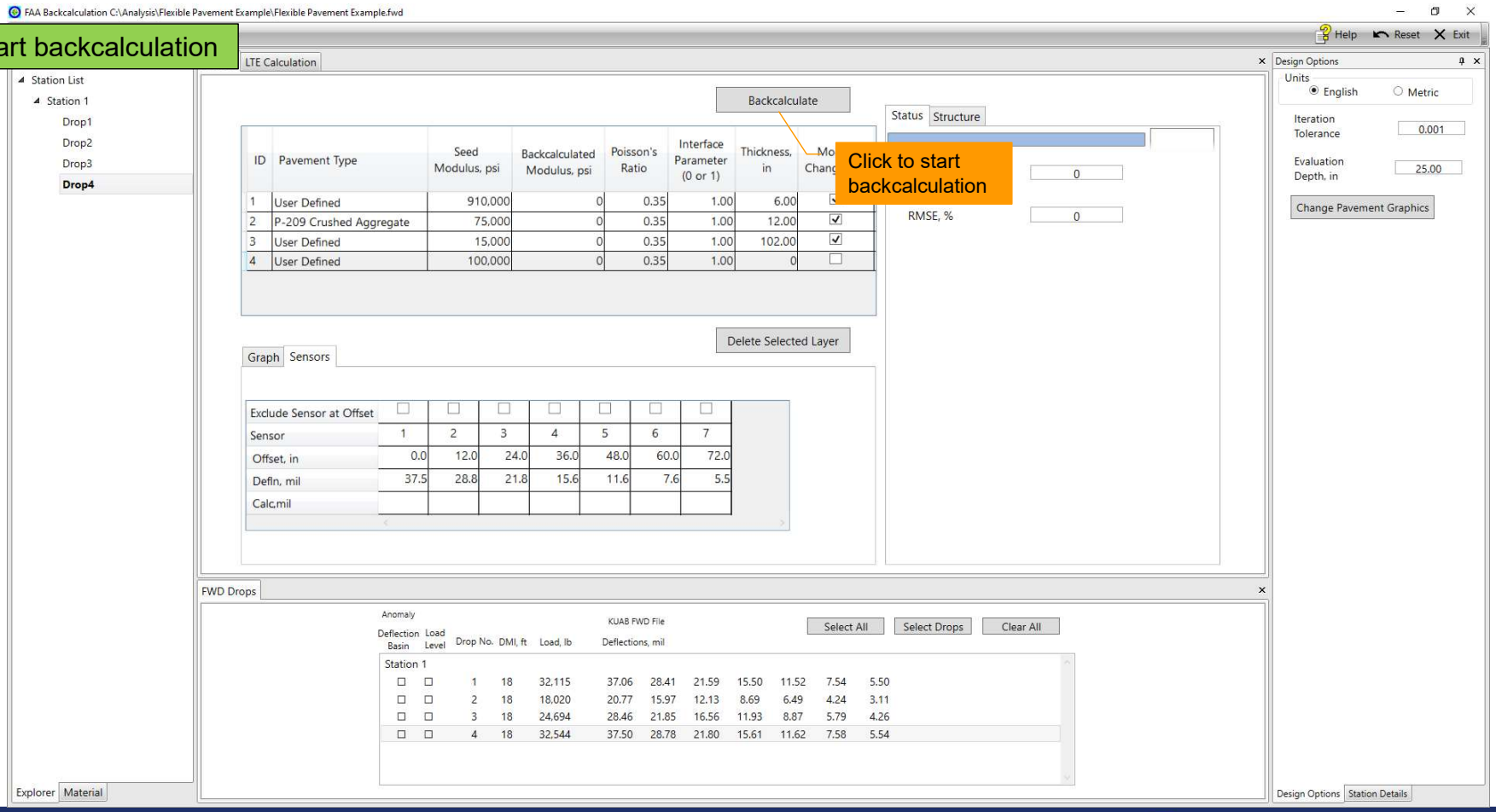
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Fix bedrock modulus value during backcalculation



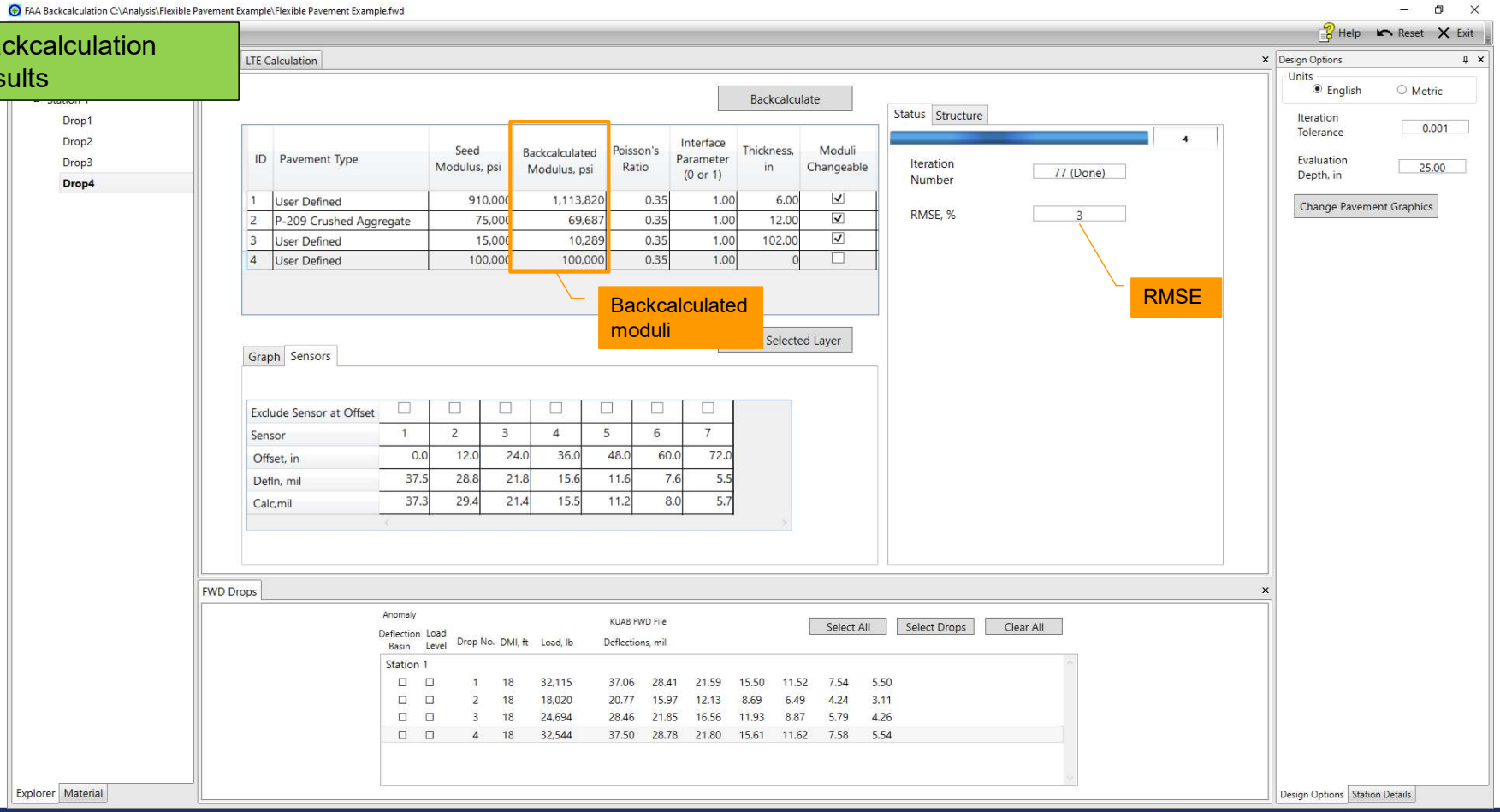
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Start backcalculation

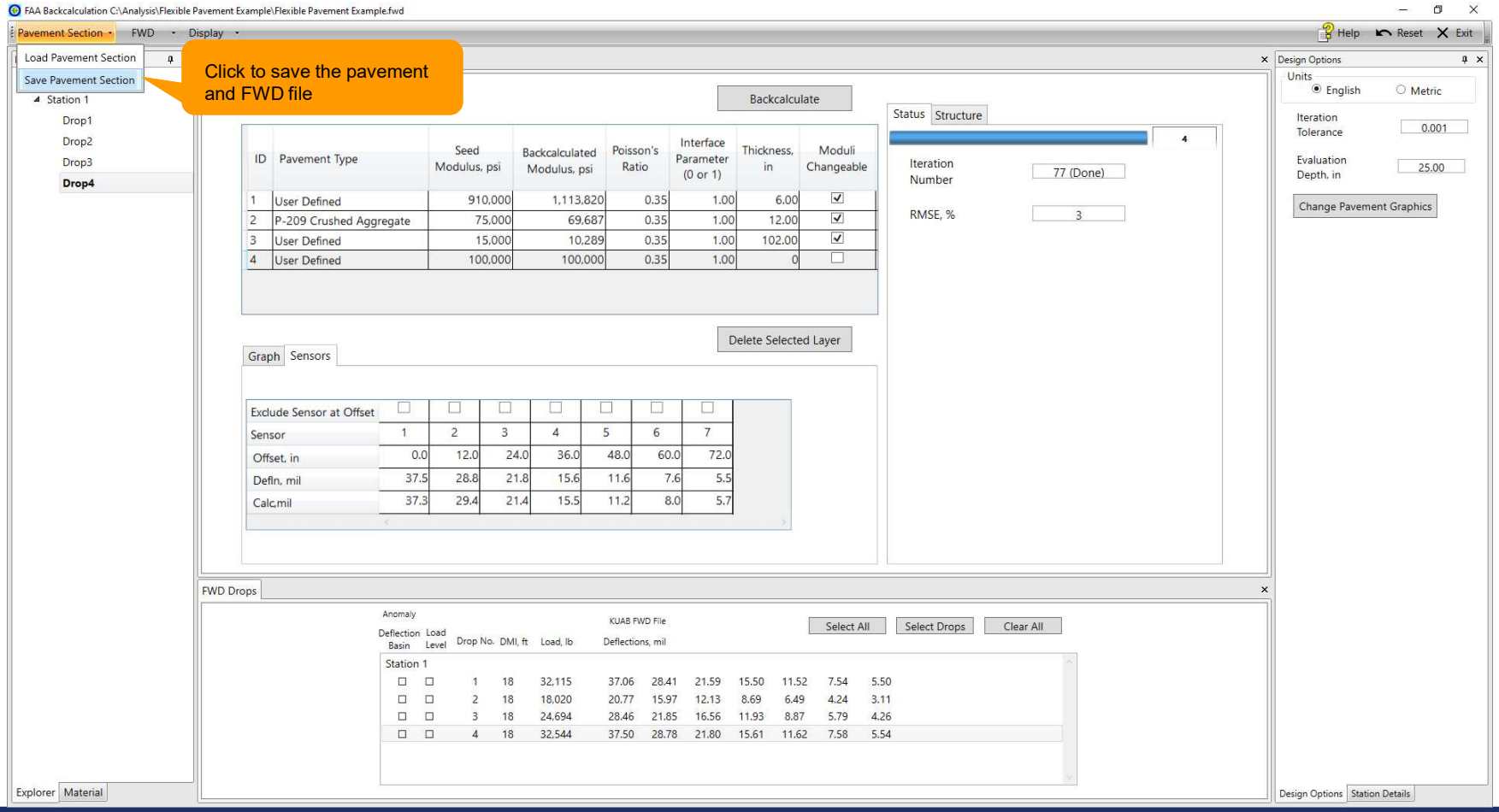


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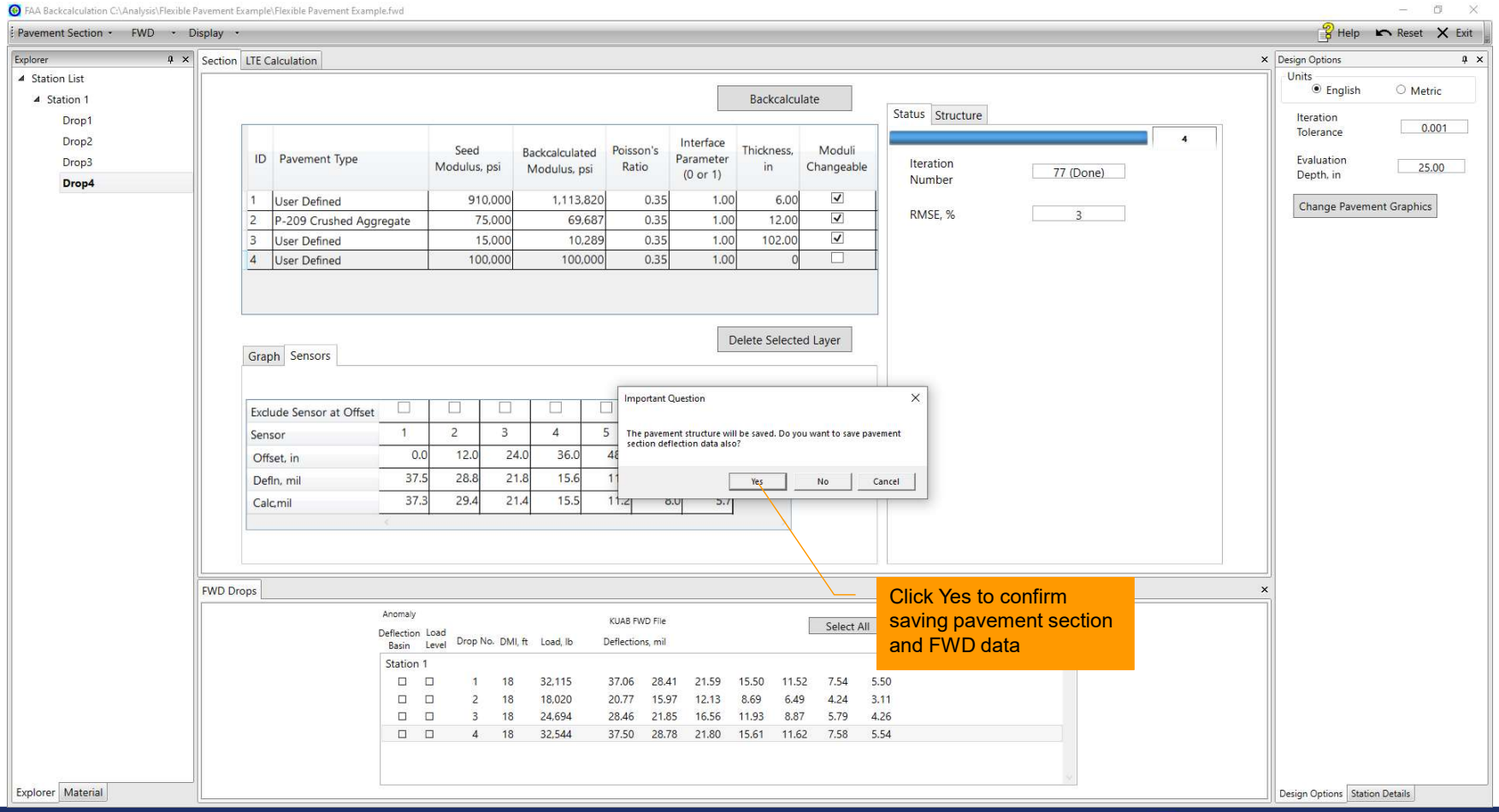
Backcalculation results



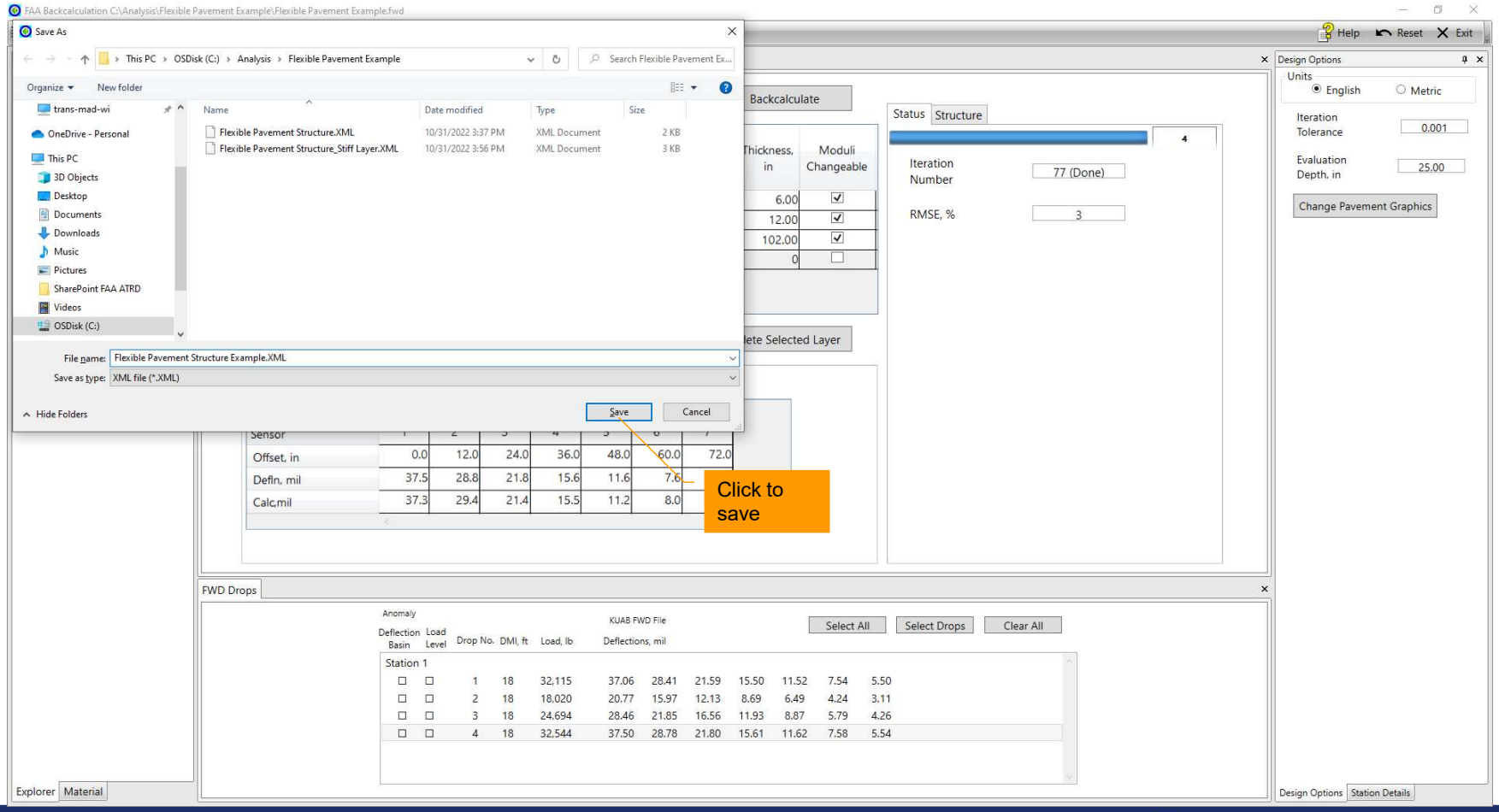
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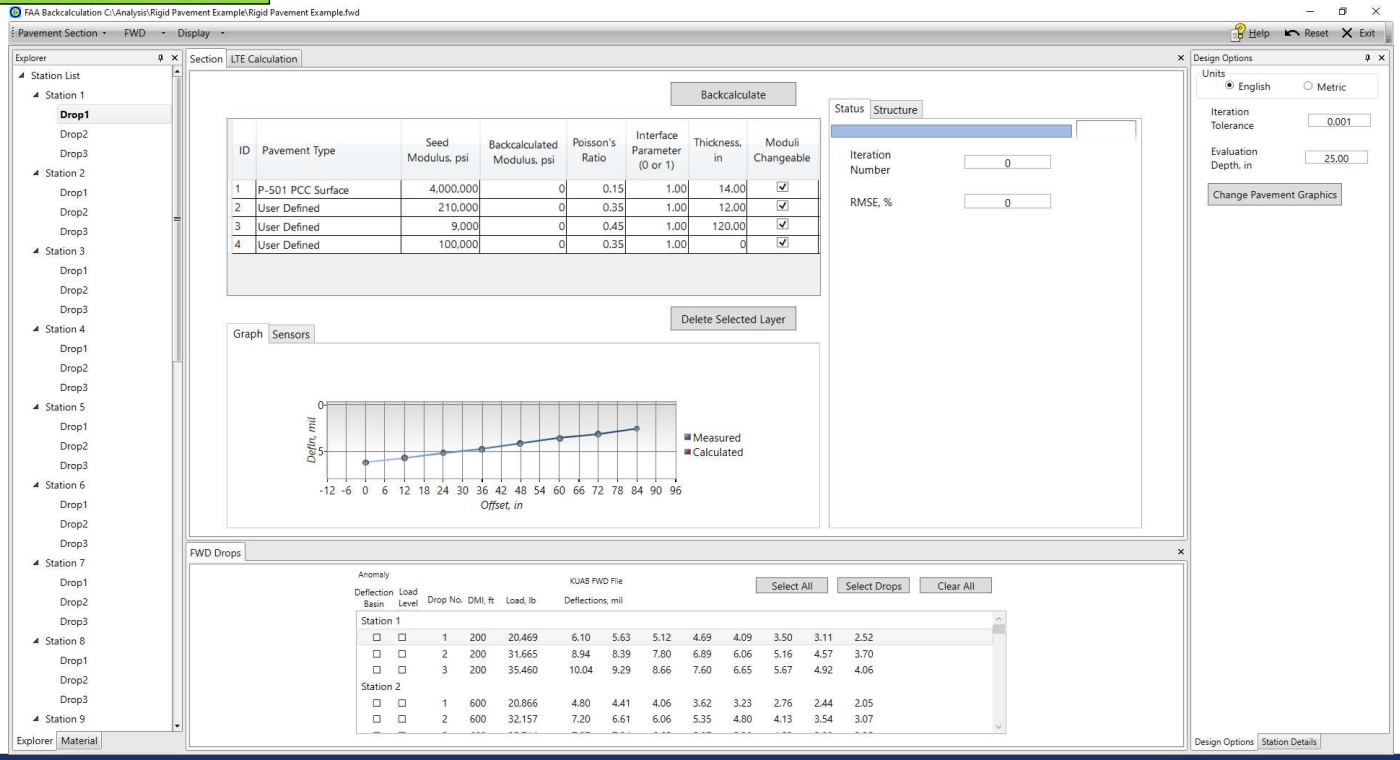
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Rigid Pavement Example

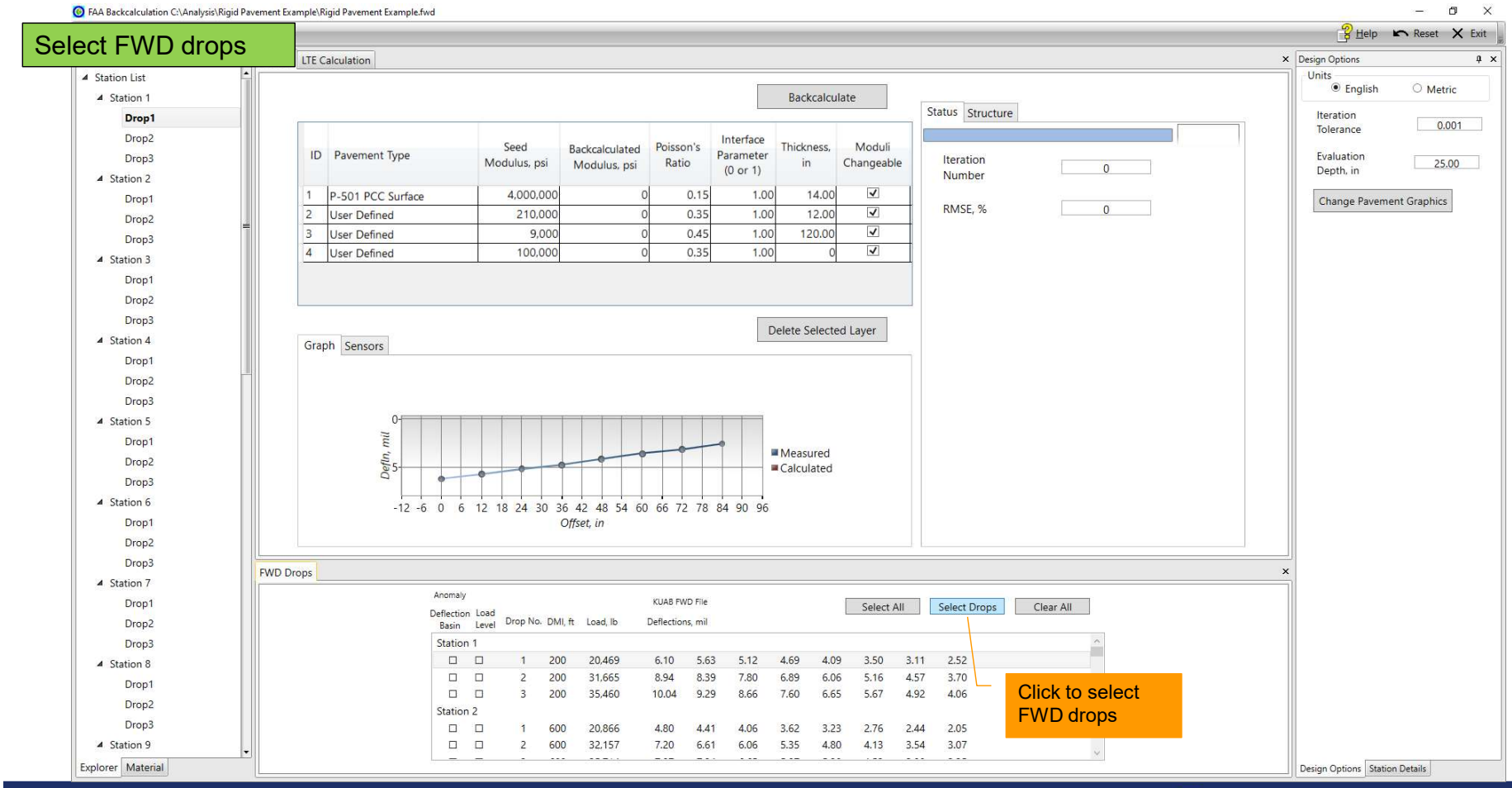
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- Load pavement section
- Load FWD file
- Assign layers and thickness
- Assign seed modulus

Material Type	Thickness (in)
P-501 cement concrete pavement	14
P-156 cement treated subgrade	12
Subgrade with a CBR of 6	

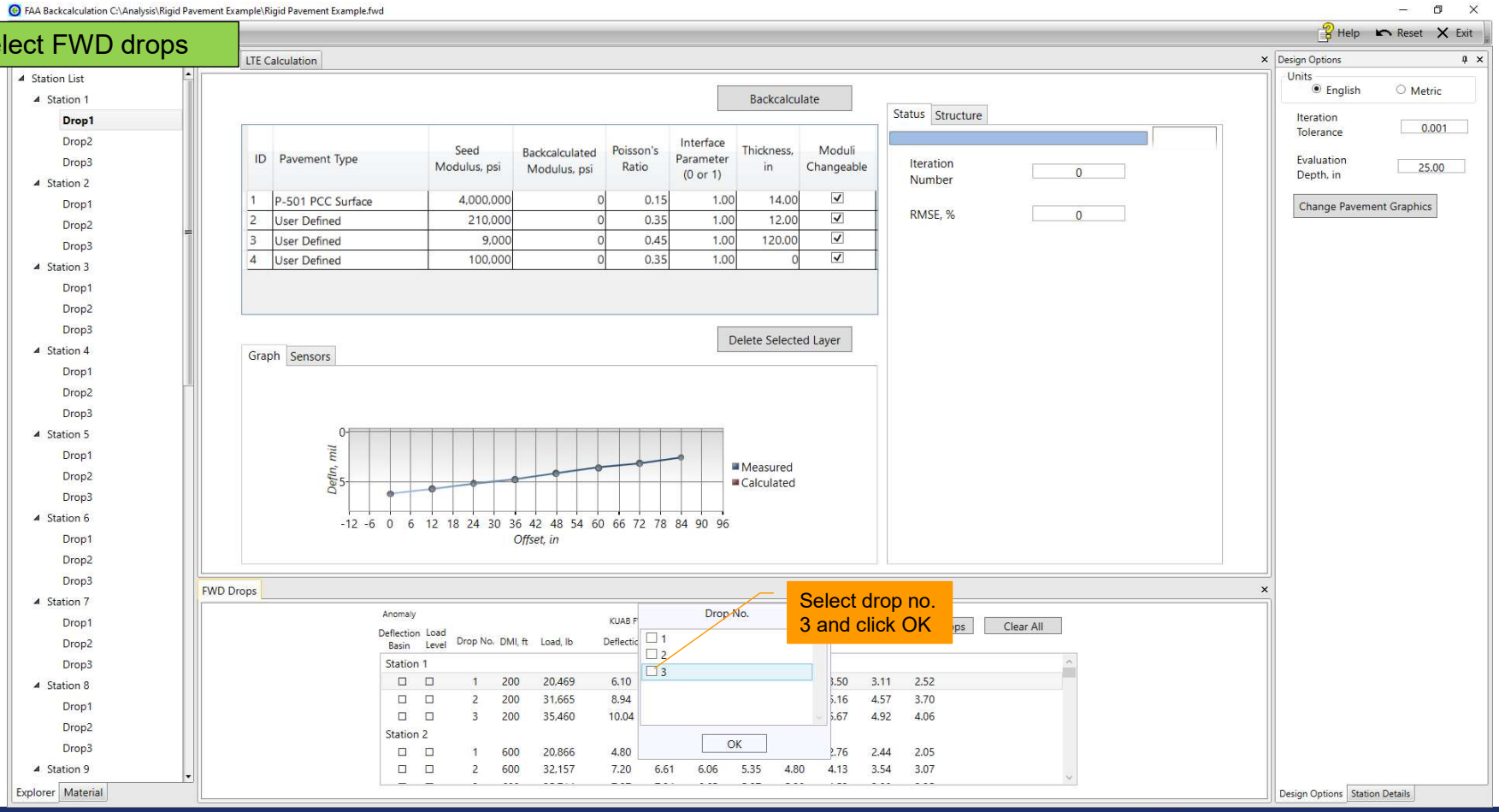


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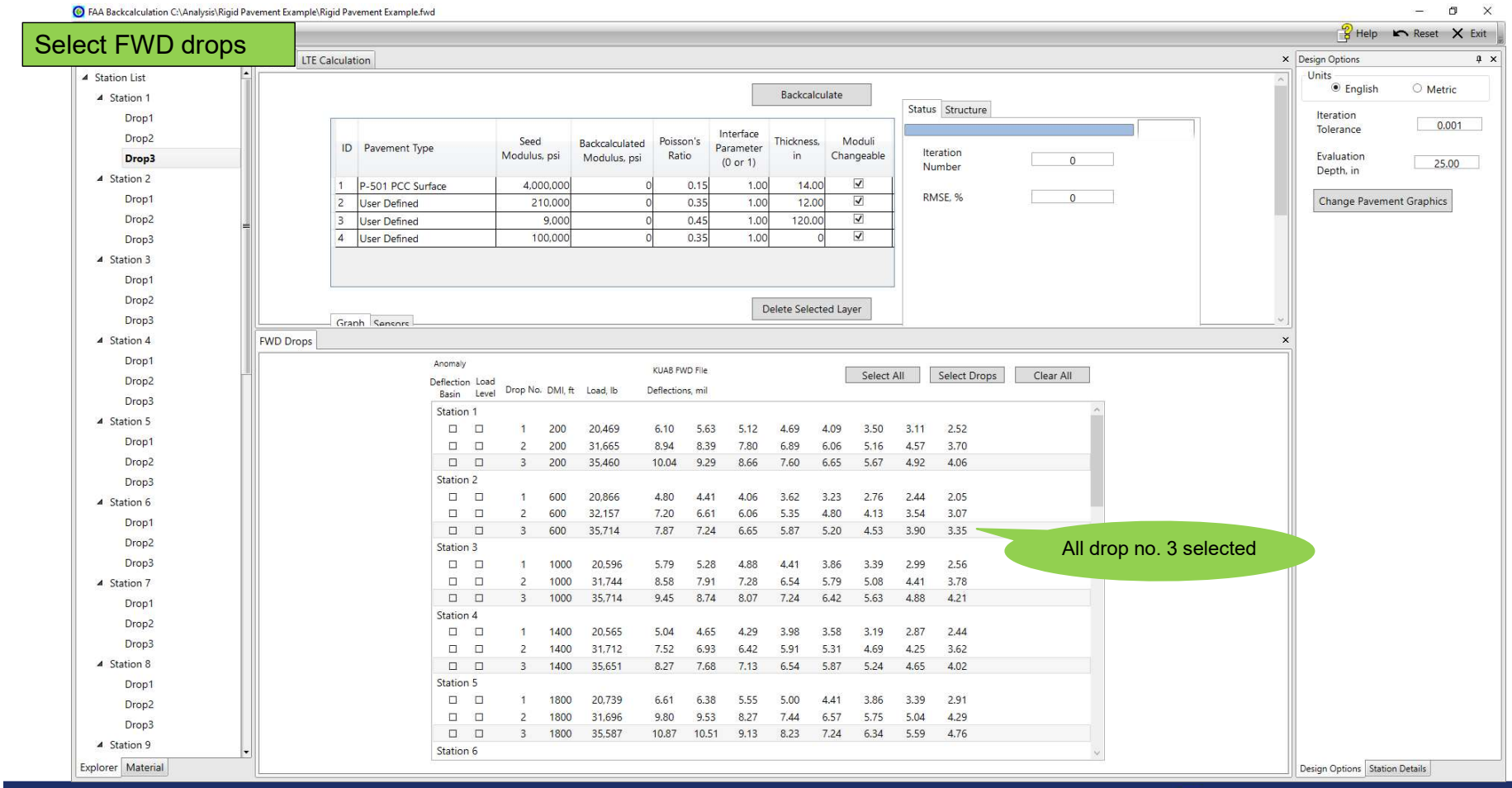


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Select FWD drops

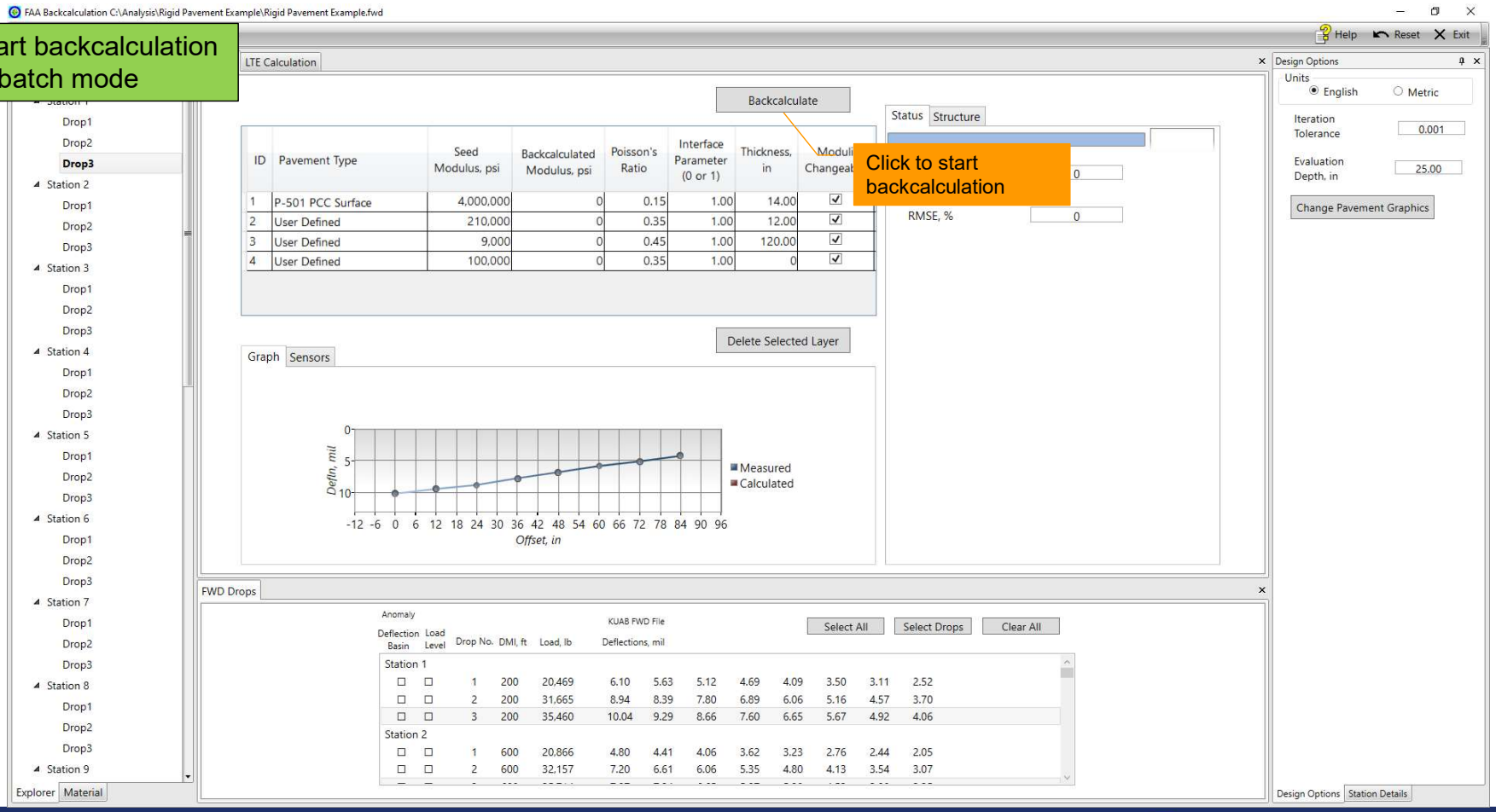


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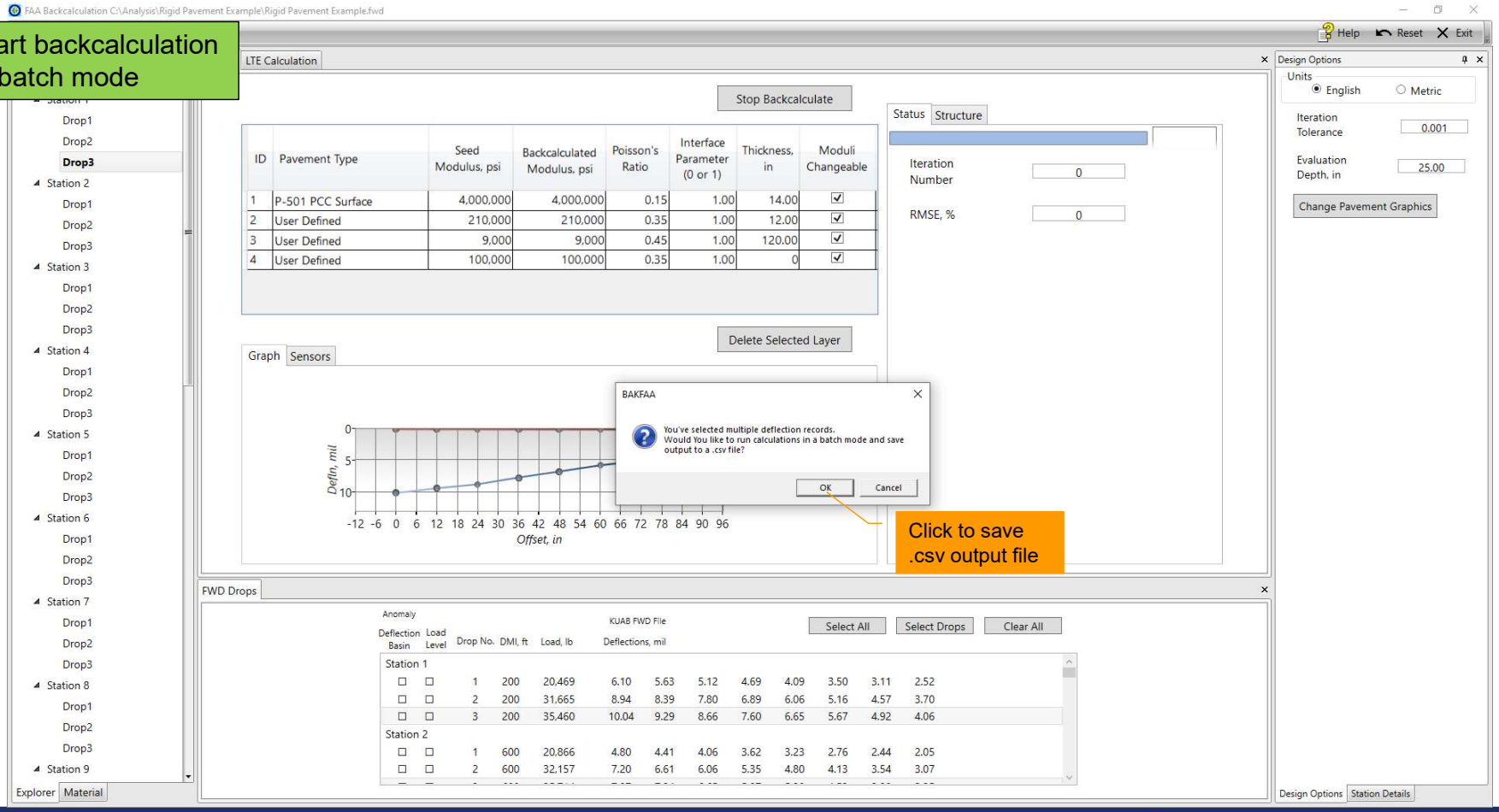
Start backcalculation in batch mode



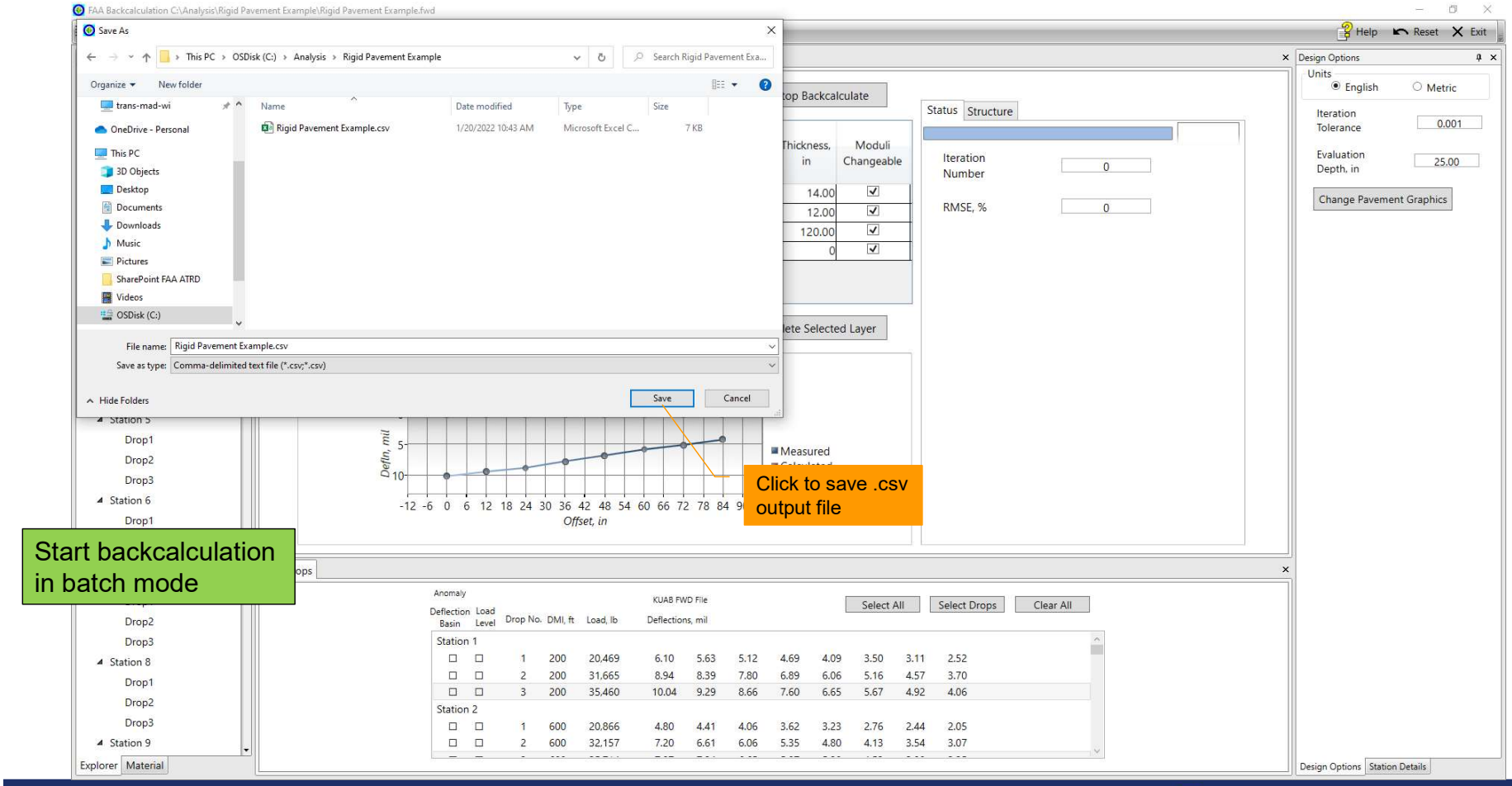
ID	Pavement Type	Seed Modulus, psi	Backcalculated Modulus, psi	Poisson's Ratio	Interface Parameter (0 or 1)	Thickness, in	Modul Changeat
1	P-501 PCC Surface	4,000,000	0	0.15	1.00	14.00	☑
2	User Defined	210,000	0	0.35	1.00	12.00	☑
3	User Defined	9,000	0	0.45	1.00	120.00	☑
4	User Defined	100,000	0	0.35	1.00	0	☑

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Start backcalculation in batch mode

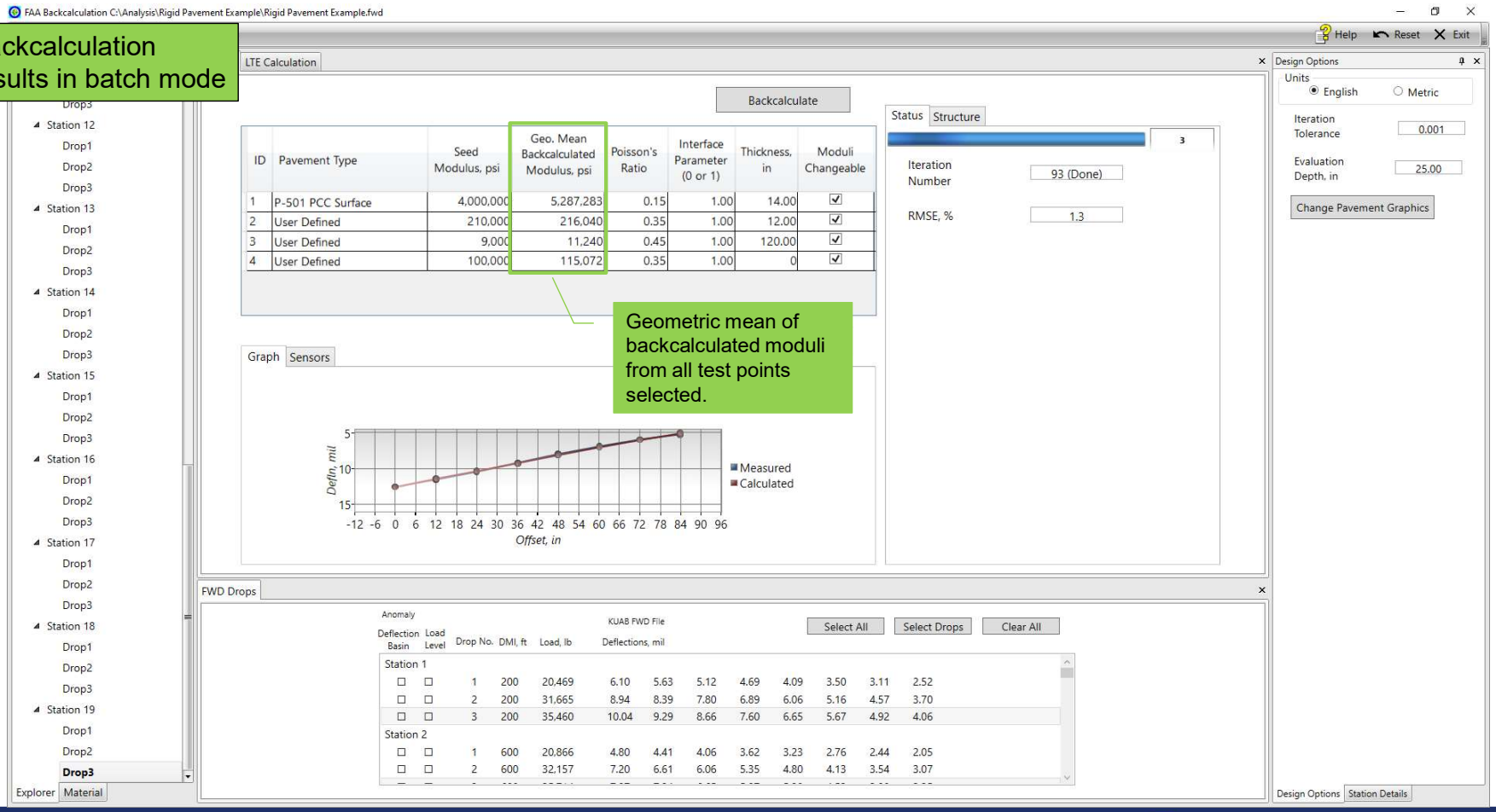


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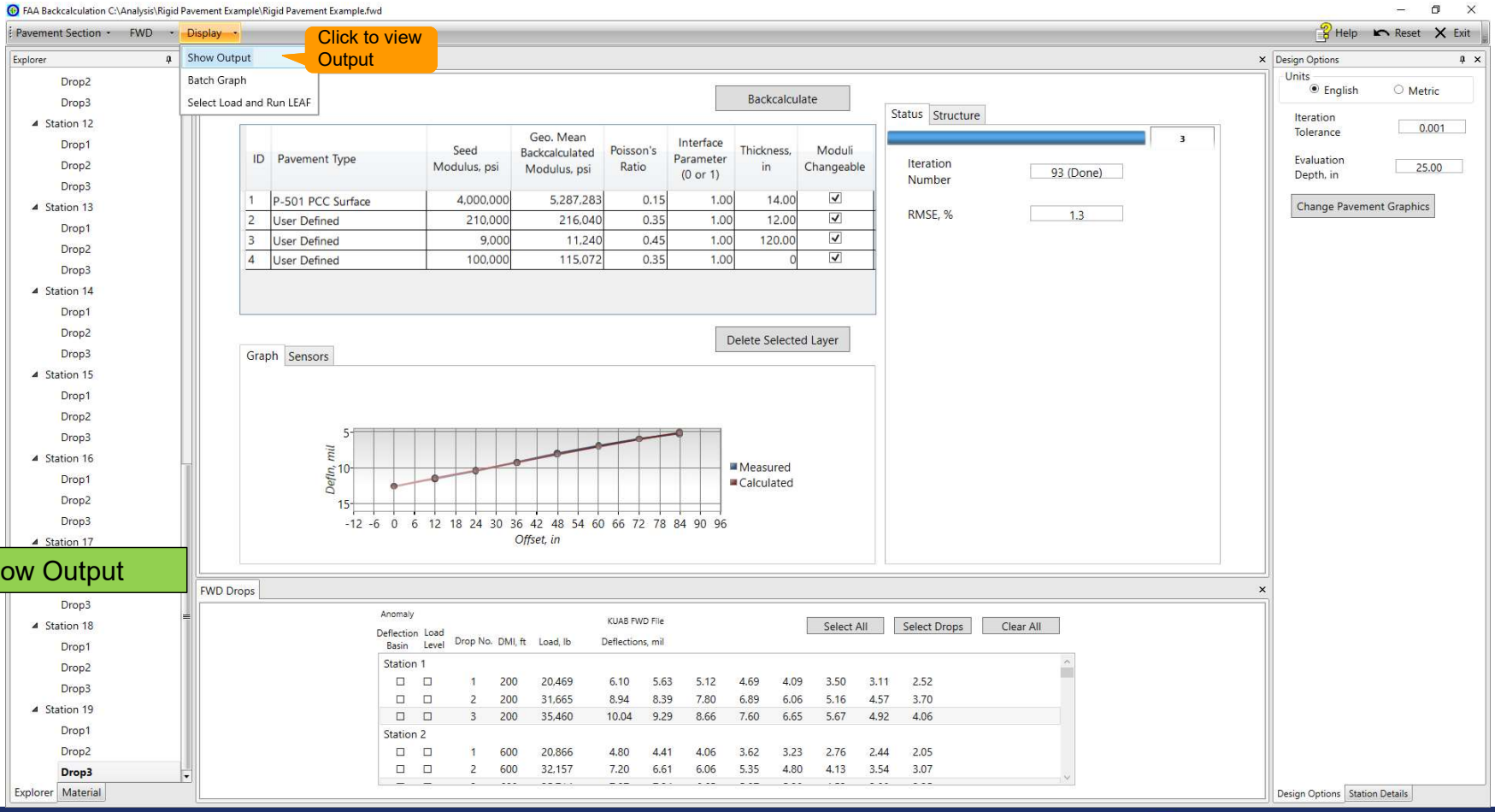


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Backcalculation results in batch mode



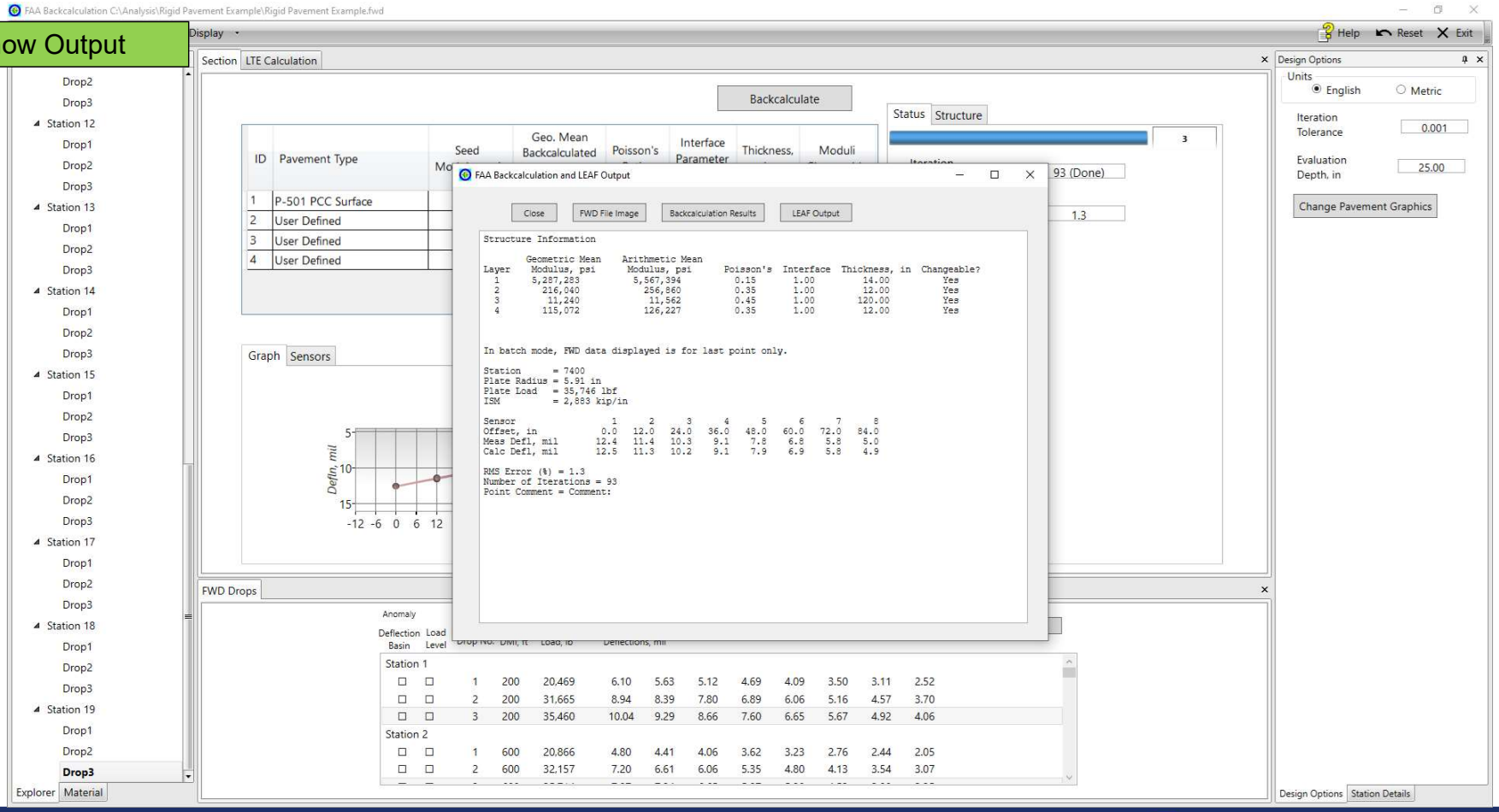
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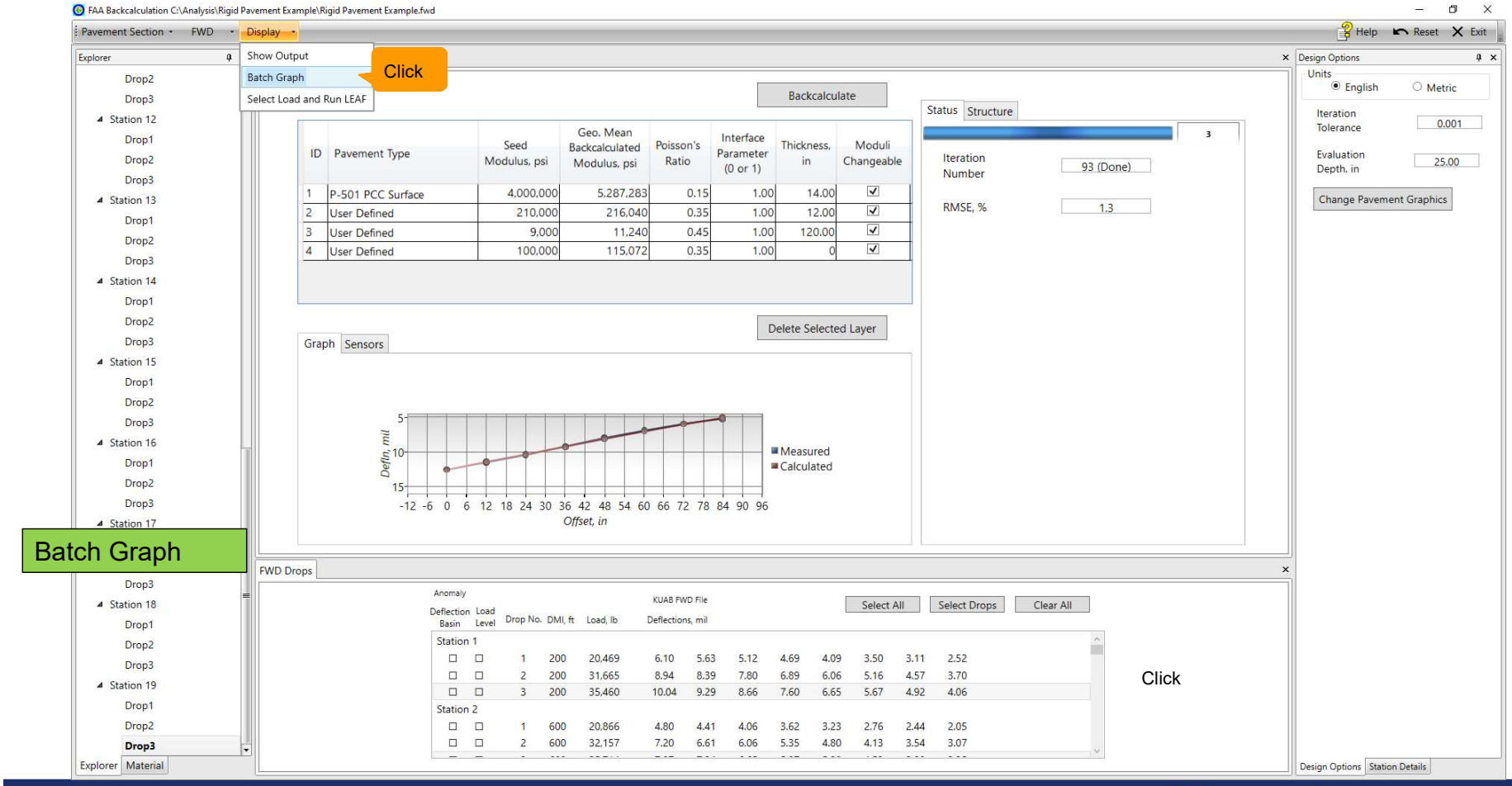
Show Output

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Show Output

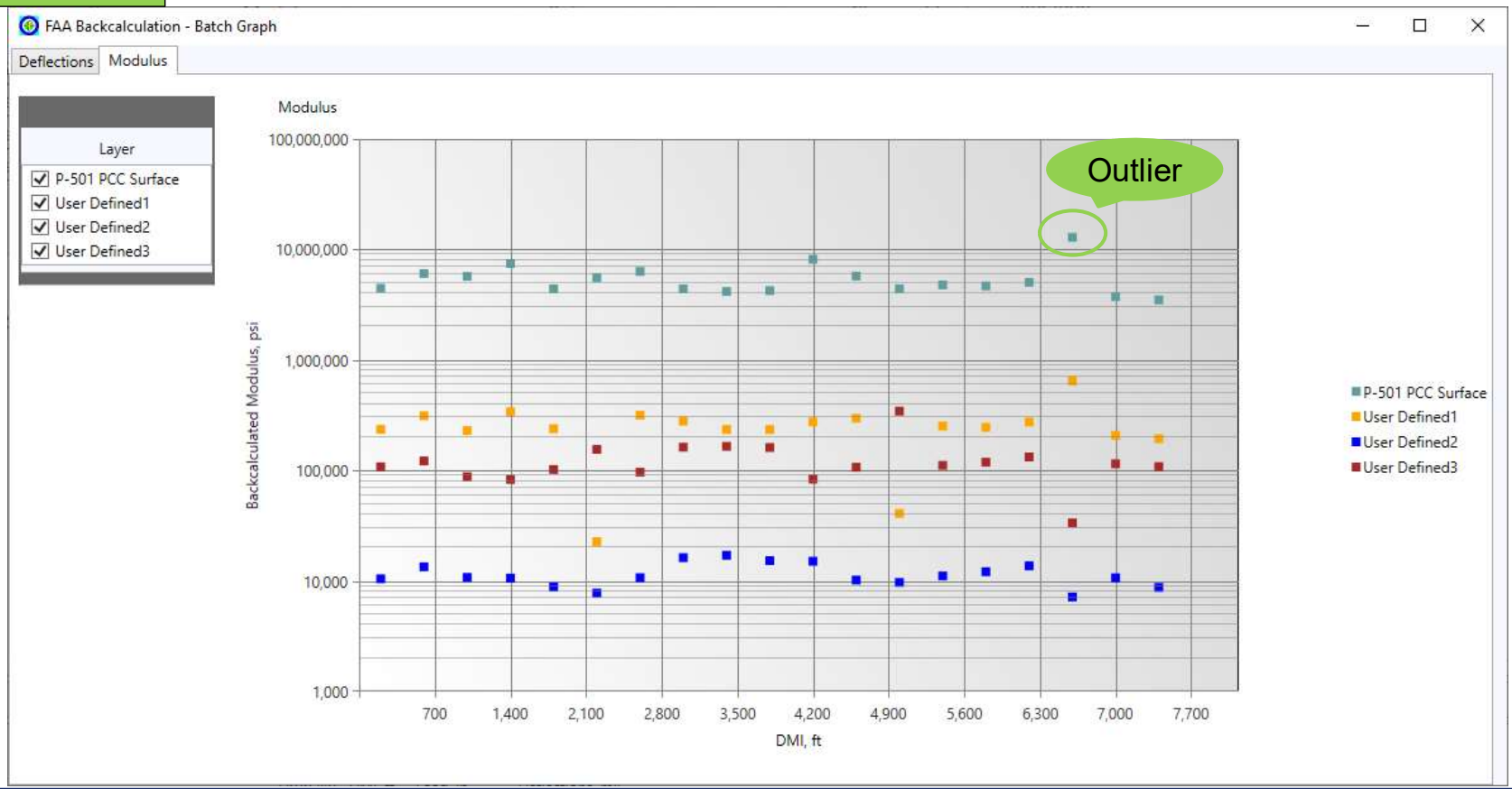


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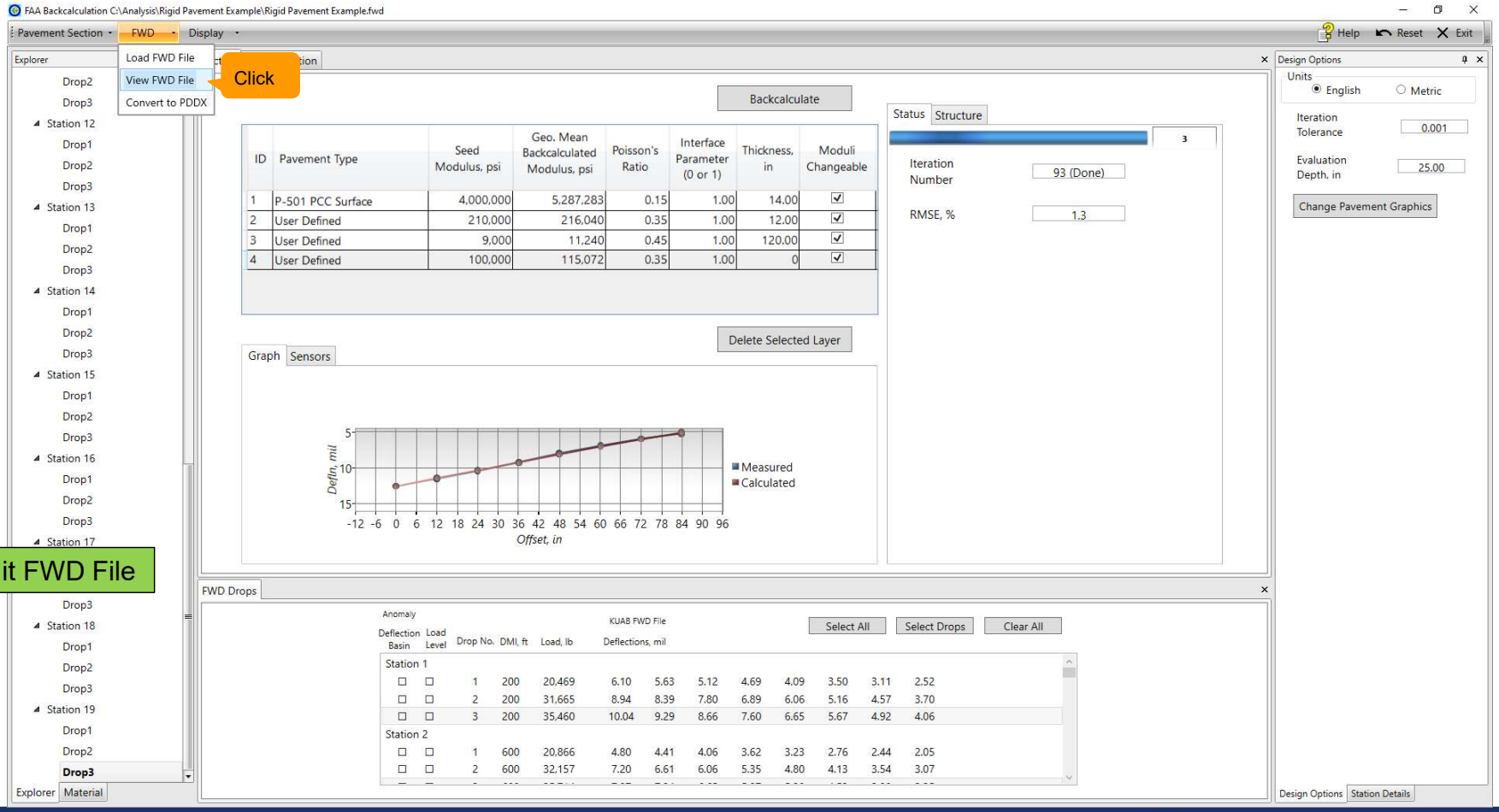


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Batch Graph



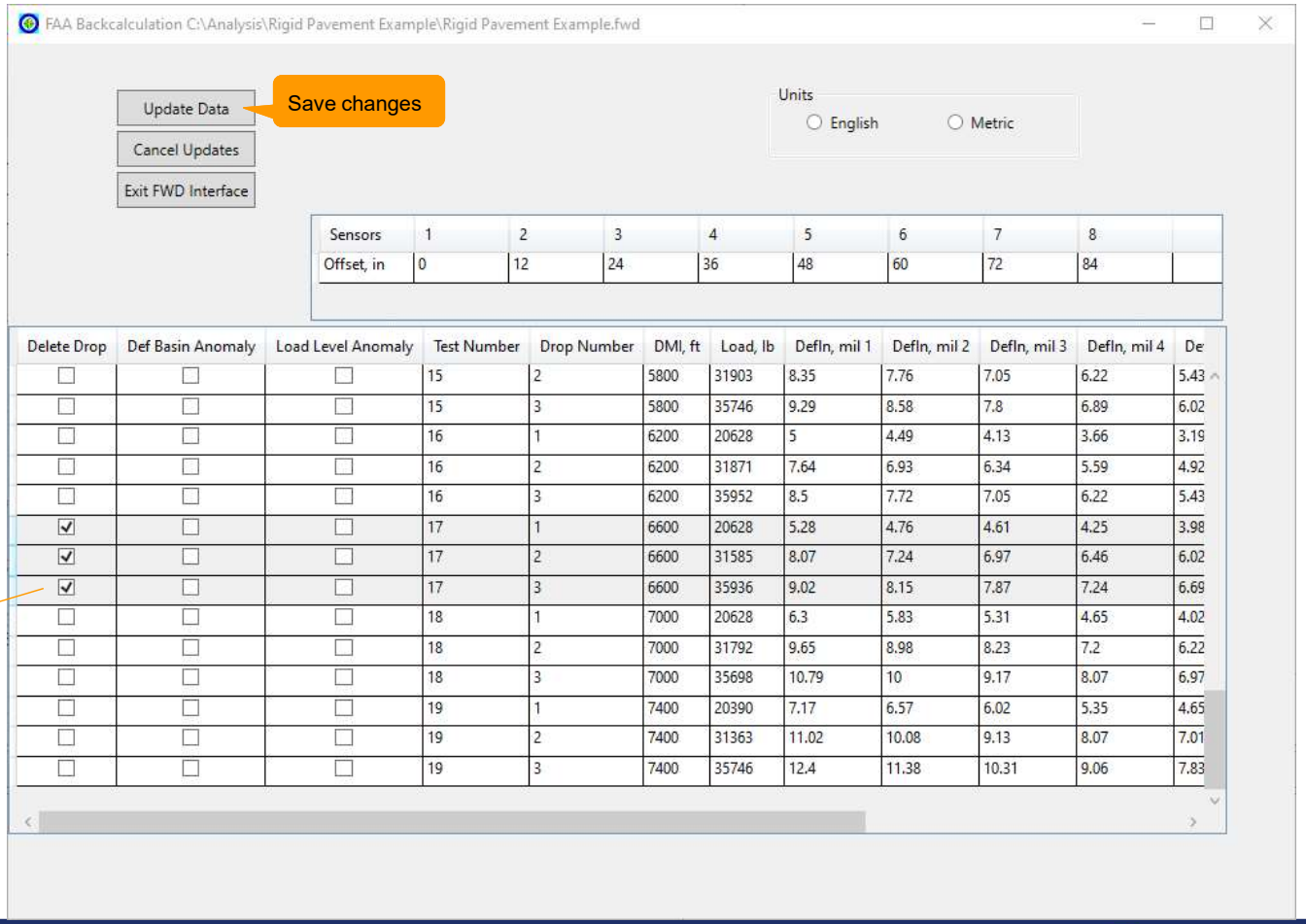
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Edit FWD File

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Edit FWD File



Click to delete outlier drops

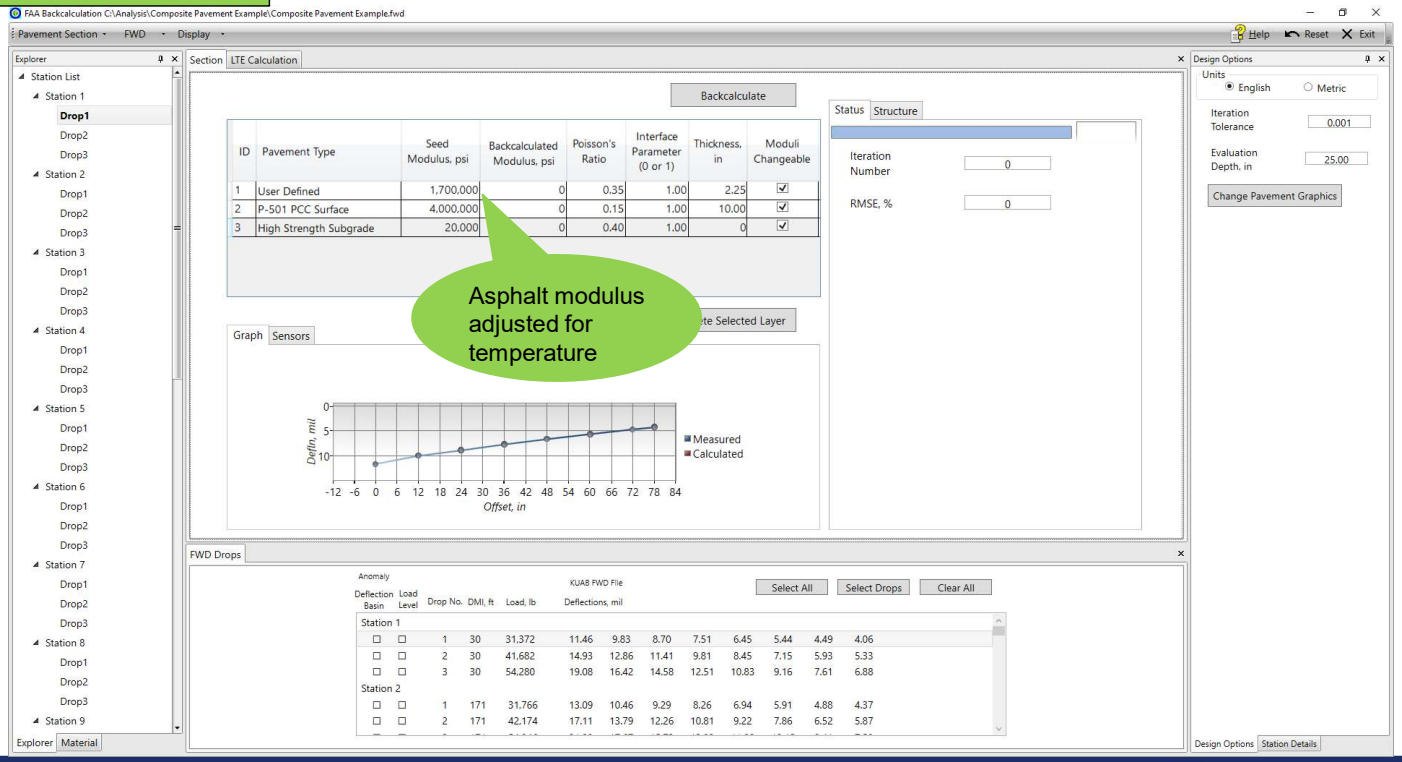
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Composite Pavement Example

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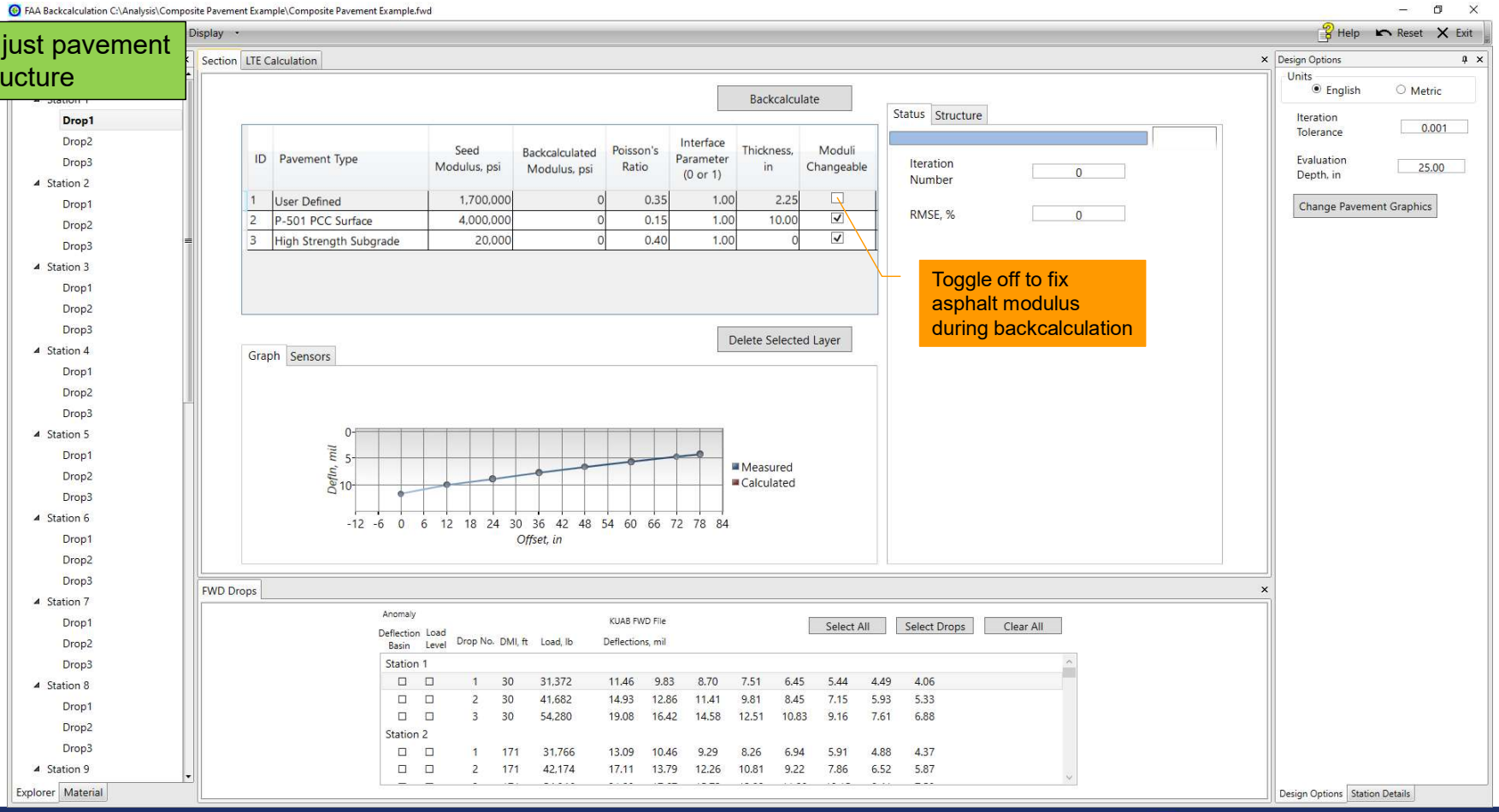
- Load pavement section
- Load FWD file
- Assign layers and thickness
- Assign seed modulus

Material Type	Thickness (in.)
P-401/P-403 hot mix asphalt	2.25
P-501 Portland cement concrete	10
Subgrade with a CBR of 15	

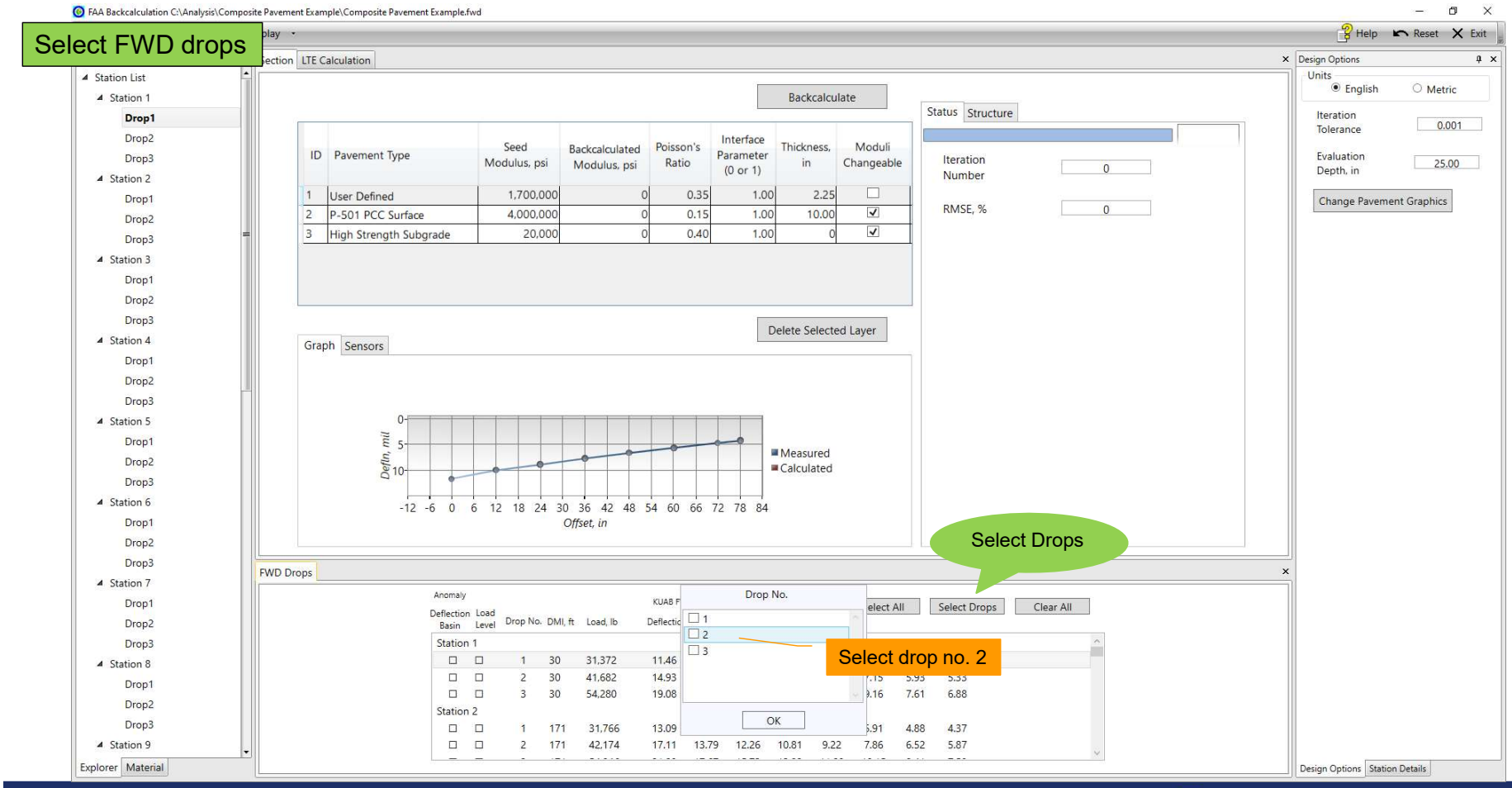


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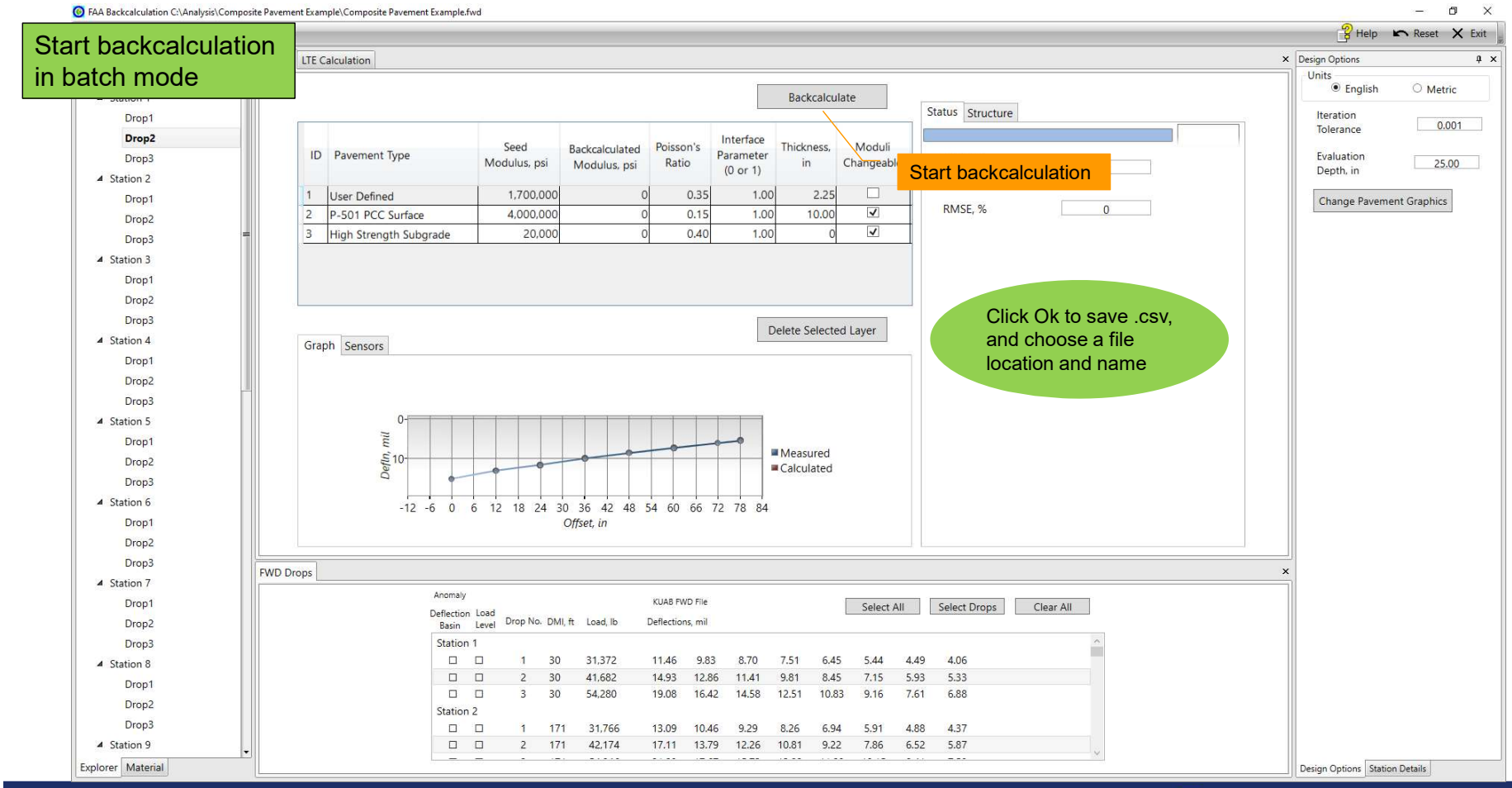
Adjust pavement structure



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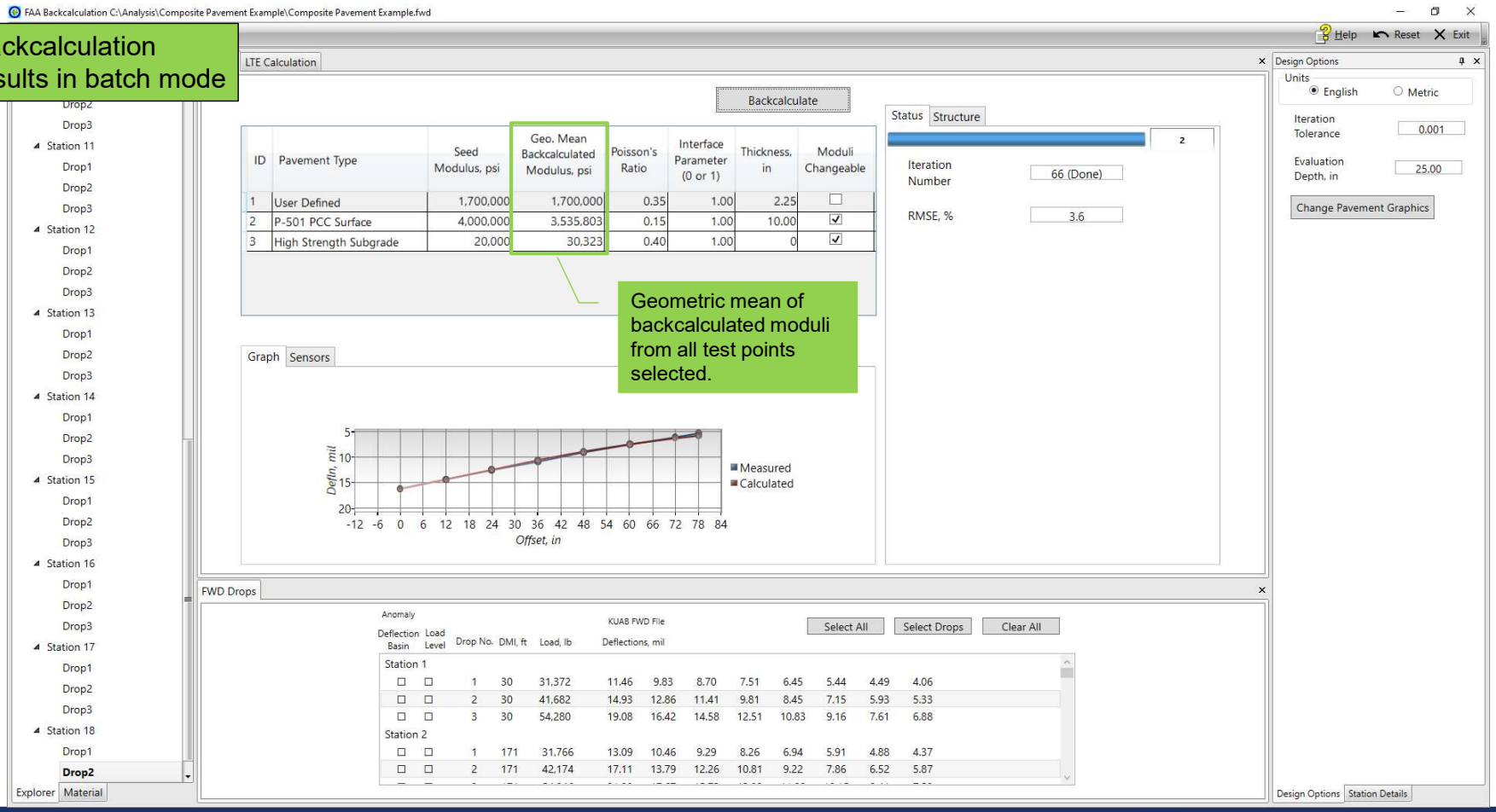


ID	Pavement Type	Seed Modulus, psi	Backcalculated Modulus, psi	Poisson's Ratio	Interface Parameter (0 or 1)	Thickness, in	Moduli Changeabl
1	User Defined	1,700,000	0	0.35	1.00	2.25	☐
2	P-501 PCC Surface	4,000,000	0	0.15	1.00	10.00	☑
3	High Strength Subgrade	20,000	0	0.40	1.00	0	☑

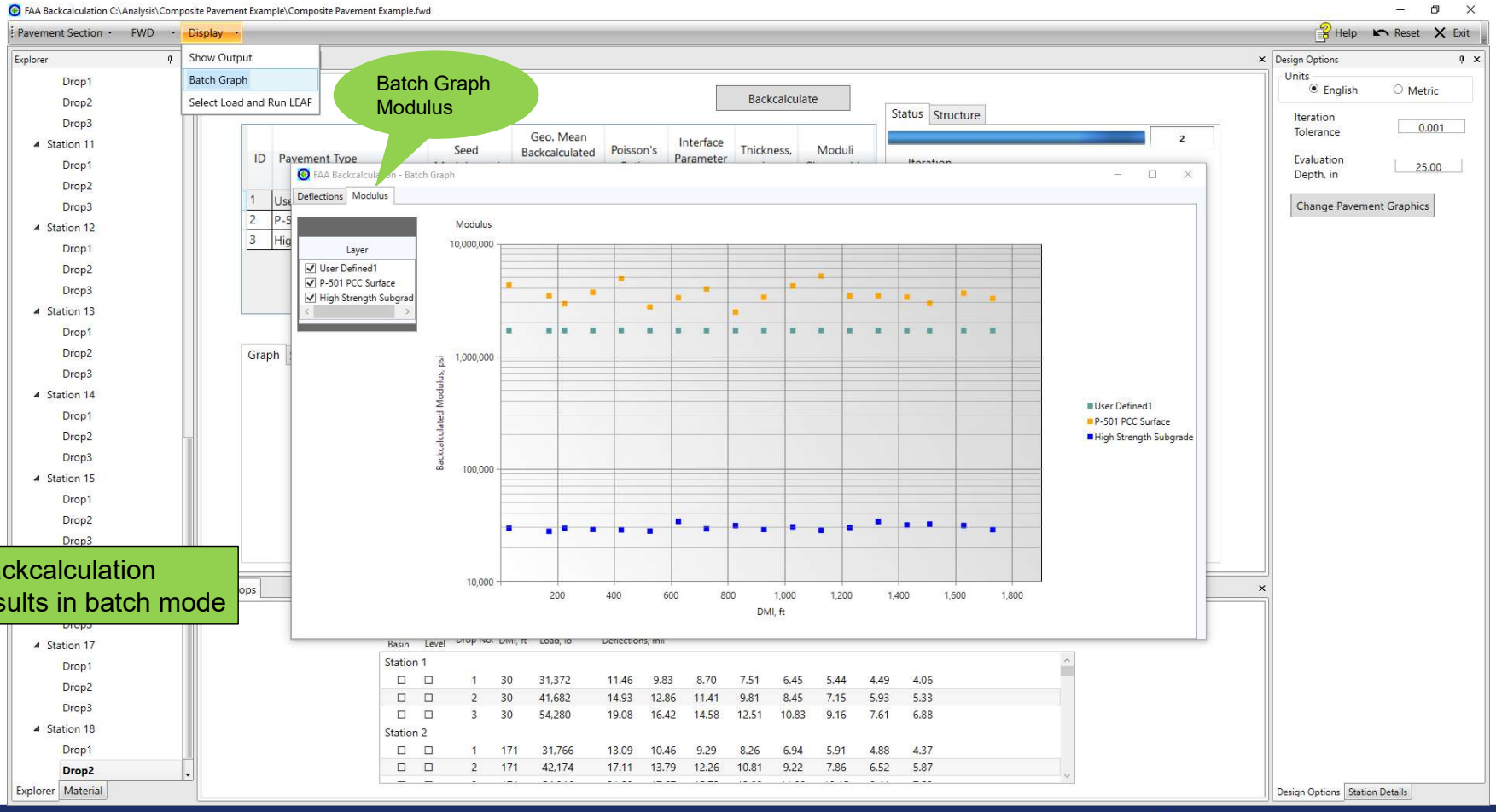
Anomaly Deflection Basin	Anomaly Load Level	Drop No.	DMI, ft	Load, lb	KUAB FWD File Deflections, mil							
Station 1												
☐	☐	1	30	31,372	11.46	9.83	8.70	7.51	6.45	5.44	4.49	4.06
☐	☐	2	30	41,682	14.93	12.86	11.41	9.81	8.45	7.15	5.93	5.33
☐	☐	3	30	54,280	19.08	16.42	14.58	12.51	10.83	9.16	7.61	6.88
Station 2												
☐	☐	1	171	31,766	13.09	10.46	9.29	8.26	6.94	5.91	4.88	4.37
☐	☐	2	171	42,174	17.11	13.79	12.26	10.81	9.22	7.86	6.52	5.87

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Backcalculation results in batch mode



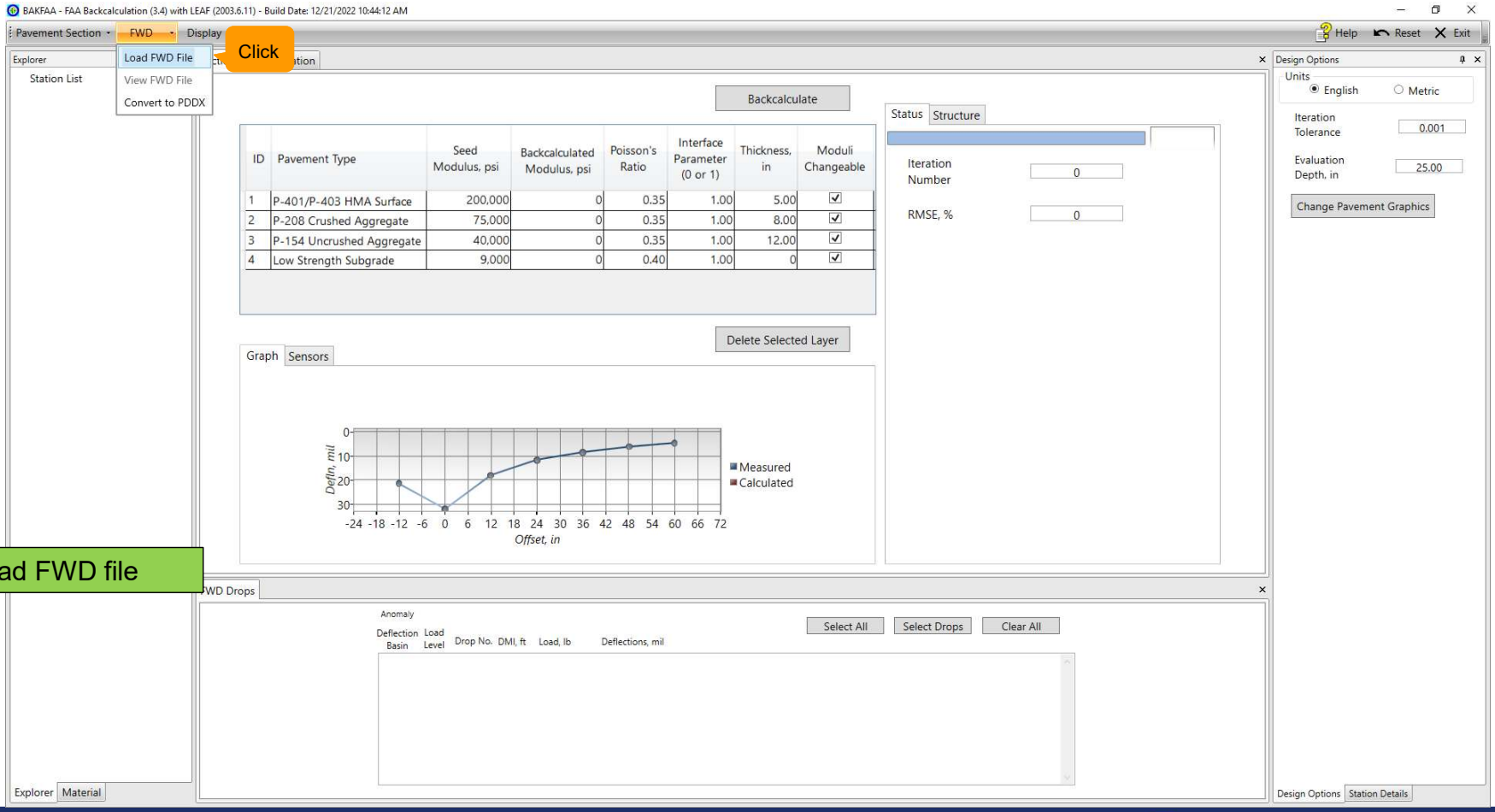
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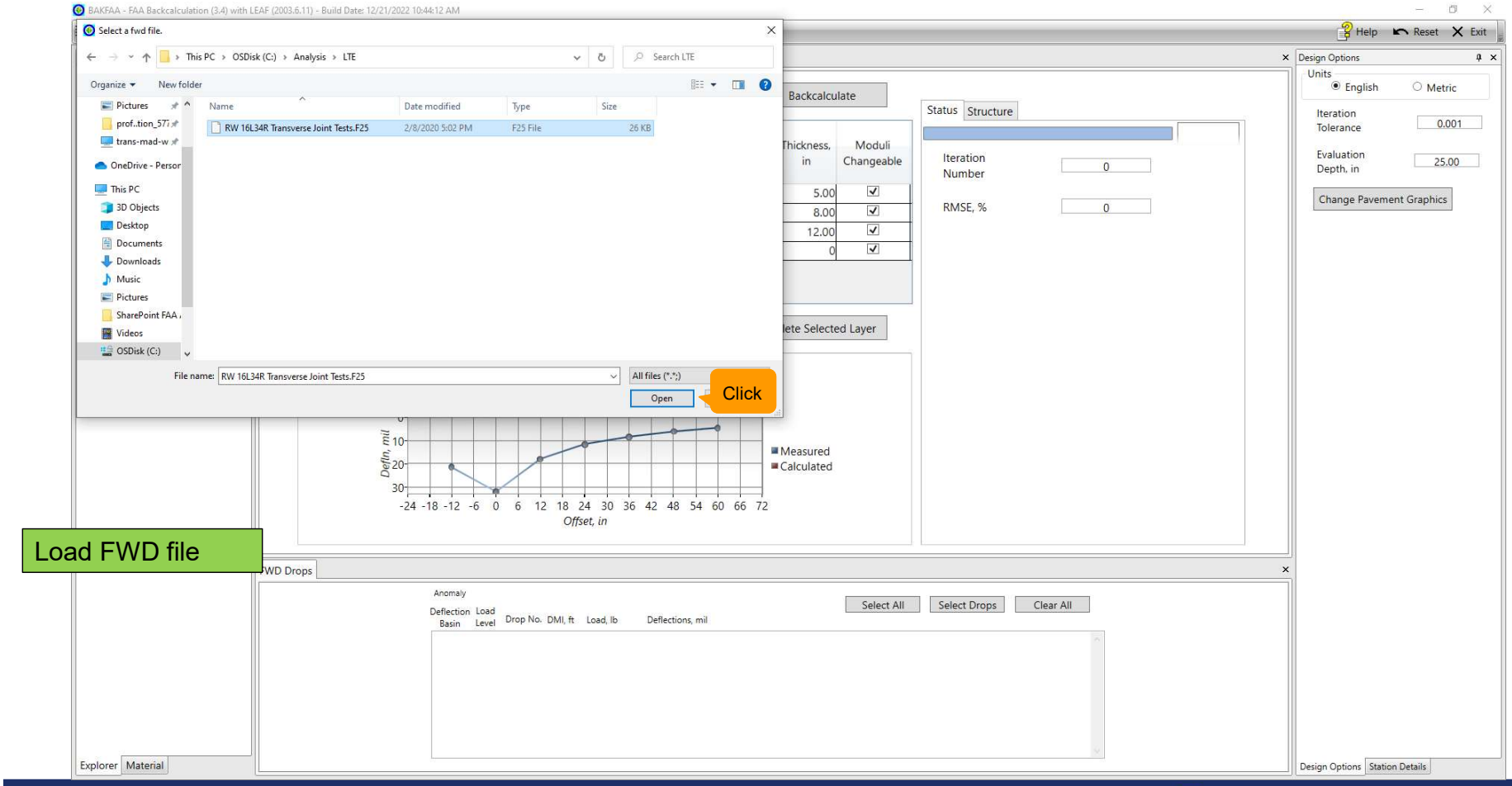
Load Transfer Efficiency (LTE) Example

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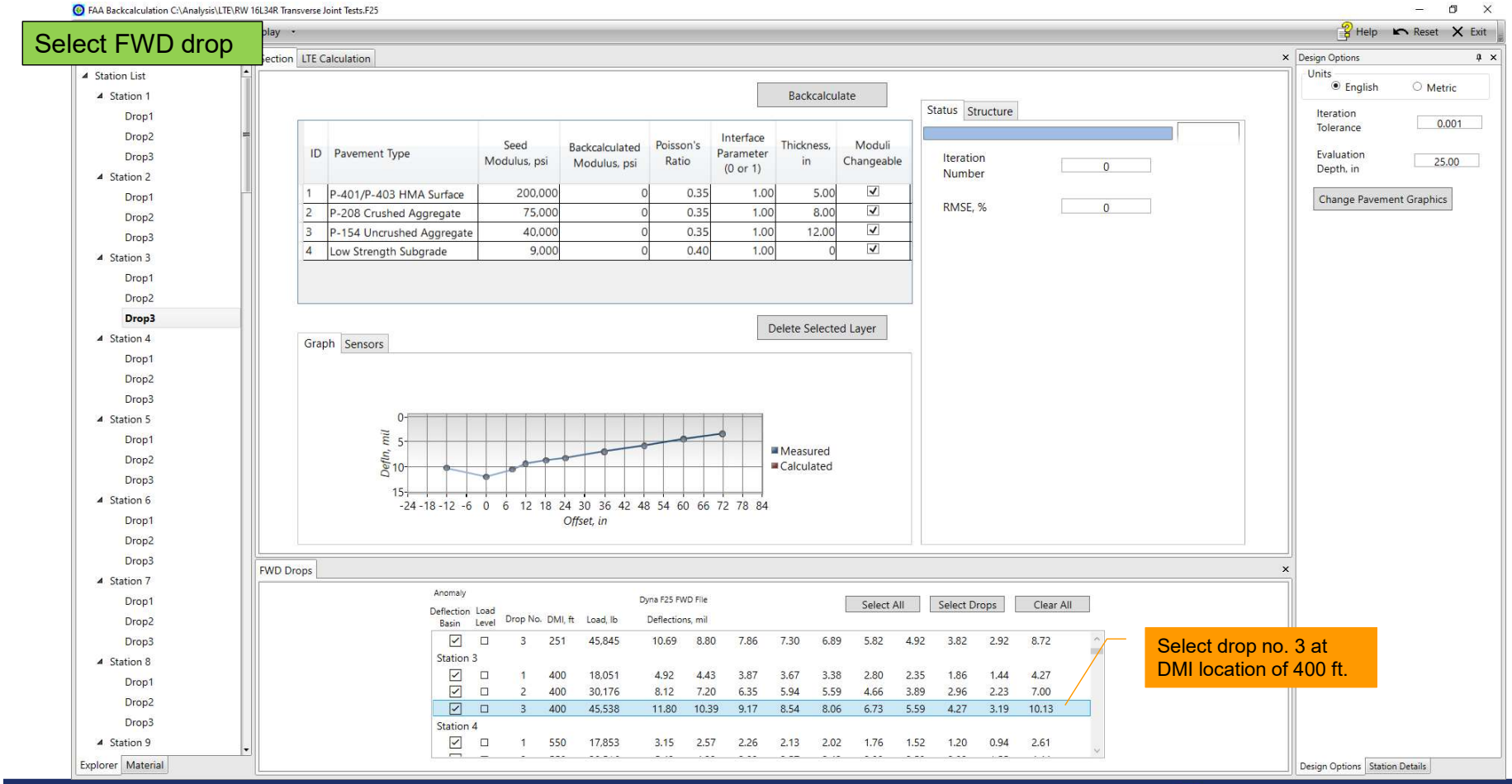


Load FWD file

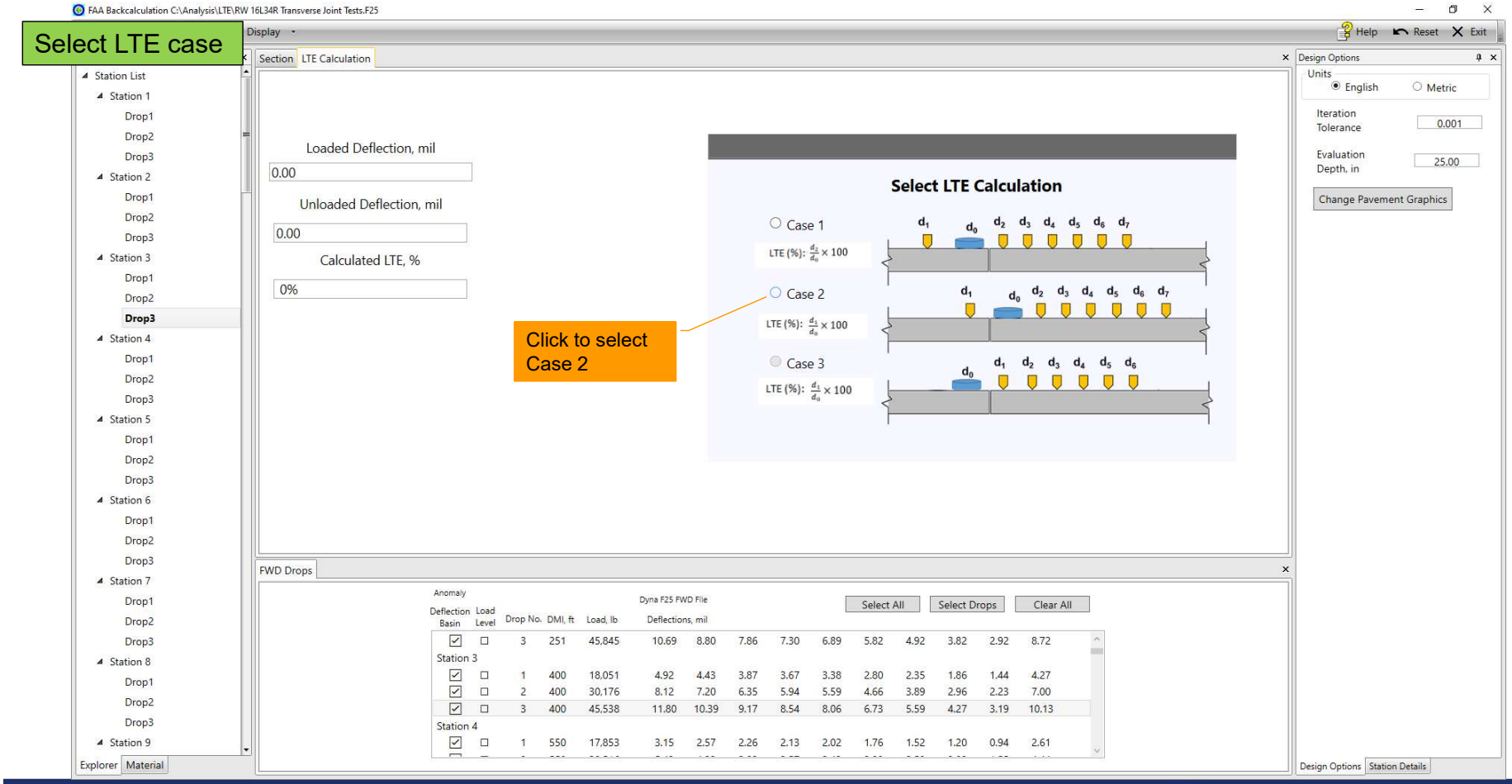
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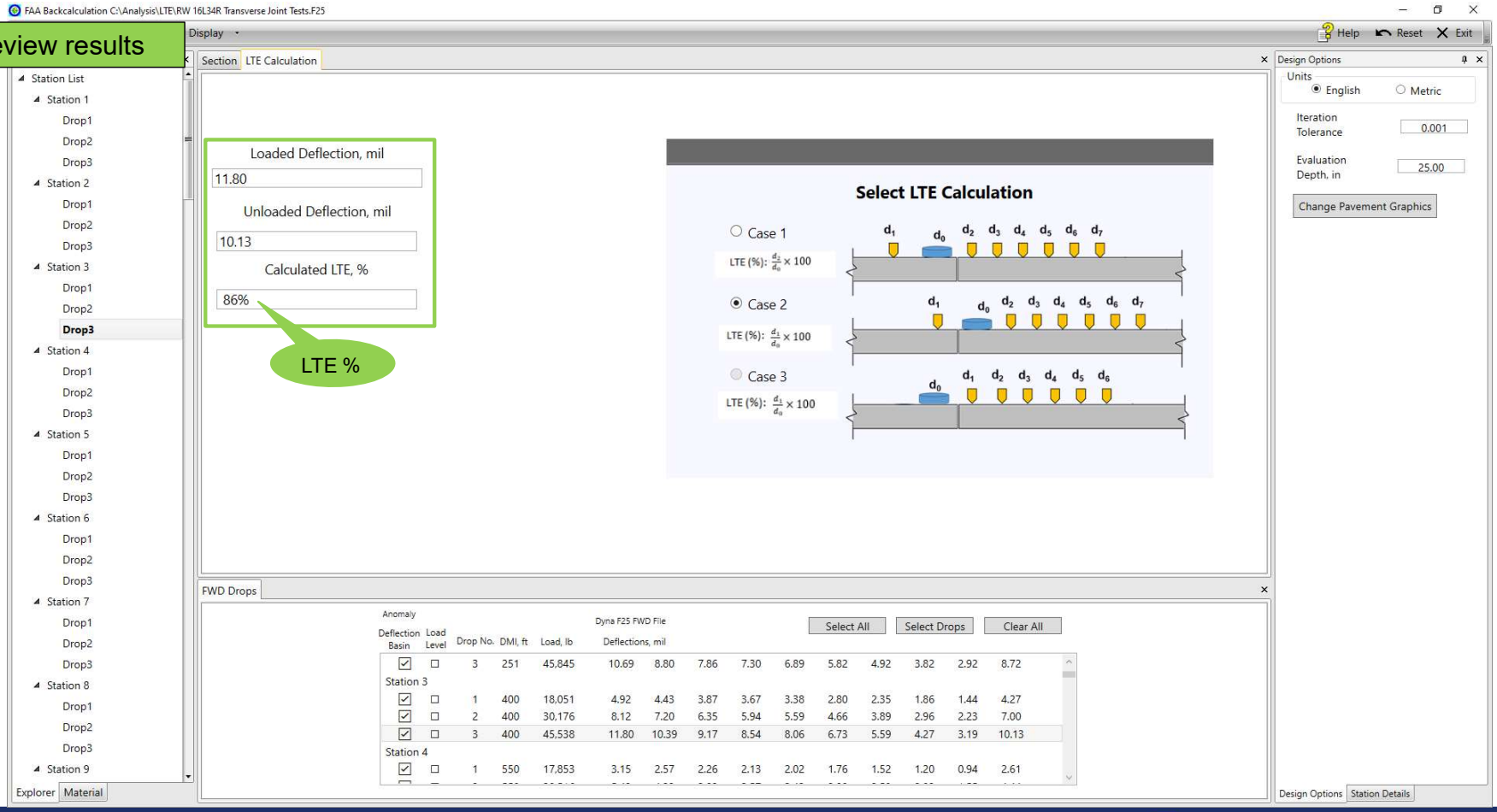


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Review results



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Contact Us

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